GENERIC V-ADIC EQUIDISTRIBUTION FOR SEMIABELIAN VARIETIES WITH MONOCRITICAL ARITHMETICALLY T-EFFECTIVE TORIC METRICS

WEI XUE

Allowed standard prerequisites (editorial).

- Automorphic tensor products, induction/base change and analytic Rankin-Selberg theory. Original complete bibliography and source-local auxiliary L-function arguments retained.

1. Introduction

Heights and equidistribution form one of the main bridges between arithmetic positivity and the geometry of special subvarieties. Weil's theory of local heights [24] shows that global height functions are assembled from local data, while Néron's local height theory [25] gives the canonical local functions behind the Néron–Tate height on abelian varieties. These ideas enter both finiteness results, such as the Mordell–Weil theorem, and density statements for algebraic points of small height.

The Bogomolov conjecture is a central example. Its basic assertion is that a non-torsion subvariety cannot contain a Zariski-dense set of points of arbitrarily small Néron–Tate height. For curves in their Jacobians it was proved by Ullmo [28] and Zhang [29]; Zhang then proved the corresponding theorem for subvarieties of abelian varieties. Zhang's proof introduced adelic line bundles and used equidistribution of small points. Yuan and Zhang later extended this adelic framework and the equidistribution theorem to quasi-projective varieties over global fields [19].

For an integrable adelic line bundle $\overline{L}$ on an irreducible variety Z, its essential minimum is

$$\mu_{\overline{L}}^{\mathrm{ess}}(Z) = \sup_{U \subseteq Z \text{ nonempty open}} \inf_{x \in U(\overline{K})} h_{\overline{L}}(x).$$

It is the smallest height threshold below which algebraic points can be Zariski dense. The formal conventions for heights and small sequences are recalled below.

More precisely, if A is an abelian variety equipped with a symmetric ample line bundle and its canonical metric, and $Z \subseteq A_{\overline{K}}$ is irreducible, the classical Bogomolov theorem may be written

$$\mu_{\overline{L}}^{\mathrm{ess}}(Z) = 0 \quad \Longleftrightarrow \quad Z \text{ is a torsion subvariety.}$$

Here a torsion subvariety is a translate of an abelian subvariety by a torsion point. In this form the conjecture exhibits geometric rigidity directly: vanishing of an arithmetic minimum forces the subvariety, up to translation by a torsion point, to arise from an algebraic subgroup.

The present paper is motivated by the following relative form of the Bogomolov conjecture.

Question 1.1. *Let $\pi : X \to Y$ be a projective flat morphism between quasi-projective varieties over a number field K. Suppose that L is relatively ample and satisfies suitable invariance conditions, for instance that (X, f, L) forms a polarized dynamical system over Y in the sense*

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of [19]. Thus one obtains a nef adelic line bundle $\overline{L}$ on $X/\mathbb{Z}$ with $\mu^{\mathrm{ess}}_{\overline{L}}(X) = 0$. For a closed $\overline{K}$-subvariety $X_1 \subseteq X$, what geometric condition on X_1 implies that, if X_1 is not special, then

$$\mu^{\mathrm{ess}}_{\overline{L}}(X_1) > 0?$$

We list four illustrative cases of this framework.

(1) The classical Bogomolov conjecture for abelian varieties: Y is a point, X is an abelian variety, and the special subvarieties are torsion subvarieties.
(2) Kühne's equidistribution theorem in families [3]: the morphism $\pi : X = \mathscr{U}_g \to Y = \mathbb{A}_g$ is the universal abelian variety, the fibers $X_{1,y}$ are irreducible curves, and the special condition is that each fiber $X_{1,y}$ is a translate of an elliptic curve.
(3) The relative Bogomolov conjecture [33]: the morphism $\pi : X = \mathcal{A} \to Y = S$ is a relative abelian scheme, one assumes $\overline{\mathbb{Z}X_1} = \mathcal{A}$, and the special condition is governed by the expected codimension inequality $\operatorname{codim}_{\mathcal{A}} X_1 \leq \dim S$.
(4) The Bogomolov conjecture for semiabelian varieties, as treated by Kühne [1]: $X = G$ is a semiabelian variety, $Y = A$ is its abelian quotient, and the special subvarieties are translates of algebraic subgroups of G by torsion points.

The full relative Bogomolov conjecture remains open. Gao proved a weaker relative Manin–Mumford statement [34]: if X_1 is not special, then its torsion points cannot be Zariski dense in X_1. Equidistribution is the main proof mechanism in Cases (1), (2), and (4), and it is the point of departure for the present work on semiabelian varieties.

Question 1.2. *Let $\pi : X \to Y$ be a projective flat morphism between quasi-projective varieties over a number field K. Let L be big and nef, and let $\overline{L}$ be a nef adelic line bundle on X. Assume that the Deligne pairing*

$$\big(\langle \overline{L}, \dots, \overline{L} \rangle, Y\big)$$

and every fiber pair $(\overline{L}|_{X_y}, X_y)$ satisfy equidistribution. What additional hypotheses on $\overline{L}$ imply equidistribution for $(\overline{L}, X)$?

Existing proofs for Cases (1) and (2) use Yuan–Zhang's equidistribution theorem [19]. In Case (4), Kühne [1] treats a canonical semiabelian metric: $X = G$ is semiabelian, $Y = A$ is its abelian quotient, and

$$\overline{L} = \overline{G}(\overline{D}^{\mathrm{can}}) \otimes \overline{\pi}^* \overline{N}.$$

Here T-effectivity of D controls the Deligne pairing and leads to equidistribution. The key difficulty is that the metric defining $\overline{L}$ is generally not quasi-canonical in the sense required by Yuan–Zhang; one may have

$$h_{\overline{L}}(\overline{G}) < \mu^{\mathrm{ess}}_{\overline{L}}(\overline{G}),$$

so Yuan and Zhang's theorem cannot be applied directly.

The present paper studies this gap for toric metrics on semiabelian compactifications. Its scope is the class of semipositive, monocritical and arithmetically T-effective toric metrics specified below; no assertion is made for arbitrary semipositive toric metrics. The precise comparison with earlier work is given immediately after the main theorems.

Setup. Let G be a semiabelian variety over a number field K, with split toric part T of dimension t and maximal abelian quotient $\pi : G \to A$ of dimension g. Let X_Σ be a normal proper toric variety whose open dense torus is T, and let $\overline{D} = (D, \{\|\cdot\|_v\})$ be a toric metrized $\mathbb{R}$-divisor on X_Σ. We write $\overline{G}$ for the toric compactification of G associated to X_Σ, with projection $\overline{\pi} : \overline{G} \to A$.

In Section 4, we construct an adelic line bundle $\overline{G}(\overline{D})$ on $\overline{G}/\mathbb{Z}$ via local trivializations valid at every place of K, extending Kühne's archimedean local trivialization to all valuations. Fix a symmetric ample line bundle $\overline{N}$ on A, equipped with its canonical metric, and define $\overline{L} := \overline{G}(\overline{D}) \otimes \overline{\pi}^* \overline{N}$.

Theorem 1.3 (Generic equidistribution)**.** *Suppose that D is big and T-effective, and that $\overline{D} = (D, \{\|\cdot\|_v\})$ is a semipositive, monocritical and arithmetically T-effective toric metrized $\mathbb{R}$-divisor in the sense of Definitions 3.12 and 4.20. Then every generic $\overline{L}$-small sequence satisfies v-adic equidistribution at every place v of K. If $\overline{D}$ is additionally quasi-canonical, the limiting measure is*

$$\mu_{\overline{L},v} = \frac{\hat{c}_1(\overline{G}(\overline{D})_v)^t \wedge \hat{c}_1(\overline{\pi}^*\overline{N}_v)^g}{\overline{G}(D)^t(\overline{\pi}^*N)^g}.$$

Theorem 1.4 (Bogomolov theorem for the paper's metric class)**.** *Let $X \subseteq G_{\overline{K}}$ be irreducible, and assume that D is ample and T-effective and that $\overline{D}$ is semipositive, monocritical and arithmetically T-effective. At every archimedean place, assume in addition either the standard $(\mathbb{P}^1)^t$-compactification used by Kühne, or that the general-fan curvature–Haar slice formula 9.5, including its lattice-index and boundary no-mass assertions, has been independently established for every subvariety occurring in the difference-morphism argument. Put*

$$\overline{L} = \overline{G}(\overline{D}) \otimes \overline{\pi}^*\overline{N}.$$

(1) *If*
$$\mu_{\overline{L}}^{\mathrm{ess}}(X) = \mu_{\overline{L}}^{\mathrm{ess}}(\overline{G}),$$
then X is a special subvariety.

(2) *If $G(\overline{K})$ has $\overline{L}$-special points, then*
$$\mu_{\overline{L}}^{\mathrm{ess}}(X) = \mu_{\overline{L}}^{\mathrm{ess}}(\overline{G})$$
if and only if X is $\overline{L}$-special.

What was known and what is new. In the notation above, the antecedents divide into four distinct inputs. Yuan–Zhang [19] provide the adelic height and generic-equidistribution framework, with direct application requiring the relevant quasi-canonical equality. Kühne [1] proves canonical semiabelian equidistribution and Bogomolov by local trivializations, Haar slices and difference morphisms. BGPS [16] isolate monocritical toric metrics and their critical tropical data on pure toric varieties. More recently, Ballay–Sombra [8] develop approximation beyond the quasi-canonical equality case, whereas Hultberg [9] computes heights, minima and intersections for toric bundles. The latter height-theoretic results are not, by themselves, the strict Bogomolov bridge used here.

The new point of Theorems 1.3 and 1.4 is the interaction between a semiabelian extension and a non-canonical toric metric. The proof compresses the metric toward the quasi-canonical model singled out by its critical tropical center, transports Kühne's Haar-slice and difference-morphism argument from the standard compactification to a compactification satisfying the stated Haar-slice input, and then combines monocritical concentration with restriction to subvarieties. This yields the metric-dependent implication

$$\mu_{\overline{L}}^{\mathrm{ess}}(X) = \mu_{\overline{L}}^{\mathrm{ess}}(\overline{G}) \quad \Longrightarrow \quad X \text{ is special},$$

with the converse under the special-point hypothesis stated in Theorem 1.4. Thus the contribution is neither a new proof of the canonical theorem nor a theorem for all semipositive metrics: it is the extension to the stated metric class, organized by its critical tropical data.

Proof strategy. The strategy is stated only after the results and their relation to the literature have been made explicit. The generic equidistribution theorem is proved internally; the Bogomolov theorem then uses the Kühne transport assembly established in Proposition 100.

The proof of the generic equidistribution theorem decomposes into four distinct stages:

(1) Reduce to the ample case by a birational modification of X_Σ.
(2) Prove the canonical metric case by extending Kühne's asymptotic estimates to arbitrary ample toric divisors.

(3) Treat quasi-canonical metrics by constructing explicit compression paths and proving the corresponding lower-bound, measure, first-variation, and arithmetic-volume estimates.
(4) Pass to monocritical metrics using tropical measure theory and arithmetical T-effectivity, reducing the limiting measure to the quasi-canonical model.

2. Conventions and terminology

Asymptotic notation. For two non-negative quantities a and b, we write $a \ll b$ if there is a constant $c > 0$ such that $a \leq cb$. If c is allowed to depend on auxiliary data, for instance an algebraic variety X, we write $a \ll_X b$. We use $\gg$ with the analogous meaning.
Number fields. Throughout the paper, $K \subseteq \overline{\mathbb{Q}}$ is a number field with ring of integers O_K, and $S = \mathrm{Spec}(O_K)$. We denote by $\Sigma(K)$ the set of places of K, and by $\Sigma_f(K)$ and $\Sigma_\infty(K)$ the non-archimedean and archimedean places, respectively. For $v \in \Sigma(K)$, let K_v be the v-adic completion of K, let $\mathbb{C}_v$ be the completion of an algebraic closure of K_v, let $O_v \subseteq \mathbb{C}_v$ be its ring of integers, let $\boldsymbol{p}_v$ be the maximal ideal of O_v, and let $k_v = O_v/\boldsymbol{p}_v$.
Generic sequences. Let X be an algebraic K-variety. A sequence $(x_i)_{i\geq 1}$ of closed points of X is called X-generic if no subsequence is contained in a proper algebraic subvariety of X. When X is clear from context, we simply say generic.
Small sequences. Let X be a quasi-projective algebraic K-variety, and let $\overline{L}$ be an integrable adelic line bundle on $X/\mathbb{Z}$. A sequence $(x_i)_{i\geq 1}$ of closed points of X is called $\overline{L}$-small if

$$h_{\overline{L}}(x_i) \to \mu_{\overline{L}}^{\mathrm{ess}}(X) \qquad (i \to \infty).$$

3. Preliminary

In this section, we summarize the basic facts that are essential for the proof.

3.1. **Valuation theory.** A global field K is either a number field or the function field of a regular projective curve over a field. We equip K with a set of places $\Sigma(K)$. A place $v \in \Sigma(K)$ consists of an absolute value $|\cdot|_v$ on K, together with a positive weight n_v.

For $K_0 = \mathbb{Q}$, the places are the usual archimedean and p-adic absolute values, with all weights equal to 1. For a function field $K_0 = k(C)$, where C is a regular projective curve over k, the places are indexed by closed points $v_0 \in C$. For $\alpha \in k(C)^\times$, we set

$$|\alpha|_{v_0} = c_k^{-\operatorname{ord}_{v_0}(\alpha)}, \qquad n_{v_0} = [k(v_0) : k],$$

where $c_k = e$ if k is infinite and $c_k = \#k$ if k is finite.

Let K/K_0 be a finite extension. A place v of K extending $v_0 \in \Sigma(K_0)$ is assigned the weight

$$N_v^K = [K_v : K_{0,v_0}] n_{v_0}, \qquad n_v^K = \frac{N_v^K}{[K : K_0]}.$$

We call N_v^K the raw local degree and n_v^K the normalized weight. Unless the superscript "raw" is displayed, every height, adelic intersection and adelic measure in this paper uses normalized weights. Raw degrees are only intermediate bookkeeping and are divided by the global degree before any height is formed. From now on we suppress the field superscript and write $n_v = n_v^K$ for the normalized weight. Then

$$\sum_{v|v_0} N_v^K = [K : K_0] n_{v_0}, \qquad \sum_{v|v_0} n_v^K = n_{v_0}, \qquad \prod_{v\in\Sigma(K)} |\alpha|_v^{n_v^K} = 1 \quad (\alpha \in K^\times).$$

3.2. **Toric varieties.** Let $T \cong \mathbb{G}_m^t$ be a split torus of dimension t over K. Let $N = \mathrm{Hom}(\mathbb{G}_m, T)$ be the lattice of cocharacters of T, let $M = \mathrm{Hom}(T, \mathbb{G}_m) = N^\vee$ be the lattice of characters, and write

$$N_{\mathbb{R}} = N \otimes \mathbb{R}, \qquad M_{\mathbb{R}} = M \otimes \mathbb{R}.$$

Definition 3.1. *Let $\mathscr{P}$ be a non-empty collection of convex subsets of $N_{\mathbb{R}}$. We call $\mathscr{P}$ a convex subdivision if*

(1) *every face of an element of $\mathscr{P}$ belongs to $\mathscr{P}$;*
(2) *any two elements of $\mathscr{P}$ are either disjoint or intersect in a common face.*

The support of $\mathscr{P}$ is

$$|\mathscr{P}| = \bigcup_{C \in \mathscr{P}} C.$$

The subdivision is complete if $|\mathscr{P}| = N_{\mathbb{R}}$.

Definition 3.2. *A polyhedral complex in $N_{\mathbb{R}}$ is a finite convex subdivision whose elements are polyhedra. It is strongly convex if all of its polyhedra are strongly convex, and conic if all of its elements are cones. A strongly convex conic polyhedral complex in $N_{\mathbb{R}}$ is called a fan.*

Let X be a proper toric variety over K with torus T, described by a complete fan Σ on $N_{\mathbb{R}}$. To each cone $\sigma \in \Sigma$ there correspond an open affine subset X_σ and an orbit $O(\sigma)$. For $\sigma = \{0\}$, the principal open subset X_0 agrees with the orbit $O(0)$, and is canonically isomorphic to T. The action of T on X is denoted by $(t, p) \mapsto t \cdot p$.

Definition 3.3. *Let Σ be a fan in $N_{\mathbb{R}}$. A function $\Psi : |\Sigma| \to \mathbb{R}$ is called a virtual support function on Σ if it is a conic H-lattice function in the sense of [13, Definition 2.6.6]; equivalently, for every cone $\sigma \in \Sigma$, there is $m_\sigma \in M$ such that*

$$\Psi(u) = \langle m_\sigma, u \rangle \qquad (u \in \sigma).$$

The m_σ are called defining vectors of Ψ. A concave virtual support function on a complete fan is called a support function.

A toric $\mathbb{R}$-divisor on the proper toric variety X is an $\mathbb{R}$-divisor invariant under the action of T. Such a divisor D defines a function $\Psi_D : N_{\mathbb{R}} \to \mathbb{R}$ whose restriction to each cone of Σ is linear. We also attach to D the stability set

$$\Delta_D = \operatorname{stab}(\Psi_D) \subseteq M_{\mathbb{R}}.$$

In the sequel, toric varieties are assumed to be proper; equivalently, Σ is a complete fan.

Proposition 3.4. *The following is [17, Propositions 4.7–4.9]. Let D be a toric $\mathbb{R}$-divisor on X. Then:*

(1) *D is ample if and only if Ψ_D is strictly concave.*
(2) *D is nef if and only if Ψ_D is concave, equivalently D is globally generated.*
(3) *D is big if and only if $\dim \Delta_D = \dim X$.*
(4) *D is pseudo-effective if and only if $\Delta_D \neq \emptyset$.*
(5) *D is T-effective if and only if $\Psi_D \leq 0$, equivalently $0 \in \Delta_D$.*

Now let K be a global field. For each place $v \in \Sigma(K)$, the algebraic torus T has an analytic space T_v^{an}, and we denote its compact torus by $\mathbb{S}_v^{an}$. At an archimedean place, this compact torus is isomorphic to $(S^1)^t$. In the non-archimedean case, it is a compact analytic group; see [13, §4.2].

Proposition 3.5. *By [13, Chapter 4.1], there is a commutative diagram*

$$\begin{array}{ccc} & & X_{0,v}^{an} \\ & {\scriptstyle \mathrm{val}_v} \swarrow & \big\downarrow {\scriptstyle \rho_0} \\ N_{\mathbb{R}} & \xrightarrow{\boldsymbol{e}} & X_0(\mathbb{R}_{\geqslant 0}) \end{array}$$

which extends to X_v^{an}:

$$\begin{array}{ccc} & & X_{v}^{an} \\ & {\scriptstyle \mathrm{val}_v} \swarrow & \big\downarrow {\scriptstyle \rho_\Sigma} \\ N_{\Sigma} & \xrightarrow{\boldsymbol{e}} & X(\mathbb{R}_{\geqslant 0}) \end{array}$$

where the spaces with corners N_Σ and $X(\mathbb{R}_{\geqslant 0})$ are compactifications of $N_\mathbb{R}$ and $X_0(\mathbb{R}_{\geqslant 0})$, respectively.

Proposition 3.6. *By [13, Definition 4.2.12], there exists a continuous proper section θ_Σ of ρ_Σ whose image is contained in $\mathbb{S}_v^{an}$.*

Definition 3.7. *A v-adically metrized $\mathbb{R}$-divisor $\overline{D} = (D, \|\cdot\|)$ on X is toric if D is a toric $\mathbb{R}$-divisor and its Green function is invariant under the action of $\mathbb{S}_v^{an}$, equivalently constant on the fibers of ρ_0. We then define*

$$\Psi_{\overline{D},v}(u) = -g_{\overline{D}}(p),$$

where $p \in X_{0,v}^{an}$ is any point with $\mathrm{val}_v(p) = u$.

A quasi-algebraic metrized $\mathbb{R}$-divisor $\overline{D} = (D, \{\|\cdot\|_v\})$ is toric if each $(D, \|\cdot\|_v)$ is toric. In this case we call $\overline{D}$ a toric metrized $\mathbb{R}$-divisor. The canonical v-adic toric metric on D is characterized by $\Psi_{\overline{D},v} = \Psi_D$.

Corner complexes and dual cells. The following convex-geometric objects will be used in Chapter 9. Let $\Psi : N_\mathbb{R} \to \mathbb{R}$ be a continuous piecewise-affine concave function. Its *corner complex* $\mathcal{C}(\Psi)$ is the polyhedral complex formed by the maximal domains on which Ψ is affine and all their common faces. Its codimension-one cells are called *walls*, and a codimension-r cell is called an *r-corner*.

If τ is a cell and $u \in \mathrm{relint}(\tau)$, define its dual cell by

$$\Delta_\tau := \partial^+\Psi(u) = \{m \in M_\mathbb{R} : \Psi(w) \leq \Psi(u) + \langle m, w - u\rangle \text{ for all } w\}.$$

This set is independent of u in the relative interior of τ. If locally $\Psi = \min_\alpha(\langle m_\alpha, \cdot\rangle + c_\alpha)$, then Δ_τ is the convex hull of the slopes of the branches active on τ. Under the usual regularity hypotheses, $\dim \Delta_\tau = \mathrm{codim}\, \tau$. Thus higher-codimension corner contributions are governed in general by the volume, or mixed volume, of the dual cells. A decomposition into independent one-wall currents is available only under the additional product-corner condition that the relevant dual cell is a Minkowski sum of independent segments.

For $\Psi = \Psi_{\overline{D},v}$, these definitions apply to the local toric weight introduced above. The critical vector $(u_v)_v$ of a monocritical metric, defined below, specifies the tropical translations used in the quasi-canonical replacement; it should not be confused with a cell of $\mathcal{C}(\Psi_{\overline{D},v})$.

Proposition 3.8. *Let D be a toric $\mathbb{R}$-divisor on X.*

1. *For a place v, the correspondence $\|\cdot\| \mapsto \Psi_{\overline{D},v}$ is a bijection between toric v-adic metrics on D and functions $\Psi : N_\mathbb{R} \to \mathbb{R}$ such that $\Psi - \Psi_D$ extends continuously to N_Σ.*
2. *The correspondence*

$$\{\|\cdot\|_v\} \longmapsto \{\Psi_{\overline{D},v}\}$$

is a bijection between quasi-algebraic toric metrics on D and families $\{\Psi_v\}_{v\in\Sigma(K)}$ satisfying the preceding continuity condition for every v, and $\Psi_v = \Psi_D$ for all but finitely many v.

Definition 3.9. *Let $\overline{D}$ be a toric metrized $\mathbb{R}$-divisor on X. For each $v \in \Sigma(K)$, the v-adic roof function of D is the concave function on Δ_D defined by*

$$\upsilon_{\overline{D},v} = \Psi_{\overline{D},v}^\vee.$$

The global roof function is

$$\upsilon_{\overline{D}} = \sum_{v\in\Sigma(K)} n_v \upsilon_{\overline{D},v}.$$

Proposition 3.10. *The following characterization is [17, Theorem 6.1]. Let D be a toric metrized $\mathbb{R}$-divisor on X, and let $\upsilon_{\overline{D}} : \Delta_D \to \mathbb{R}$ be its global roof function. Then:*

1. *$\overline{D}$ is ample if and only if Ψ_D is strictly concave on Σ, each $\Psi_{\overline{D},v}$ is concave, and $\upsilon_{\overline{D}}(x) > 0$ for all $x \in \Delta_D$.*

(2) $\overline{D}$ *is nef if and only if each* $\Psi_{\overline{D},v}$ *is concave and* $\upsilon_{\overline{D}}(x) \geq 0$ *for all* $x \in \Delta_D$.
(3) $\overline{D}$ *is big if and only if* $\dim \Delta_D = \dim X$ *and* $\upsilon_{\overline{D}}(x) > 0$ *for some* $x \in \Delta_D$.
(4) $\overline{D}$ *is pseudo-effective if and only if* $\upsilon_{\overline{D}}(x) \geq 0$ *for some* $x \in \Delta_D$.
(5) $\overline{D}$ *is effective if and only if* $0 \in \Delta_D$ *and* $\upsilon_{\overline{D},v}(0) \geq 0$ *for every* v.
(6) $\overline{D}$ *is vertically semipositive if and only if each* $\Psi_{\overline{D},v}$ *is concave.*

Proposition 3.11. *By [18, Theorem 3.9 and Corollary 3.10], let* $\overline{D}$ *be a toric metrized* $\mathbb{R}$*-divisor on* X*. Then*

$$\mu^{\mathrm{ess}}_{\overline{D}}(X) = \mu^{\mathrm{abs}}_{\overline{D}}(X_0) = \max_{x \in \Delta_D} \upsilon_{\overline{D}}(x).$$

Let $\Delta_{D,\max}$ be the locus where $\upsilon_{\overline{D}}$ attains its maximum.

Definition 3.12. *A vertically semipositive toric metrized* $\mathbb{R}$*-divisor* $\overline{D}$ *is called monocritical if, for every* $x \in \Delta_{D,\max}$*, the origin is a vertex of the subdifferential* $\partial\upsilon_{\overline{D}}(x)$.

The critical point of $\overline{D}$ *is the family* $(u_v)_v \in \bigoplus_v \partial\upsilon_{\overline{D},v}(x)$ *determined by the decomposition*

$$0 = \sum_v n_v u_v \qquad \text{in} \qquad \partial\upsilon_{\overline{D}}(x) = \sum_v n_v \partial\upsilon_{\overline{D},v}(x).$$

Remark 1. *The critical point of* $\overline{D}$ *is independent of the choice of* $x \in \Delta_{D,\max}$.

The monocritical condition is therefore a uniqueness condition on the decomposition of the zero slope at the maximum locus of the global roof function.

3.3. **Semi-abelian varieties.**

Definition 3.13. *A semiabelian variety* G *over a field* K *is a connected smooth commutative algebraic* K*-group fitting into an exact sequence*

$$0 \longrightarrow T \longrightarrow G \longrightarrow A \longrightarrow 0,$$

where T *is a* K*-torus and* A *is an abelian variety over* K*. This exact sequence defines a Yoneda extension class*

$$\eta_G \in \mathrm{Ext}^1_K(A, T).$$

Proposition 3.14. *(The Weil-Barsotti formula) There is a canonical isomorphism*

$$\mathrm{Ext}^1_K(A, \mathbb{G}_m) \cong A^\vee(K).$$

If T is split and $T \simeq \mathbb{G}_m^t$, we frequently use

$$\mathrm{Ext}^1_K(A, \mathbb{G}_m^t) \cong (A^\vee)^t(K).$$

Proposition 3.15. *The following is [10, Lemma 1]. Let* G *and* G' *be semiabelian varieties over* K*, with toric parts* T, T' *and abelian quotients* A, A'*. For any homomorphism* $\varphi : G \to G'$*, there are unique homomorphisms* $\varphi_{\mathrm{tor}} : T \to T'$ *and* $\varphi_{\mathrm{ab}} : A \to A'$ *such that*

$$\begin{array}{ccccccccc} 0 & \longrightarrow & T & \longrightarrow & G & \longrightarrow & A & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle \varphi_{\mathrm{tor}}} & & \downarrow{\scriptstyle \varphi} & & \downarrow{\scriptstyle \varphi_{\mathrm{ab}}} & & \\ 0 & \longrightarrow & T' & \longrightarrow & G' & \longrightarrow & A' & \longrightarrow & 0 \end{array}$$

is a morphism of exact sequences. Moreover, the induced map

$$\mathrm{Hom}(G, G') \longrightarrow \mathrm{Hom}(T, T') \times \mathrm{Hom}(A, A')$$

is injective, and its image consists of the pairs $(\varphi_{\mathrm{tor}}, \varphi_{\mathrm{ab}})$ *satisfying*

$$(\varphi_{\mathrm{tor}})_* \eta_G = (\varphi_{\mathrm{ab}})^* \eta_{G'}.$$

Proposition 3.16. *The following quotient construction is recalled from [10, Chapter 2]. Let G be a semiabelian variety over K with split toric part $T \simeq \mathbb{G}_m^t$ and abelian quotient A. Let X_Σ be a toric variety with principal torus $X_0 = T$. Then the categorical quotient*

$$\overline{G} := G_{X_\Sigma} := (G \times_K X_\Sigma)/T$$

exists in the category of K-schemes and is a proper K-variety. If (M, ρ) is a T-linearized line bundle on X_Σ, then

$$G(M, \rho) := (G \times_K M)/T$$

is a line bundle on $\overline{G}$. Since T-linearized line bundles on X_Σ correspond to toric divisors, we write this quotient line bundle as $\overline{G}(D)$ when M corresponds to D.

Proposition 3.17. *If X_Σ is smooth, there is a split exact sequence*

$$0 \longrightarrow \mathrm{Pic}(A) \longrightarrow \mathrm{Pic}(\overline{G}) \longrightarrow \mathrm{Pic}(X_\Sigma) \longrightarrow 0.$$

This follows from the seesaw theorem, the vanishing $\mathrm{Pic}^0(X_\Sigma) = 0$, and the fact that every line bundle on X_Σ admits a T-linearization. The quotient construction $(G \times_K M)/T$ gives a section of $\mathrm{Pic}(\overline{G}) \to \mathrm{Pic}(X_\Sigma)$.

3.4. **Adelic line bundles.** We recall the adelic language used in [19]. The theory of adelic line bundles first appeared in Zhang's work on small points and adelic metrics. In this framework the Néron–Tate height of an algebraic point on an abelian variety over a number field is interpreted as the normalized arithmetic degree of an adelic line bundle at that point. Adelic line bundles are limits of hermitian line bundles in a suitable topology, and their arithmetic invariants are obtained as limits of the corresponding invariants on models.

The original theory was formulated for projective varieties over number fields or one-variable function fields. Yuan and Zhang extended it to quasi-projective varieties over finitely generated fields. More precisely, if F is a finitely generated field over $\mathbb{Q}$ or over a constant field, and X is a quasi-projective variety over F, they construct adelic line bundles on X, develop their intersection theory, study volumes of effective sections, and define the associated heights.

Using this theory, Yuan and Zhang proved an equidistribution theorem for small points on quasi-projective varieties over number fields, generalizing the equidistribution theorems of Szpiro–Ullmo–Zhang, Chambert-Loir, and Yuan. Kühne proved an independent semiabelian equidistribution theorem in the canonical metric case and used it, together with the work of Dimitrov–Gao–Habegger, to obtain uniformity results related to the Mordell conjecture.

Let X be a quasi-projective variety over the number field K.

Definition 3.18. *Let U be a quasi-projective variety over $\mathbb{Z}$. An object of $\widehat{\mathcal{P}ic}(U/\mathbb{Z})$ is a pair*

$$\big(\mathcal{L}, (\mathcal{X}_i, \overline{\mathcal{L}}_i, \ell_i)_i\big)$$

where:

(1) *$\mathcal{L}$ is a line bundle on U;*
(2) *$\mathcal{X}_i$ is a projective model of U over $\mathbb{Z}$;*
(3) *$\overline{\mathcal{L}}_i$ is a hermitian $\mathbb{Q}$-line bundle on $\mathcal{X}_i$;*
(4) *$\ell_i : \mathcal{L} \to \mathcal{L}_i|_U$ is an isomorphism in $\mathcal{P}ic(U)_\mathbb{Q}$.*

The sequence is required to satisfy the Cauchy condition of [19, §2.5.1]. Adelic line bundles on $X/\mathbb{Z}$ are objects of

$$\widehat{\mathcal{P}ic}(X/\mathbb{Z}) := \varinjlim_U \widehat{\mathcal{P}ic}(U/\mathbb{Z}),$$

where U ranges over quasi-projective models of X over $\mathbb{Z}$.

Yuan and Zhang extend arithmetic intersection theory from projective varieties over number fields to quasi-projective varieties over finitely generated fields. In particular, if d denotes the absolute dimension of a model of X, they construct an arithmetic intersection map

$$\widehat{\mathrm{Pic}}(X/\mathbb{Z})^d_{\mathrm{int}} \longrightarrow \mathbb{R},$$

where the subscript int denotes integrable adelic line bundles; see [19, Definition 2.5.2 and Proposition 4.1.1].

There is also a fully faithful analytification functor

$$\widehat{\mathcal{P}ic}(X/\mathbb{Z}) \longrightarrow \widehat{\mathcal{P}ic}(X^{r\text{-}an}),$$

where the target is the category of metrized line bundles on the reified analytic space, with isometries as morphisms [19, Proposition 3.5.1]. For projective varieties over $\mathbb{Q}$, the essential image is described by the coherent condition [19, Proposition 3.5.2]. In the quasi-projective case we will only use the induced v-adic metrized line bundles

$$\widehat{\mathcal{P}ic}(X/\mathbb{Z}) \longrightarrow \widehat{\mathcal{P}ic}(X_v^{an}).$$

Definition 3.19. *A v-metrized line bundle on X is a pair $(L, \|\cdot\|_v)$, where L is a line bundle on X and $\|\cdot\|_v$ is a metric on $L^{an}_{\mathbb{C}_v}$. Thus, for each open subset $U \subseteq X$, each section $\boldsymbol{s} : U \to L$, and every $f \in O_X(U)$, we require:*

(1) *the function*
$$\|\boldsymbol{s}\| = \|\cdot\|_v \circ \boldsymbol{s}^{an}_{\mathbb{C}_v} : U^{an}_{\mathbb{C}_v} \to \mathbb{R}_{\geq 0}$$
is continuous;
(2) *if $\boldsymbol{s}$ vanishes nowhere on U, then $\|\boldsymbol{s}\|$ has no zeros on $U^{an}_{\mathbb{C}_v}$;*
(3) $\|f\boldsymbol{s}\| = |f|\,\|\boldsymbol{s}\|$;
(4) *the metric is $\mathrm{Gal}(\mathbb{C}_v/K_v)$-invariant.*

Definition 3.20. *Let X be a projective variety over K. If v is archimedean, a v-metrized line bundle $(L, \|\cdot\|)$ is semipositive if $dd^c c_1(L, \|\cdot\|) \geq 0$. If v is non-archimedean, it is semipositive if its metric is the uniform limit of formally semipositive v-metrics. An adelic line bundle on $X/\mathbb{Z}$ is vertically semipositive if its v-adic realization is semipositive for every place v of K.*

Remark 2. *Yuan and Zhang also define semipositive v-metrized line bundles in the quasi-projective case; see [19, p. 105]. We use the same terminology for vertical semipositivity in that setting.*

We next recall the arithmetic quantities used to formulate smallness and equidistribution.

Definition 3.21. *Let Z be a closed $\overline{K}$-subvariety of X, and let $\overline{L}$ be an integrable adelic line bundle on $X/\mathbb{Z}$. Let Z' be the image of $Z \to X$, and let $\widetilde{L}$ be the image of $\overline{L}$ under the canonical map*

$$\widehat{\mathrm{Pic}}(X/\mathbb{Z}) \longrightarrow \widehat{\mathrm{Pic}}(X/K).$$

The height of Z with respect to $\overline{L}$ is

$$h_{\overline{L}}(Z) := h_{\overline{L}}(Z') := \frac{(\overline{L}|_{Z'})^{\dim Z+1}}{(\dim Z + 1)\deg_{\widetilde{L}}(Z'/K)}.$$

If $x \in X(\overline{K})$ and x' is the corresponding closed point of X, then

$$h_{\overline{L}}(x) := h_{\overline{L}}(x') := \frac{\overline{L}|_{x'}}{\deg(x')}.$$

The essential minimum is

$$\mu^{\mathrm{ess}}_{\overline{L}}(X) = \sup_{U \subseteq X} \inf_{x \in U(\overline{K})} h_{\overline{L}}(x),$$

where U ranges over non-empty Zariski open subschemes of X. The absolute minimum is

$$\mu_{\overline{L}}^{\mathrm{abs}}(X) = \inf_{x \in X(\overline{K})} h_{\overline{L}}(x).$$

An integrable adelic line bundle $\overline{L}$ is called quasi-canonical if

$$h_{\overline{L}}(X) = \mu_{\overline{L}}^{\mathrm{ess}}(X) \qquad \textit{and} \qquad \deg_{\widetilde{L}}(X/K) > 0.$$

Let $\overline{L}$ be a vertically semipositive adelic line bundle on $X/\mathbb{Z}$ with $\deg_{\widetilde{L}}(X/K) > 0$.

Definition 3.22. *For a place v of K, we say that $\overline{L}$ has the v-adic equidistribution property if, for every generic $\overline{L}$-small sequence $(x_i)_{i\in\mathbb{N}}$ in $X(\overline{K})$, the probability measures $(\delta_{x_i,v})_{i\in\mathbb{N}}$ associated with the v-adic Galois orbits of the x_i converge. In this case the limit measure is unique; it is called the canonical measure associated with $\overline{L}$, and is denoted by $\mu_{\overline{L},v}$.*

This equidistribution property is closely related to Bogomolov-type statements. For abelian varieties, equidistribution together with the Faltings–Zhang morphism gives Zhang's proof of Bogomolov [29]. Similarly, Kühne [1] uses the description of the canonical measure at archimedean places and the Faltings–Zhang morphism to prove Bogomolov for semiabelian varieties. In general, equidistribution plus a Bogomolov non-density statement yields strict equidistribution, and strict equidistribution for semiabelian varieties implies the Manin–Mumford theorem in this setting.

We close this section by recalling some standard equidistribution theorems.

Example 3. *(Szpiro–Ullmo–Zhang, Chambert-Loir, and Yuan; see [23, 30, 31].) If X is projective over a global field K, and if $\bar{L}$ is vertically semipositive and quasi-canonical with $\widetilde{L}$ ample, then $\bar{L}$ has the v-adic equidistribution property for all places v of K. The limiting measure is*

$$\frac{1}{\deg_{\widetilde{L}}(X/K)} \hat{c}_1(\bar{L}_v)^{\wedge \dim X}.$$

Example 4. *(Yuan–Zhang; [19, Theorem 5.4.3].) Let $\overline{L}$ be a nef and quasi-canonical adelic line bundle on $X/\mathbb{Z}$ such that $\deg_{\widetilde{L}}(X/K) > 0$, where $\widetilde{L}$ is the image of $\overline{L}$ under the canonical map*

$$\widehat{\mathrm{Pic}}(X/\mathbb{Z}) \to \widehat{\mathrm{Pic}}(X/K).$$

Then $\overline{L}$ has the v-adic equidistribution property for every place v of K, and the limiting measure is

$$\frac{1}{\deg_{\widetilde{L}}(X/K)} \hat{c}_1(\overline{L}_v)^{\wedge \dim X}.$$

The preceding examples are quasi-canonical. This condition does not hold in general: even for projective varieties, varying the metric on $\overline{L}$ can lead to

$$h_{\overline{L}}(X) < \mu_{\overline{L}}^{\mathrm{ess}}(X).$$

For toric varieties, Burgos Gil, Philippon, Rivera-Letelier, and Sombra give a necessary and sufficient condition for a toric metrized divisor to have the v-adic equidistribution property on algebraic points of the principal orbit X_0, for every place v of K. This condition is strictly weaker than quasi-canonicity:

Example 5. *By [16], let X be a proper toric variety, and let $\overline{D}$ be a vertically semipositive toric metrized $\mathbb{R}$-divisor with D big. Then $\overline{D}$ is monocritical if and only if it has the v-adic equidistribution property on algebraic points of X_0, for every place v of K. Moreover, BGPS describe the limiting measures in terms of the critical points of $\overline{D}$.*

Kühne gives semiabelian examples in which $\overline{L}$ is not quasi-canonical but still has the equidistribution property, with canonical measure formally given by the normalized Monge–Ampère measure:

Example 6. *In Kühne's construction [1], choose* $(\mathbb{P}^1)^t$ *as a compactification of* T*. He constructs an adelic line bundle*

$$\overline{L} = \overline{M_{\overline{G}}} \otimes \overline{\pi}^* \overline{N}$$

on $\overline{G}$*, and proves that* $\overline{L}$ *has the* v*-adic equidistribution property for every place* v *of* K*, with limiting measure equal to the normalized Monge–Ampère measure of* $\overline{L}$*.*

Remark 7. *The adelic line bundle constructed by Kühne is*

$$\overline{G}(\overline{D}^{\mathrm{can}}) \otimes \overline{\pi}^* \overline{N},$$

where $D = \sum D_i$ *is the sum of the divisors corresponding to the rays in the fan of* $(\mathbb{P}^1)^t$*.*

4. Local trivialization

In this section, we extend Kühne's local trivialization at archimedean places to all places of K, using p-adic theta functions and the Raynaud–Bosch–Lütkebohmert uniformization of abelian varieties over complete algebraically closed non-archimedean fields. We then use the local trivialization to define the adelic line bundle associated with a toric metrized $\mathbb{R}$-divisor and to compare the essential minima of this adelic line bundle on G with the absolute minimum of the corresponding toric metrized divisor on T.

Theorem 4.1 (Non-archimedean local trivialization)**.** *Let* v *be a non-archimedean place of* K*. There is a finite collection* $\{(U_j, \psi_j)\}_{j \in J}$ *such that the* U_j *form an open covering of* $A^{an}_{\mathbb{C}_v}$*, and*

$$\psi_j : \overline{G}^{an}_{\mathbb{C}_v}|_{U_j} \longrightarrow X^{an}_{\Sigma, \mathbb{C}_v}$$

are analytic maps for which

$$\overline{\pi}^{an}_{\mathbb{C}_v} \times \psi_j : \overline{G}^{an}_{\mathbb{C}_v}|_{U_j} \longrightarrow U_j \times X^{an}_{\Sigma, \mathbb{C}_v}$$

is an isomorphism of analytic spaces. Moreover, for every toric $\mathbb{R}$*-Cartier divisor* D*,*

$$\overline{G}(D)^{\mathrm{can}}_v|_{U_j} = \psi_j^*(\overline{D}^{\mathrm{can}}_v).$$

For all $j, j' \in J$ *and all*

$$y \in \overline{G}^{an}_{\mathbb{C}_v}|_{U_j \cap U_{j'}},$$

one has

$$\mathrm{val}_v \circ \psi_j(y) = \mathrm{val}_v \circ \psi_{j'}(y).$$

We now recall the analytic inputs used to construct the maps ψ_j.

We first recall p-adic theta functions and Raynaud–Bosch–Lütkebohmert uniformization for abelian varieties. The following form of the uniformization theorem can be found in [15]. In Kühne's notation, the two uniformization statements below correspond to [1, Section 3.2].

Let $\mathbf{K}$ be a complete algebraically closed non-archimedean valued field with valuation ring R, and let A be an abelian variety over $\mathbf{K}$.

Theorem 4.2 (Raynaud–Bosch–Lütkebohmert uniformization I)**.** *There exists a topological universal covering* $p : E^{an} \to A^{an}$*. If we choose a base point* $0 \in E^{an}$ *over the identity, then* E^{an} *has a unique structure of analytic group with identity* 0*, and* p *is a homomorphism. Moreover,* E^{an} *is the analytification of a group scheme* E*.*

Theorem 4.3 (Raynaud–Bosch–Lütkebohmert uniformization II)**.** *There are exact sequences*

$$0 \longrightarrow \mathbf{T} \longrightarrow E \xrightarrow{q} \mathbf{B} \longrightarrow 0$$

and

$$0 \longrightarrow M' \longrightarrow E^{an} \xrightarrow{p} A^{an} \longrightarrow 0,$$

where $\mathbf{T} = \mathrm{Spec}(\mathbf{K}[M])$ *is a split* $\mathbf{K}$*-torus,* $\mathbf{B}$ *is an abelian variety with good reduction, and* M' *is a lattice in* E^{an}*.*

Theta functions and descent data. We use the Raynaud theta formalism in the form developed in [1, Sections 3.3–3.6]. A character $u \in M$ pushes the Raynaud extension out to a rigidified Picard-zero line bundle E_u on B, with canonical section e_u. The resulting tropicalization is characterized by

$$\langle u, \mathrm{trop}(x)\rangle = -\log \|e_u(x)\|_{E_u}.$$

The period lattice M' maps to a full lattice in $N_{\mathbb{R}}$, and the fiber over 0 is the generic fiber of the formal Raynaud extension.

Rigidified line bundles on A are described by Raynaud descent triples (L, λ, c); see [1, Theorem 3.6]. On a formal affinoid chart where the Raynaud extension splits, a section of q^*L has a unique Fourier expansion

$$f = \sum_{u\in M} a_u \otimes e_u.$$

The theta transformation law of [1, Proposition 3.13] is exactly the condition that this section descend through M' to A. For a rational section s of a rigidified algebraically trivial line bundle, the associated theta function f_s satisfies

$$\mathrm{div}(f_s) = p^{-1}\big(\mathrm{div}(s)^{an}\big).$$

These are the only theta-function facts used in the local-trivialization argument. We record them, together with the toric extension criterion, in the next proposition.

We next recall the skeleton of $A^{an}_{\mathbf{K}}$, constructed on pp. 124–125 of [14].

Lemma 4.16. *There is a skeleton $\Delta(A)$ homeomorphic to a real torus, and it is a strong deformation retract of A^{an}. Moreover, A^{an} admits a finite cover by contractible analytic open subsets.*

Remark 8. *For any divisor D of A, the lemma allows us to choose an open contractible subset of A^{an} disjoint from the analytic divisor D^{an}.*

We use these facts and the theta functions above to construct the local trivializations.

Proposition 9 (Analytic inputs for the local trivialization)**.** *The proof of Theorem 4.1 uses the following standard facts in the precise forms stated here.*

(1) Raynaud–Bosch–Lutkebohmert uniformization gives

$$0 \to M'_v \to E_v^{an} \xrightarrow{p_v} A^{an}_{\mathbb{C}_v} \to 0$$

and the associated Raynaud extension. Over a sufficiently small contractible analytic open $U \subset A^{an}_{\mathbb{C}_v}$, the covering map p_v admits an analytic section.

(2) For a rational section s of a rigidified algebraically trivial line bundle on A, Raynaud theta descent produces a function $f_{s,v}$ satisfying

$$\mathrm{div}(f_{s,v}) = p_v^{-1}(\mathrm{div}(s)^{an}).$$

(3) A morphism from the principal torus to a normal toric variety extends over the affine chart X_σ precisely when all characters χ^m, $m \in \sigma^\vee \cap M$, pull back to analytic functions with effective boundary divisor. This is the standard affine-chart criterion for toric morphisms, applied in the analytic category.

Proof of Theorem 4.1. Let $\Sigma(1) = \{\rho_1, \dots, \rho_d\}$, let v_ℓ be the primitive generator of ρ_ℓ, and let D_ℓ be the corresponding T-invariant prime divisor on X_Σ. Write the extension class of G in the chosen character basis as

$$\eta_G = (H_1, \dots, H_t) \in A^\vee(\overline{K})^t.$$

The cocycle description of the associated toric fibration gives

$$\overline{\pi}^* H_i = \sum_{\ell=1}^{d} \langle e_i^\vee, v_\ell\rangle\, \overline{G}(D_\ell), \qquad 1 \le i \le t.$$

Choose an analytic open subset $U \subset A^{an}_{\mathbb{C}_v}$ which is contractible and disjoint from the analytic divisors of rational sections s_i of $H_i^\vee$. By Proposition 9, the covering $p_v : E_v^{an} \to A^{an}_{\mathbb{C}_v}$ has an analytic section over U; denote the corresponding lift of π by

$$\widetilde{\pi}_U : \overline{G}^{an}_{\mathbb{C}_v}|_U \longrightarrow E_v^{an}.$$

The section s_i determines a theta function $f_{s_i,v}$ with

$$\operatorname{div}(f_{s_i,v}) = p_v^{-1}\big(\operatorname{div}(s_i)^{an}_{\mathbb{C}_v}\big).$$

The cocycle formula gives a rational function g_i on $\overline{G}$ with

$$\operatorname{div}(g_i) = \sum_{\ell=1}^{d} \langle e_i^\vee, v_\ell\rangle\, \overline{G}(D_\ell) + \overline{\pi}^* \operatorname{div}(s_i).$$

After multiplying s_i by constants, we may assume $g_i(e_G) = 1$ and $\|f_{s_i,v}(e_E^{an})\| = 1$.

On the open semiabelian part over U, define

$$\psi_U^{(i)}(y) = \frac{g_i(y)}{f_{s_i,v}(\widetilde{\pi}_U(y))}, \qquad \psi_U = (\psi_U^{(1)}, \dots, \psi_U^{(t)}) : G^{an}_{\mathbb{C}_v}|_U \to T^{an}_{\mathbb{C}_v}.$$

The denominator has no zero over U, and hence $\psi_U^{(i)}$ is an invertible analytic function on $G^{an}_{\mathbb{C}_v}|_U$. For each $a \in U$, the divisor of $\psi_U^{(i)}$ on the compactified fiber $\overline{G}_a \simeq X_{\Sigma,\mathbb{C}_v}$ is exactly the divisor of the torus character $\chi^{e_i^\vee}$. Thus ψ_U identifies the principal torus in the fiber with $T^{an}_{\mathbb{C}_v}$, up to multiplication by nonzero constants.

We now extend over the toric boundary. Let $X_\sigma = \operatorname{Spec} \mathbb{C}_v[\sigma^\vee \cap M]$ be an affine toric chart. For $m = \sum_i m_i e_i^\vee \in \sigma^\vee \cap M$, the pullback of χ^m is

$$\prod_{i=1}^{t} \big(\psi_U^{(i)}\big)^{m_i}.$$

Its boundary divisor is

$$\sum_{\ell=1}^{d} \langle m, v_\ell\rangle\, \overline{G}(D_\ell),$$

which is effective on the chart because $m \in \sigma^\vee \cap M$. By the affine-chart criterion in Proposition 9, the map $(\overline{\pi}, \psi_U)$ extends uniquely to an analytic morphism

$$\Psi_U : \overline{G}^{an}_{\mathbb{C}_v}|_U \longrightarrow U \times X^{an}_{\Sigma,\mathbb{C}_v}.$$

The same character calculation applied to the inverse torus coordinates gives an inverse morphism. Hence Ψ_U is an isomorphism of analytic spaces.

The finite cover is obtained by choosing finitely many such contractible opens and, on each of them, rational sections s_i whose divisors avoid the open. Compactness of $A^{an}_{\mathbb{C}_v}$ gives a finite subcover $\{U_j\}_{j\in J}$, and we write the second component of Ψ_{U_j} as ψ_j.

It remains to verify overlap compatibility. On $U_j \cap U_{j'}$, write

$$s_j^{(i)} = t_{jj'}^{(i)} s_{j'}^{(i)}$$

with $t_{jj'}^{(i)}$ an analytic unit. The corresponding rational functions and theta functions satisfy

$$g_j^{(i)} = (t_{jj'}^{(i)} \circ \pi) g_{j'}^{(i)}, \qquad f_{s_j^{(i)},v} = (t_{jj'}^{(i)} \circ p_v) f_{s_{j'}^{(i)},v}.$$

If the chosen local lifts of U_j and $U_{j'}$ differ on a connected component of the overlap by an element of the period lattice M'_v, the theta transformation law adds the usual Raynaud automorphy factor. The same factor occurs in the descent cocycle defining the semiabelian extension class η_G; hence it cancels in the quotient $g_i/f_{s_i,v}$. After this cancellation the displayed

formulas give the actual change of local coordinates. Since $p_v(\widetilde{\pi}_U(y)) = \pi(y)$, the two unit factors have the same absolute value after evaluation at y. Therefore

$$\left|\frac{\psi_j^{(i)}(y)}{\psi_{j'}^{(i)}(y)}\right|_v = 1, \qquad 1 \le i \le t.$$

Equivalently, the transition multiplier lies in the compact torus, so it has valuation zero. The toric tropicalization map is invariant under compact torus multiplication. Hence

$$\mathrm{val}_v \circ \psi_j(y) = \mathrm{val}_v \circ \psi_{j'}(y)$$

on overlaps.

Finally, under Ψ_U, the line bundle $\overline{G}(D)$ is identified with the pullback of the toric line bundle $O(D)$ on the second factor. When D is equipped with the canonical toric metric, that metric is a function of the toric valuation and is invariant under compact torus multiplication. The overlap compatibility therefore identifies the canonical metrics:

$$\overline{G}(D)_v^{can}|_{U_j} = \psi_j^*(\overline{D}_v^{can}).$$

□

Lemma 10 (Good-place model for the induced canonical metric)**.** *For all but finitely many non-archimedean places v, the semiabelian extension G, the abelian quotient A, the toric variety X_Σ, the fan Σ, the toric divisor D, and the local data used in Theorem 4.1 extend over O_{K_v}. At such a place the local trivializations are induced by formal trivializations of the corresponding toric fibration, their transition functions lie in the maximal compact torus, and the canonical toric metric on $O(D)$ is the model metric coming from the toric model.*

Proposition 4.17 (Adelicity of the induced metric)**.** *Let $\overline{D}$ be a toric metrized $\mathbb{R}$-divisor on X_Σ. On $\overline{\pi}^{-1}(U_j)$, define the induced local metric on $\overline{G}(D)$ by*

$$\|\widetilde{s}(y)\|_{\overline{G}(\overline{D}),v} = \|s(\psi_{j,v}(y))\|_{\overline{D},v},$$

where s is a local section of $O(D)$ and $\widetilde{s}$ is the corresponding local section of $\overline{G}(D)$. These local formulas glue to a continuous v-adic metric on $\overline{G}(D)$. If $\overline{D}$ is adelic, then the induced metric is adelic.

Proof. The only point to check for gluing is independence of j. On an overlap, Theorem 4.1 shows that the two toric coordinates differ by multiplication by an element of the compact torus. Toric metrics are invariant under this compact torus. Hence the displayed local formula is independent of the chart. Continuity is local and follows from continuity of the toric metric and of the analytic maps $\psi_{j,v}$.

We next prove adelicity. First suppose that D is an integral toric divisor. Write

$$\overline{D} = \overline{D}^{can} + \varphi, \qquad \varphi = (\varphi_v)_v,$$

where each φ_v is a compact-torus-invariant continuous function on $X_{\Sigma,\mathbb{C}_v}^{an}$, and $\varphi_v = 0$ for all but finitely many places. The induced metric is obtained from the induced canonical metric by the finite family of functions

$$\varphi_v \circ \psi_{j,v}.$$

Thus it remains to prove that the induced canonical metric is adelic.

By Lemma 10, outside a finite set of non-archimedean places the local trivializations are formal trivializations of a toric fibration over O_{K_v}, and the canonical toric metric is the model metric on the toric model. Pulling back that model metric through the formal trivializations gives the model metric on $\overline{G}(D)$. Therefore the induced canonical metric is a model metric at almost all places, and hence is adelic. Adding finitely many continuous changes preserves adelicity. The case of toric metrized $\mathbb{R}$-divisors follows by linearity and uniform approximation by rational linear combinations of integral toric divisors. □

Remark 11. *Lemma 10 is the precise good-place fact used here. No separate conjectural assertion about canonical metrics at every good-reduction place is needed for the adelicity proof.*

Proposition 4.18 (Semipositivity of the induced metric)**.** *If $\overline{D}$ is vertically semipositive then the adelic line bundle $\overline{G}(\overline{D})$ associated with $\overline{D}$ is vertically semipositive and $\overline{G}(D)$ is nef.*

Proof. First consider the underlying line bundle. For a toric metrized $\mathbb{R}$-divisor, vertical semipositivity implies that the associated local metric functions are concave; in particular the virtual support function of D is concave, and hence D is nef on X_Σ.

Let N be an ample symmetric line bundle on A. If D is ample, then by Kühne's ampleness criterion [1, Lemma 3, Chapter 2], $\overline{G}(D) \otimes \overline{\pi}^* N$ is ample, and therefore $\overline{G}(D)$ is nef. If D is only nef, choose an ample toric divisor D'. Since the nef cone is the closure of the ample cone, $D + \epsilon D'$ is ample for every $\epsilon > 0$. Hence

$$\overline{G}(D) + \epsilon\big(\overline{G}(D') \otimes \overline{\pi}^* N\big)$$

is ample for every $\epsilon > 0$, and letting $\epsilon \to 0$ shows that $\overline{G}(D)$ is nef.

For the metric, on each chart U_j the induced metric is the pullback of the semipositive toric metric $\overline{D}_v$. Pullback by an analytic morphism preserves semipositivity, both at archimedean and non-archimedean places. Since the chartwise metrics glue by Proposition 4.17, $\overline{G}(\overline{D})$ is vertically semipositive. □

Corollary 4.19. *The line bundle*

$$L = \overline{G}(D) \otimes \overline{\pi}^* N$$

is big if D is nef and big and N is an ample symmetric line bundle on A.

Proof. By Proposition 4.18, $\overline{G}(D)$ is nef. Since the pullback of the ample symmetric line bundle N is nef, L is nef. It therefore suffices to show $L^{g+t} > 0$. Let η be the generic point of A. Then

$$L^{g+t} = \overline{G}(D)^t (\overline{\pi}^* N)^g = N^g\big(\overline{G}(D)|_\eta\big) = N^g D^t > 0.$$

□

Definition 4.20 (Arithmetic T-effectivity)**.** *A toric metrized $\mathbb{R}$-divisor $\overline{D}$ on X_Σ is called arithmetically T-effective if there exist a finitely supported family of real numbers $(c_v)_v \in \bigoplus_v \mathbb{R}$ such that the distinguished toric section $s_0 = s_D$ of $O(D)$, corresponding to $0 \in \Delta_D$, satisfies*

$$\sum_v n_v c_v = \mu_{\overline{D}}^{\mathrm{abs}}(T)$$

and

$$\|s_0\|_{v,\sup} \le e^{-c_v} \qquad \text{for every place } v.$$

We call s_0 a $\mu_{\overline{D}}^{\mathrm{abs}}(T)$-small distinguished section. Here, as in the preceding definition of T-effectivity, the term toric invariant section always means this distinguished section corresponding to $0 \in \Delta_D$, rather than an arbitrary character eigensection.

Example 12. *If D is T-effective and $\overline{D}$ is the canonical toric metrized $\mathbb{R}$-divisor, then $\overline{D}$ is arithmetically T-effective. Indeed $\mu_{\overline{D}}^{\mathrm{abs}}(T) = 0$, and the canonical toric section s_D satisfies $\|s_D\|_{v,\sup} \le 1$ at every place.*

Remark 13. *The distinguished section s_0 is nowhere vanishing on the open torus T. This is the point of the definition that will be used below: after passing to the semiabelian compactification, this section gives a global lower bound for the height on G.*

4.1. **Meaning and practical criteria for arithmetic T-effectivity.** The preceding condition is an attainment condition, not merely an auxiliary positivity assumption. Let $\vartheta_{\overline{D},v}$ and $\vartheta_{\overline{D}} = \sum_v n_v \vartheta_{\overline{D},v}$ be the local and global roof functions. The toric sup-norm identity and the successive-minima formula give

$$-\log \|s_0\|_{v,\sup} = \vartheta_{\overline{D},v}(0), \qquad \mu^{\mathrm{abs}}_{\overline{D}}(T) = \max_{x\in\Delta_D} \vartheta_{\overline{D}}(x)$$

[13, Chapters 4–5] [18, Theorem 3.9 and Corollary 3.10]. Consequently the following are equivalent:

$$\overline{D} \text{ is arithmetically } T\text{-effective} \iff \vartheta_{\overline{D}}(0) = \mu^{\mathrm{abs}}_{\overline{D}}(T) \iff 0 \in \operatorname*{argmax}_{\Delta_D} \vartheta_{\overline{D}}.$$

Indeed the local inequalities imply $c_v \le \vartheta_{\overline{D},v}(0)$; after summing, both sides are trapped between $\mu^{\mathrm{abs}}_{\overline{D}}(T)$ and the maximum of the roof. Conversely one takes $c_v = \vartheta_{\overline{D},v}(0)$.

This characterization gives several quick tests. It is enough that $0 \in \partial^+\vartheta_{\overline{D}}(0)$; at an interior differentiability point this reduces to $\nabla\vartheta_{\overline{D}}(0) = 0$. It also holds when Δ_D is centrally symmetric and the roof is even, for canonical metrics, and for the quasi-canonical normal form used later in this paper. In the rational piecewise-affine case it is a finite linear check on the affine cells of the roof.

The condition is not automatic. On $\mathbb{P}^1$, take $\Delta_D = [0,1]$ and a semipositive toric metric with global roof

$$\vartheta_{\overline{D}}(x) = C - (x-\alpha)^2$$

for an irrational $\alpha \in (0,1)$. The unique maximizer is not a lattice point; in particular it is not 0, so the distinguished section cannot realize the absolute minimum. This example exhibits the arithmetic integrality obstruction that is invisible to semipositivity and bigness.

More generally, if arbitrary toric eigensections are allowed, the corresponding coordinate-free condition is

$$\operatorname*{argmax}_{\Delta_D} \vartheta_{\overline{D}} \cap M \neq \varnothing.$$

Such a witness fixes m, not automatically the origin. The definition above deliberately uses s_0, because the monocritical argument in Section 9 requires the origin to lie on the maximizing face.

This also connects the condition with the general Dirichlet problem in Arakelov geometry. Chen–Moriwaki ask whether an adelic $\mathbb{R}$-Cartier divisor is $\mathbb{R}$-linearly equivalent to an effective one and study both criteria and obstructions [39, 40]; their adelic-curve framework relates absolute minima to asymptotic minimal slopes [41, Chapter 7]. Arithmetic T-effectivity is a torus-equivariant, threshold-sharp version of this principle: the effective witness is restricted to the distinguished toric eigenline, remains nonvanishing on the open group, and realizes exactly the absolute minimum. Thus it is stronger than the ordinary Dirichlet property, but it is precisely adapted to the global height separation used here.

What is necessary is correspondingly limited. Arithmetic T-effectivity is necessary for the present proof mechanism if one insists on deriving the optimal lower bound from the single distinguished section. It is not proved to be necessary for the equidistribution or Bogomolov conclusion itself.

We next recall the theta-function calculation which identifies the local metric contribution of a rigidified Picard-zero line bundle with the corresponding Néron local function. The non-archimedean argument uses the tropical theta formalism to play the role of the hermitian form in the archimedean case.

As in the toric case, there is a canonical continuous section of

$$\mathrm{trop} : E^{an} \to N_{\mathbb{R}}.$$

Choose a local section $V \to E^{an}$ such that the formal Raynaud extension splits over V, so $q^{-1}(V) \cong T \times V$. Then $\mathrm{trop}^{-1}(v) = U_v \times V$, where U_v is the inverse image of v under

$\mathrm{trop} : T^{an} \to N_{\mathbb{R}}$. Every analytic function h on $U_v \times V$ has a unique Laurent expansion

$$h = \sum_{u \in M} a_u \chi^u$$

with $a_u \in \mathscr{O}(V)$ and

$$|a_u|_{\sup} e^{-\langle u,v \rangle} \to 0$$

outside finite subsets of M.

Definition 4.21. *Define $\sigma(v)$ to be the multiplicative seminorm on $\mathscr{O}(U_v \times V)$ given by*

$$|h(\sigma(v))| = \max_{u \in M} \{|a_u|_{\sup} e^{-\langle u,v \rangle}\}.$$

This seminorm is independent of the choice of V and of the splitting, and it depends continuously on v; see [35, Chapter 4.1].

Fix a rigidified line bundle L_A on A, and let (L, λ, c) be a corresponding triple for L_A. Let $f \in H^0(E^{an}, q^*L)$ be a nonzero theta function.

Definition 4.22. *The tropicalization of f is the function*

$$f_{\mathrm{trop}} : N_{\mathbb{R}} \to \mathbb{R}$$

defined by

$$f_{\mathrm{trop}}(v) = -\log \|f(\sigma(v))\|_{q^*L}.$$

Remark 14. *This definition extends to the theta function f_D associated with any Cartier divisor D: if $D = D_1 - D_2$ with D_1, D_2 ample, then $f_{D,\mathrm{trop}}$ is defined as the corresponding difference of tropicalizations.*

Proposition 4.23. *Assume that L_A is ample, and put*

$$c_{\mathrm{trop}}(u') = -\log \|c(u')\|_L.$$

Then, for all $u'_1, u'_2 \in M'$,

$$c_{\mathrm{trop}}(u'_1 + u'_2) - c_{\mathrm{trop}}(u'_1) - c_{\mathrm{trop}}(u'_2) = [u'_1, \lambda(u'_2)].$$

Moreover, for $u' \in M'$ and $v \in N_{\mathbb{R}}$,

$$f_{\mathrm{trop}}(v) = f_{\mathrm{trop}}(v + u') + c_{\mathrm{trop}}(u') + \langle \lambda(u'), v \rangle.$$

Proposition 4.24 (Canonical Picard-zero theta metrics)**.** *Let A be an abelian variety over a number field K, let $H \in \mathrm{Pic}^0(A)$ be a rigidified line bundle, and let s be a rational section of H. Equip H with its canonical adelic Picard-zero metric, characterized by the rigidification at the origin and by the requirement that the canonical square isomorphism, equivalently the isomorphisms $[n]^*H \simeq H^{\otimes n}$, be isometries.*

For every place v, let $f_{s,v}$ be the theta representative of s on the Raynaud/Kühne analytic uniformization, normalized by the rigidification of H. Then $f_{s,v}$ computes the canonical local norm of s:

$$\|s(x)\|^{can}_{H,v} = \|f_{s,v}(\widetilde{x})\|_v$$

for every $x \notin |\operatorname{div}(s)|$ and every lift $\widetilde{x}$ of x. Hence

$$\lambda_{s,v}(x) := -\log \|f_{s,v}(\widetilde{x})\|_v$$

is the normalized Neron local function attached to the canonically metrized pair (H, s). It is independent of the choice of lift $\widetilde{x}$, and

$$\sum_v n_v \lambda_{s,v}(x) = h_{\overline{H}}(x).$$

The normalization is compatible with principal divisors: if $\operatorname{div}(s) = \operatorname{div}(g)$, then $\lambda_{s,v}(x) = -\log |g(x)|_v$ up to the usual adelic constant, whose global sum is zero by the product formula.

Proof. We spell out the normalization because this is the point at which the Raynaud–Kühne theta expression has to be identified with the canonical Picard-zero metric. Choose a Cartier divisor $D = \mathrm{div}(s)$ representing H. On the Raynaud analytic uniformization at a non-archimedean place v, let $f_{D,v}$ be the theta representative of D, normalized by the rigidification at the origin. Following the tropical theta formalism recalled above, set

$$W_{D,v}(x) = -\log \|f_{D,v}(\widetilde{x})\|_v - f_{D,v}^{\mathrm{trop}}\big(\mathrm{trop}(\widetilde{x})\big) + f_{D,v}^{\mathrm{trop}}(0).$$

The transformation law for theta functions attached to a triple (L, λ, c), together with the corresponding transformation law for tropical theta functions, shows that $W_{D,v}(x)$ is independent of the lift $\widetilde{x}$. Explicitly, replacing $\widetilde{x}$ by $\widetilde{x} + u'$, $u' \in M'$, changes the first term by the automorphy factor $c(u')e_{\lambda(u')}(\widetilde{x})$, while the tropical term changes by the logarithm of the same factor. Hence the two changes cancel. This is precisely the compatibility recorded in the theta transformation formulas of [15, Section 3] and in the tropical theta calculation of [35, Section 4].

The same construction is additive in D. If $D = \mathrm{div}(g)$ is principal, then the associated theta representative is the pullback of g, up to the constant fixed by the rigidification, and therefore

$$W_{D,v}(x) = -\log |g(x)|_v$$

up to a v-constant. Functoriality of the Raynaud extension and of the theta descent data gives

$$W_{[2]^* D,v}(x) = W_{D,v}(2x).$$

Thus $W_{D,v}$ satisfies the standard characteristic properties of the normalized Neron local function attached to D. At archimedean places the same statement is the classical theta-function comparison with Neron functions, as in [6, Chapter 13, Theorem 1.1]; the hermitian quadratic correction is the archimedean analogue of the tropical correction.

Now use that $H \in \mathrm{Pic}^0(A)$. In the Raynaud triple this means that the homomorphism λ is zero, or equivalently that the Chern form of the canonical metric vanishes. Consequently the tropical correction $f_{D,v}^{\mathrm{trop}}$ is constant after the above rigidified normalization; at archimedean places the hermitian form attached to D is zero. Hence the normalized local Neron function is simply

$$\lambda_{s,v}(x) = -\log \|f_{s,v}(\widetilde{x})\|_v.$$

By uniqueness of the canonical rigidified Picard-zero metric, this is $-\log \|s(x)\|_{H,v}^{can}$. Summing over all places gives the height of the canonically metrized line bundle $\overline{H}$, and changing the rational section by a rational function only changes the local functions by $-\log |g(x)|_v$, whose weighted adelic sum is zero by the product formula. □

Remark 15 (Source status of the theta–Neron comparison)**.** *The normalization in Proposition 4.24 is fixed in two independent layers. First, Zhang's canonical adelic metric construction [7, Theorem 2.2], in the Picard-zero form recalled by Chambert-Loir [5, §1 and final appendix], gives the unique canonical adelic metric on a rigidified algebraically trivial line bundle for which the square isomorphism and all multiplication isomorphisms are isometries. At complex places this is the unique flat hermitian metric compatible with the rigidification; at non-archimedean places it is the v-adic canonical metric obtained by Tate's limiting construction.*

Second, the theta functions used here are local expressions of that same canonical metric. At non-archimedean places, Raynaud–Bosch–Lutkebohmert uniformization and theta functions are recalled in [15, Section 3]; the line-bundle/triple correspondence is [15, Theorem 3.10], the descent criterion for theta functions is [15, Proposition 3.19 and Definition 3.20], and the Fourier/model-metric compatibility is recorded around [15, Remark 3.18 and Proposition 3.19]. At archimedean places this is the classical theta/Neron-function comparison, e.g. [6, Chapter 13, Theorem 1.1]. With this convention, the function $-\log \|s(x)\|_{H,v}^{can}$ is the normalized Neron local function attached to (H, s), and its adelic sum is by definition the height of the canonically metrized Picard-zero bundle $\overline{H}$.

Proposition 4.25 (Theta functions and Neron heights)**.** *Let A be an abelian variety over a number field K, let $H \in \mathrm{Pic}^0(A)$ be rigidified, and let s be a rational section of H. If $x \in A(\overline{K})$ does not lie in $|\mathrm{div}(s)|$, then*

$$h_{\overline{H}}(x) = \sum_v -n_v \log \big\| f_{s,v}(\widetilde{\pi_v}(x)) \big\|_v.$$

Here $f_{s,v}$ is the normalized theta representative from Proposition 4.24, and $\widetilde{\pi_v}(x)$ is any lift of x to the analytic universal cover. The summand is independent of that lift.

Proof. This is the global form of Proposition 4.24, corresponding to Kühne's theta-height comparison [1, Theorem 4.7]. Indeed the local functions

$$-\log \| f_{s,v}(\widetilde{\pi_v}(x)) \|$$

are the normalized Neron local functions associated with (H, s). Their adelic sum is therefore the canonical height attached to the canonically metrized Picard-zero bundle $\overline{H}$. The product formula removes the ambiguity coming from replacing s by a rational multiple. □

Corollary 16 (Picard-zero heights and the Neron pairing)**.** *Let N be a symmetric ample line bundle on A, and let $\phi_N : A \to A^\vee$ be the associated polarization, with the convention*

$$\phi_N(q) = T_q^* N \otimes N^{-1}.$$

Let $\langle \cdot, \cdot \rangle_N$ be the Neron–Tate bilinear form normalized by

$$h_{\overline{N}}(y + z) = h_{\overline{N}}(y) + h_{\overline{N}}(z) + 2\langle y, z \rangle_N.$$

Suppose $H \in \mathrm{Pic}^0(A)_{\mathbb{R}}$, $a > 0$, and $q \in A(\overline{K}) \otimes_{\mathbb{Z}} \mathbb{R}$ satisfy $\phi_N(q) = aH$. Then

$$h_{\overline{H}}(y) = \frac{2}{a} \langle y, q \rangle_N$$

for every $y \in A(\overline{K})$.

Proof. First assume that H is integral and that $q \in A(\overline{K})$. With the above convention, $\phi_N(q) = T_q^* N \otimes N^{-1}$, rigidified at the origin. The canonical height of this rigidified Picard-zero bundle is therefore

$$h_{\overline{\phi_N(q)}}(y) = h_{\overline{N}}(y + q) - h_{\overline{N}}(y) - h_{\overline{N}}(q) = 2\langle y, q \rangle_N.$$

If $\phi_N(q) = aH$, then

$$a\, h_{\overline{H}}(y) = h_{\overline{\phi_N(q)}}(y),$$

and the formula follows. The real case follows by linearity of Picard-zero heights and of the Neron–Tate bilinear form. □

Remark 17 (Neron local-height convention)**.** *The preceding proposition uses the convention that the theta norm associated with a rigidified Picard-zero bundle is the normalized Neron local function. Remark 15 records the source chain: Zhang's canonical adelic metric, Chambert-Loir's Picard-zero formulation, and the Raynaud/Kühne theta expression of the corresponding local norm.*

Corollary 4.26 (Global torsion cancellation for Picard-zero theta factors)**.** *With the normalization of Proposition 4.24, if $x \in A(\overline{K})$ is torsion and $x \notin |\,\mathrm{div}(s)|$, then*

$$\sum_v n_v \big(-\log \| f_{s,v}(\widetilde{x}) \|_v \big) = 0.$$

Proof. By Proposition 4.25, the displayed sum is the canonical height $h_{\overline{H}}(x)$ attached to the rigidified Picard-zero bundle H. This height is linear on the torsion-free quotient and vanishes on torsion points. □

Remark 18 (Local source ledger)**.** *The construction in this section uses three standard analytic tools. At non-archimedean places the analytic uniformization is Raynaud–Bosch–Lutkebohmert uniformization [35], recalled in the form needed here by [15]; the analytic skeleton and the contractible cover are taken from Berkovich's construction [14]; and the tropicalization of theta functions follows the tropical theta formalism in [15]. At archimedean places we use Kühne's local trivialization [1, Lemma 11]. The statements below record exactly the formal consequences of these tools used later in the proof.*

For the theta-descent input, the precise source chain is [1, Sections 3.2–3.3 and Theorem 3.6] for Raynaud uniformization and line-bundle descent, [1, Propositions 3.8–3.9] for Picard-zero translation, [1, Section 3.6 and Proposition 3.13] for Fourier expansions and the theta transformation law, and [1, Theorem 4.7] for the global theta-height comparison. Proposition 9 isolates exactly the portion used in Theorem 4.1; Proposition 4.24 fixes the metric normalization needed later.

Proposition 19 (Semiabelian tropicalization)**.** *For every place v, the Raynaud/Kühne construction gives a canonical real-valued tropicalization homomorphism*

$$\mathrm{trop}_{G,v} : G^{an}_{\mathbb{C}_v} \longrightarrow N_{\mathbb{R}}$$

attached to the split torus $T \subset G$. Equivalently, for each character $u \in M$, pushing out the extension $0 \to T \to G \to A \to 0$ by $\chi^u : T \to \mathbb{G}_m$ gives a rigidified algebraically trivial line bundle H_u on A, and

$$\langle u, \mathrm{trop}_{G,v}(y)\rangle = -\log \|e_u(y)\|_{H_u,v}$$

for the canonical Raynaud/Kühne metric. This map extends the ordinary tropicalization of $T^{an}_{\mathbb{C}_v}$ and is a group homomorphism. Consequently, if $q \in G(\overline{K})$ is torsion, then

$$\mathrm{trop}_{G,v}(q) = 0 \in N_{\mathbb{R}}.$$

Lemma 20 (Coordinate tropicalization)**.** *Let $\psi_{j,v}$ be one of the Raynaud/Kühne local toric coordinate systems constructed in Theorem 4.1. If $y \in G^{an}_{\mathbb{C}_v}$ is in the chart, then*

$$\mathrm{val}_v(\psi_{j,v}(y)) = \mathrm{trop}_{G,v}(y) \in N_{\mathbb{R}}.$$

Proof. It is enough to pair with every character $u \in M$. On a Raynaud formal splitting in the non-archimedean case, the map $\psi_{j,v}$ is the toric coordinate in the product $T^{an} \times U$; hence the assertion is exactly the local description of tropicalization as the first projection followed by the ordinary torus tropicalization. At archimedean places this is the corresponding unitary local coordinate in Kühne's local trivialization.

Equivalently in the notation used in the proof of Theorem 4.1, the pullback of the character χ^u through $\psi_{j,v}$ is the local expression of the pushout vector $e_u(y)$ in the rigidified Picard-zero bundle H_u. The canonical Raynaud/Kühne metric in this trivialization is the ordinary absolute value of that coefficient. Therefore

$$\big\langle u, \mathrm{val}_v(\psi_{j,v}(y))\big\rangle = -\log|\chi^u(\psi_{j,v}(y))|_v = -\log\|e_u(y)\|_{H_u,v} = \big\langle u, \mathrm{trop}_{G,v}(y)\big\rangle.$$

The overlap calculation in the proof of Theorem 4.1 shows that transition multipliers lie in the compact torus, hence have valuation zero. Thus the equality is independent of the chosen chart. □

Lemma 21 (Torsion lifts have zero toric valuation)**.** *Let $q \in G(\overline{K})_{\mathrm{tor}}$. For every place v and every local trivialization chart $\psi_{j,v}$ defined at $\pi(q)$, the toric coordinate of q has valuation zero:*

$$\mathrm{val}_v(\psi_{j,v}(q)) = 0 \in N_{\mathbb{R}}.$$

Consequently, for every $t \in T(\overline{K})$,

$$\mathrm{val}_v(\psi_{j,v}(q\,t)) = \mathrm{val}_v(t),$$

where the right hand side is the ordinary valuation of the split torus T.

Proof. By Proposition 19, $\mathrm{trop}_{G,v}(q) = 0$. Lemma 20 identifies this tropicalization with $\mathrm{val}_v(\psi_{j,v}(q))$ in every local chart, giving the first assertion. Multiplication by $t \in T$ is multiplication in the toric coordinate of the chart, so valuation additivity on the split torus gives

$$\mathrm{val}_v(\psi_{j,v}(q\,t)) = \mathrm{val}_v(\psi_{j,v}(q)) + \mathrm{val}_v(t) = \mathrm{val}_v(t).$$

□

Proposition 4.27 (Valuation and height package)**.** *Let $\overline{D}$ be a toric metrized $\mathbb{R}$-divisor on X_Σ, and let $\overline{G}(\overline{D})$ be the adelic line bundle on $\overline{G}$ obtained by pulling back the local toric metrics through the local trivializations $\psi_{j,v}$. Let $x \in G(\overline{K})$, and let $O_v(x)$ denote the finite set of its v-adic Galois conjugates in $G(\mathbb{C}_v)$. For every place v, choose, for each $y \in O_v(x)$, an index $j(y)$ with $\pi(y) \in U_{j(y),v}$, and put*

$$\nu_{x,v} = \frac{1}{\#O_v(x)} \sum_{y \in O_v(x)} \delta_{\mathrm{val}_v(\psi_{j(y),v}(y))}.$$

Then:

$$\nu_{x,v} \text{ is independent of the choices of } j(y),$$

$$\sum_v n_v \int_{N_\mathbb{R}} u \, d\nu_{x,v}(u) = \big(-h_{\overline{H}_1}(\pi(x)), \dots, -h_{\overline{H}_t}(\pi(x))\big),$$

and

$$h_{\overline{G}(\overline{D})}(x) = -\sum_v n_v \int_{N_\mathbb{R}} \Psi_{\overline{D},v} \, d\nu_{x,v}.$$

Moreover, if $\pi(x)$ is torsion and $x = q \cdot t$, where $q \in G(\overline{K})_{\mathrm{tor}}$ and $t \in T(\overline{K})$, then

$$h_{\overline{G}(\overline{D})}(x) = h_{\overline{D}}(t).$$

Consequently, for $\overline{L} = \overline{G}(\overline{D}) \otimes \overline{\pi}^* \overline{N}$, with $\overline{N}$ symmetric ample and canonically metrized on A, one has $h_{\overline{L}}(x) = h_{\overline{D}}(t)$ on such a torsion fiber.

Proof. The independence of the choices follows from Theorem 4.1 at non-archimedean places and from Kühne's archimedean trivialization [1, Lemma 11]: on an overlap the two charts have the same valuation map. Hence the measure $\nu_{x,v}$ is well-defined.

Let $e_1^\vee, \dots, e_t^\vee$ be the character basis used in the construction of the functions g_i/f_{s_i}. On a chart $U_{j,v}$, the i-th coordinate of $\mathrm{val}_v(\psi_{j,v}(y))$ is

$$\langle e_i^\vee, \mathrm{val}_v(\psi_{j,v}(y)) \rangle = -\log |g_i(y)|_v + \log \|f_{s_i,v}(\widetilde{\pi_v}(y))\|.$$

After averaging over $O_v(x)$ and summing over v, the terms $-\log|g_i(y)|_v$ vanish by the product formula. The remaining theta term is identified with $-h_{\overline{H}_i}(\pi(x))$ by Proposition 4.25. This gives the displayed formula for the center of the adelic valuation measures.

For the height identity, use the canonical toric section of D, which is nowhere vanishing on the open torus. By construction of the metric $\overline{G}(\overline{D})$,

$$-\log \|\widetilde{s_D}(y)\|_{\overline{G}(\overline{D}),v} = -\Psi_{\overline{D},v}(\mathrm{val}_v(\psi_{j(y),v}(y))).$$

Averaging over the v-adic conjugates and summing over all places gives the height formula. Linearity then gives the same identity for toric metrized $\mathbb{R}$-divisors.

Finally suppose $\pi(x)$ is torsion. Over $\overline{K}$, the fiber above $\pi(x)$ contains a torsion point q; hence $x = q \cdot t$ for some $t \in T(\overline{K})$. By Lemma 21, the valuation measure of x is the toric valuation measure of t. Therefore the above height formula reduces exactly to the toric height $h_{\overline{D}}(t)$. The canonical height of $\pi(x)$ with respect to $\overline{N}$ is zero, so the same identity holds for $\overline{L} = \overline{G}(\overline{D}) \otimes \overline{\pi}^* \overline{N}$. □

Corollary 22 (Torsion-lift height invariance)**.** *Let $q \in G(\overline{K})_{\mathrm{tor}}$ and $t \in T(\overline{K})$. For every toric metrized $\mathbb{R}$-divisor $\overline{D}$,*

$$h_{\overline{G(D)}}(q\,t) = h_{\overline{D}}(t).$$

If

$$\overline{L} = \overline{G}(\overline{D}) \otimes \overline{\pi}^*\overline{N}$$

with $\overline{N}$ symmetric ample and canonically metrized on A, then

$$h_{\overline{L}}(q\,t) = h_{\overline{D}}(t).$$

Proof. Apply Proposition 4.27 to $x = q\,t$. Since q is torsion, $\pi(q\,t) = \pi(q)$ is torsion, and the canonical height of $\pi(q)$ is zero. □

Lemma 23 (Height lower bound from arithmetic T-effectivity)**.** *Let $\overline{D}$ be arithmetically T-effective, and let*

$$\overline{L} = \overline{G}(\overline{D}) \otimes \overline{\pi}^*\overline{N},$$

where $\overline{N}$ is a symmetric ample line bundle on A equipped with its canonical metric. Then, for every $x \in G(\overline{K})$,

$$h_{\overline{L}}(x) \geq \mu^{\mathrm{abs}}_{\overline{D}}(T).$$

Proof. Choose the small distinguished section s_0 and constants $(c_v)_v$ from Definition 4.20. Since s_0 is torus-invariant, it is nowhere vanishing on the open torus T. Hence its induced section $\widetilde{s}_0$ of $\overline{G}(D)$ is nowhere vanishing on G. For every v-adic conjugate y of x, choose a local trivialization $\psi_{j,v}$ at $\pi(y)$. By construction of the induced metric,

$$\|\widetilde{s}_0(y)\|_{\overline{G(D)},v} = \|s_0(\psi_{j,v}(y))\|_{\overline{D},v} \leq \|s_0\|_{v,\sup} \leq e^{-c_v}.$$

Therefore

$$h_{\overline{G(D)}}(x) = \sum_v -\frac{n_v}{\#O_v(x)} \sum_{y \in O_v(x)} \log \|\widetilde{s}_0(y)\|_{\overline{G(D)},v} \geq \sum_v n_v c_v = \mu^{\mathrm{abs}}_{\overline{D}}(T).$$

The canonical height attached to the symmetric ample bundle $\overline{N}$ is nonnegative, so

$$h_{\overline{L}}(x) = h_{\overline{G(D)}}(x) + h_{\overline{N}}(\pi(x)) \geq \mu^{\mathrm{abs}}_{\overline{D}}(T).$$

□

Lemma 24 (Torsion fibers approximate the toric minimum)**.** *Assume that D is big and $\overline{D}$ is vertically semipositive. Let $\overline{L} = \overline{G}(\overline{D}) \otimes \overline{\pi}^*\overline{N}$, with $\overline{N}$ as above. For every nonempty open subset $U \subseteq G$ and every $\epsilon > 0$, there is a point $x \in U(\overline{K})$ such that*

$$h_{\overline{L}}(x) \leq \mu^{\mathrm{abs}}_{\overline{D}}(T) + \epsilon.$$

Proof. The image $\pi(U)$ contains a nonempty open subset of A, since $\pi : G \to A$ is smooth and surjective. Torsion points are Zariski dense in A, so choose $a \in A(\overline{K})_{\mathrm{tor}} \cap \pi(U)$. The torsion point a has a torsion lift $q \in G(\overline{K})_{\mathrm{tor}}$: if $na = 0$, choose any $z \in G(\overline{K})$ above a, then $nz \in T$, and divisibility of the torus over $\overline{K}$ allows one to multiply z by a point of T to obtain a lift killed by n.

Translation by q^{-1} identifies the open set

$$U \cap \pi^{-1}(a)$$

with a nonempty open subset $W \subseteq T$. By the toric successive-minima formula [18, Theorem 3.9 and Corollary 3.10],

$$\mu^{\mathrm{ess}}_{\overline{D}}(X_\Sigma) = \mu^{\mathrm{abs}}_{\overline{D}}(T),$$

and hence every nonempty open subset of T contains points whose $\overline{D}$-height is at most $\mu^{\mathrm{abs}}_{\overline{D}}(T) + \epsilon$. Choose such a point $t \in W(\overline{K})$, and put $x = qt \in U(\overline{K})$. Since $\pi(x) = a$ is torsion, Corollary 22 gives

$$h_{\overline{L}}(x) = h_{\overline{D}}(t) \leq \mu^{\mathrm{abs}}_{\overline{D}}(T) + \epsilon.$$

□

Proposition 4.28 (Section 4 minima formula)**.** *If D is big and T-effective, $\overline{D}$ is vertically semipositive and arithmetically T-effective, and $\overline{N}$ is a symmetric ample line bundle with its canonical metric on A, then for*

$$\overline{L} = \overline{G}(\overline{D}) \otimes \overline{\pi}^* \overline{N}$$

one has

$$\mu_{\overline{L}}^{\mathrm{ess}}(\overline{G}) = \mu_{\overline{L}}^{\mathrm{abs}}(G) = \mu_{\overline{D}}^{\mathrm{ess}}(X_\Sigma) = \mu_{\overline{D}}^{\mathrm{abs}}(T).$$

Proof. By Lemma 23,

$$\mu_{\overline{L}}^{\mathrm{abs}}(G) \geq \mu_{\overline{D}}^{\mathrm{abs}}(T).$$

On the other hand, let $V \subseteq \overline{G}$ be a nonempty open subset. Since G is dense in $\overline{G}$, the intersection $V \cap G$ is a nonempty open subset of G. Lemma 24 gives, for every $\epsilon > 0$, a point $x \in (V \cap G)(\overline{K})$ with

$$h_{\overline{L}}(x) \leq \mu_{\overline{D}}^{\mathrm{abs}}(T) + \epsilon.$$

Taking the infimum over V, then the supremum over all such V, and then letting $\epsilon \to 0$, gives

$$\mu_{\overline{L}}^{\mathrm{ess}}(\overline{G}) \leq \mu_{\overline{D}}^{\mathrm{abs}}(T).$$

Since always $\mu_{\overline{L}}^{\mathrm{ess}}(\overline{G}) \geq \mu_{\overline{L}}^{\mathrm{abs}}(G)$, we obtain

$$\mu_{\overline{L}}^{\mathrm{ess}}(\overline{G}) = \mu_{\overline{L}}^{\mathrm{abs}}(G) = \mu_{\overline{D}}^{\mathrm{abs}}(T).$$

Finally, the toric equality

$$\mu_{\overline{D}}^{\mathrm{ess}}(X_\Sigma) = \mu_{\overline{D}}^{\mathrm{abs}}(T)$$

is the toric successive-minima formula [18, Theorem 3.9 and Corollary 3.10]. □

5. Canonical metric case

5.1. **Auxiliary n-division tower.** We fix the auxiliary semiabelian tower used throughout this section. Let

$$\eta_G = (H_1, \ldots, H_t) \in A^\vee(\overline{K})^t \simeq \mathrm{Ext}^1_{\overline{K}}(A, T)$$

be the extension class of G. For every $n \geq 1$, choose

$$H_i^{(n)} \in A^\vee(\overline{K}), \qquad 1 \leq i \leq t,$$

such that $nH_i^{(n)} = H_i$. After passing to a finite extension $K_n \supset K$ over which these data are defined, let G_n be the semiabelian variety over K_n with extension class

$$\eta_{G_n} = (H_1^{(n)}, \ldots, H_t^{(n)}).$$

There is an isogeny

$$\varphi_n : G_n \longrightarrow G_{K_n}$$

whose toric part is multiplication by n and whose abelian part is the identity on A_{K_n}.

We again write $\overline{G}_n$ for the toric compactification of G_n associated with the base change X_{Σ,K_n}, and

$$\overline{\pi}_n : \overline{G}_n \longrightarrow A_{K_n}$$

for the induced projection. The homomorphism φ_n extends to the corresponding toric compactifications; throughout the section we denote this extension by the same symbol

$$\varphi_n : \overline{G}_n \longrightarrow \overline{G}_{K_n}.$$

We use the same symbols D and N for their base changes to K_n. This causes no ambiguity in the toric part, since D_{K_n} has the same virtual support function as D. Put

$$L_n = G_n(D) \otimes \pi_n^* N_{K_n}, \qquad \overline{L}_n^{\mathrm{can}} = \overline{G}_n(\overline{D}^{\mathrm{can}}) \otimes \overline{\pi}_n^* \overline{N}_{K_n}.$$

Finally, fix once and for all a place $v_0 \in \Sigma(K)$. For a real number κ_n, let $O_{\overline{G}}(\kappa_n)$ denote the trivial adelic line bundle on $\overline{G}$ whose metric is trivial at all places except v_0, and whose v_0-metric is defined by

$$-\log \|1\|_{v_0} = \kappa_n.$$

Its pullback by φ_n is the constant vertical correction used below.

5.2. **Lower bound and semipositivity.** We first record the standard canonical lower bound.

Theorem 25 (Fundamental height inequality)**.** *Let $\overline{M}$ be a vertically semipositive integrable adelic line bundle on a projective variety Y, with $\deg_M(Y) > 0$. With the normalization of heights used in this paper,*

$$\mu^{\mathrm{abs}}_{\overline{M}}(Y) \le h_{\overline{M}}(Y) \le \mu^{\mathrm{ess}}_{\overline{M}}(Y).$$

Lemma 26 (Constant correction for horizontal semipositivity)**.** *In the auxiliary semiabelian setting of this section, let $\overline{M}_n$ be a vertically semipositive adelic line bundle with nef underlying line bundle on $\overline{G}_n$. If, for some $\delta_n \to 0$,*

$$\mu^{\mathrm{abs}}_{\overline{M}_n}(\overline{G}_n) \ge -C\delta_n, \qquad h_{\overline{M}_n}(\overline{G}_n) \ge -C\delta_n,$$

then there is a constant vertical correction

$$\overline{M}_n \otimes \varphi_n^* O_{\overline{G}}(\kappa_n), \qquad 0 \le \kappa_n \le C'\delta_n,$$

which is horizontally semipositive. This is the constant-correction form used in Kühne's lower-bound argument before the arithmetic volume estimate.

Lemma 27 (Canonical chart lower bound)**.** *Assume that D is an integral ample T-effective toric divisor. After replacing D by a positive multiple and dividing the resulting height inequalities by that multiple, there are lattice points*

$$m_0 = 0,\ m_1, \dots, m_r \in \Delta_D \cap M$$

such that the effective toric divisors

$$D + \operatorname{div} \chi^{m_j} \qquad (0 \le j \le r)$$

have empty common intersection. For every $n \ge 1$ and every $x \in \overline{G}_n(\overline{K})$, one can choose j with

$$h_{\overline{G}_n(\overline{D}^{\mathrm{can}})}(x) \ge -h_{\overline{H}^{(n)}_{m_j}}(\pi_n(x)),$$

where $H^{(n)}_{m_j}$ is the Picard class on A induced by m_j for the extension G_n.

Proof. The existence of $m_0, \dots, m_r$ is the standard toric separation argument: for a sufficiently divisible multiple of D, the polytope contains lattice points whose corresponding toric sections have no common zero. The case $m_0 = 0$ is the T-effective section.

Fix x. Choose j such that the section associated with $D + \operatorname{div} \chi^{m_j}$ does not vanish at the toric coordinate of x. On this chart, the induced section on $\overline{G}_n$ is a section of $\overline{G}_n(D)$ twisted by the Picard-zero class $H^{(n)}_{m_j}$ on the abelian quotient. The canonical toric metric makes the effective toric section small on the corresponding affine chart. Using the height identity in Proposition 4.27, together with the theta-function interpretation of the Picard-zero term from Proposition 4.25, gives

$$h_{\overline{G}_n(\overline{D}^{\mathrm{can}})}(x) + h_{\overline{H}^{(n)}_{m_j}}(\pi_n(x)) \ge 0.$$

This is the required inequality. □

Lemma 28 (Bounding the Picard-zero translation terms)**.** *Let $m_1, \dots, m_r$ be fixed as in Lemma 27. There is a constant $C > 0$, independent of n, such that for every j, every $n \ge 1$, and every $y \in A(\overline{K})$,*

$$h_{\overline{N}}(y) - h_{\overline{H}^{(n)}_{m_j}}(y) \ge -Cn^{-2}.$$

Proof. Let $\phi_N : A \to A^\vee$ be the polarization attached to N. Since ϕ_N is an isogeny, after replacing K by a finite extension and after clearing denominators there are a positive integer a and points $q_j \in A(\overline{K})$ such that

$$\phi_N(q_j) = aH_{m_j}.$$

Choose $q_{j,n} \in A(\overline{K})$ with $nq_{j,n} = q_j$. Then

$$\phi_N(q_{j,n}) = aH^{(n)}_{m_j}.$$

Let $\langle\cdot,\cdot\rangle_N$ be the Neron–Tate bilinear form associated with N. By Corollary 16,

$$h_{\overline{H}^{(n)}_{m_j}}(y) = \frac{2}{a}\langle y, q_{j,n}\rangle_N.$$

Hence, by completing the square for the positive semidefinite quadratic form $h_{\overline{N}}$,

$$h_{\overline{N}}(y) - h_{\overline{H}^{(n)}_{m_j}}(y) \geq -a^{-2}h_{\overline{N}}(q_{j,n}).$$

Since $h_{\overline{N}}(q_{j,n}) = n^{-2}h_{\overline{N}}(q_j)$, the desired uniform bound follows. □

Proposition 29 (Canonical lower bound)**.** *Assume that D is ample and T-effective. There is a constant $C = C(G, D, N) > 0$ such that for all $n \geq 1$,*

$$\mu^{\mathrm{abs}}_{\overline{L}_n^{\mathrm{can}}}(\overline{G}_n) \geq -Cn^{-2}$$

and

$$-Cn^{-2} \leq h_{\overline{L}_n^{\mathrm{can}}}(\overline{G}_n) \leq 0.$$

Consequently there is $0 \leq \kappa_n \leq Cn^{-2}$ such that

$$\overline{L}_n^{\mathrm{can}} \otimes \varphi_n^* O_{\overline{G}}(\kappa_n)$$

is horizontally semipositive.

Proof. We first treat the case in which D is an integral ample toric divisor. The general ample $\mathbb{R}$-divisor case follows by writing the support function of D as a positive real linear combination of support functions of ample integral toric divisors and using additivity of the corresponding heights.

For every $x \in \overline{G}_n(\overline{K})$, choose j by Lemma 27. Combining that lemma with Lemma 28 gives

$$h_{\overline{L}_n^{\mathrm{can}}}(x) = h_{\overline{G}_n(\overline{D}^{\mathrm{can}})}(x) + h_{\overline{N}}(\pi_n(x)) \geq -Cn^{-2}$$

for a constant C independent of n. Hence

$$\mu^{\mathrm{abs}}_{\overline{L}_n^{\mathrm{can}}}(\overline{G}_n) \geq -Cn^{-2}.$$

The dense open semiabelian part $G_n \subset \overline{G}_n$ contains a Zariski-dense family of points of height 0: take torsion points on the abelian quotient and toric points of canonical toric height 0, and use Proposition 4.27. Equivalently, Proposition 4.28 applied to the canonical toric metric gives $\mu^{\mathrm{ess}}_{\overline{L}_n^{\mathrm{can}}}(G_n) = 0$. Thus $\mu^{\mathrm{ess}}_{\overline{L}_n^{\mathrm{can}}}(\overline{G}_n) \leq 0$. By Theorem 25,

$$-Cn^{-2} \leq h_{\overline{L}_n^{\mathrm{can}}}(\overline{G}_n) \leq \mu^{\mathrm{ess}}_{\overline{L}_n^{\mathrm{can}}}(\overline{G}_n) \leq 0.$$

Finally apply Lemma 26 with $\delta_n = n^{-2}$. After enlarging C, it gives $0 \leq \kappa_n \leq Cn^{-2}$ and the asserted horizontal semipositivity of

$$\overline{L}_n^{\mathrm{can}} \otimes \varphi_n^* O_{\overline{G}}(\kappa_n).$$

□

Remark 30. *The assertion "$\phi_N : A(K) \to A^\vee(K)$ is surjective" is false in general. What is used in the argument is surjectivity of the isogeny as a morphism, equivalently on $\overline{K}$-points, after replacing K by a finite extension if necessary.*

5.3. **The three Kühne-type estimates.** The canonical proof needs the following three estimates. They make explicit the Kühne-type estimates used below.

Remark 31 (Source-theorem ledger for the asymptotic estimates)**.** *The two load-bearing analytic estimates in this subsection are the first variation estimate and the arithmetic volume comparison. In Kühne's proof of semiabelian equidistribution, the corresponding statements in the published JEMS version are [1, Lemma 4.4] and [1, Lemma 4.6]. Lemma 4.4 identifies the linear term in the height variation with integration against the local Chambert-Loir measure and bounds the remaining terms by a quadratic error whose constant grows linearly with the auxiliary isogeny parameter.*

The arithmetic-volume input has a slightly longer source chain, and we record it explicitly. First, [1, Lemma 2.5(a),(b)] gives the constant-shift formula and continuity properties for the χ-arithmetic volume. Second, [1, Lemma 2.6] proves that, for semipositive metrized line bundles $\widetilde{L}, \widetilde{M}$ on a d-dimensional arithmetic variety,

$$\widehat{\mathrm{vol}}_\chi(\widetilde{L} - \widetilde{M}) \geq \widetilde{L}^{d+1} - (d+1)\widetilde{L}^d \cdot \widetilde{M}.$$

Kühne proves this by invoking Ikoma's arithmetic-volume form of Yuan's bigness theorem [4, Theorem 3.5.3 and Remark 3.5.4], ultimately resting on Yuan's arithmetic bigness theorem [31]. Third, [1, Lemma 4.6] applies this bigness input to an integrable perturbation, written as a difference of semipositive metrics, and expands the remaining high-order intersections to obtain the required quadratic volume lower bound.

We do not use this package as an unqualified black box. Kühne works with the standard compactification $(\mathbb{P}^1)^t$, whereas here the toric compactification is X_Σ and the toric divisor D is arbitrary. The new verification needed in this paper is the bridge from the source package to the present setting: the projection formula, the top-degree numerical reduction, and the intersection scaling under the auxiliary isogenies give the same $O(n)$ error in the canonical case, and the compressed pullback $f_m = f \circ [m]$ adds exactly the additional $O(m)$ factor isolated below.

Lemma 32 (Top-degree numerical reduction)**.** *Let $\overline{G} = G \times^T X_\Sigma$ be the toric compactification used in this paper, with $\dim T = t$ and $\dim A = g$. Let $M_1, \ldots, M_r$ be line bundles on A, and let $D_1, \ldots, D_s$ be toric Cartier divisors on X_Σ, with $r + s = g + t$. Then the top-degree numerical intersection on $\overline{G}$, and likewise on every auxiliary $\overline{G}_n$,*

$$\deg\big(\pi^* M_1 \cdots \pi^* M_r \cdot \overline{G}(D_1) \cdots \overline{G}(D_s)\big)\,,$$

is zero unless $r = g$ and $s = t$. This is only a numerical statement about a complete intersection; no vanishing of the corresponding Chow class is asserted.

Proof. If $r > g$, push the intersection to the base. Suppose $s > t$. For a smooth fan, the relative toric Chow-ring presentation [2, Theorem 1.2(iii)] expresses every excess relative divisor factor through first Chern classes of the character line bundles defining the associated toric bundle. These line bundles lie in $\mathrm{Pic}^0(A)$, so their first Chern classes are numerically trivial and the resulting top-degree number is zero. For a normal fan, take a smooth toric refinement and use the projection formula. The same argument applies after the auxiliary base extension and isogeny. Since $r + s = g + t$, the only remaining case is $(r, s) = (g, t)$. □

Remark 33. *No arbitrary adelic-intersection or Chow-class vanishing is used in the sequel. The height-variation and arithmetic-volume arguments use the explicit projection formula and the normalized scaling estimate below. The preceding lemma is used only for top degrees and total masses of mixed semipositive measures.*

Lemma 34 (Normalized scaling in the perturbation family)**.** *Retain the preceding notation. The scaling assertion used below is the following restricted normalized assertion. Let $F_n = \varphi_n^* O_{\overline{G}}(f)$, where $O_{\overline{G}}(f)$ is an integrable adelic line bundle with trivial underlying algebraic class, and let $d = g + t$. In the quadratic core of the error estimates, namely in a normalized*

intersection expression with two perturbing factors F_n and the remaining factors coming from the reference bundle, division by L_n^d leaves at most one explicit factor n. If the test metric is first compressed, $f_m = f \circ [m]$, so that $F_{m,n} = \varphi_n^* O_{\overline{G}}(f_m)$, the same quadratic core leaves at most one explicit factor n and at most one explicit factor m. Higher-order error monomials are reduced to this quadratic core in Lemma 35. The constants are uniform for $m \le n$ in the range used below.

Proof. We first argue for model metrics and write $M_n = \overline{G}_n(D)$. The exact scaling comes from the projection formula, not merely from counting algebraic toric factors. Since

$$\varphi_n^* \overline{G}(D) = nM_n, \qquad \varphi_n^* \pi^* N = \pi_n^* N, \qquad \deg(\varphi_n) = n^t,$$

the projection formula gives, for every $j \le g$,

$$F_n^2 \cdot M_n^{d-1-j} \cdot (\pi_n^* N)^j = n^{\,t-(d-1-j)} \big(O_{\overline{G}}(f)^2 \cdot \overline{G}(D)^{d-1-j} \cdot (\pi^* N)^j \big).$$

Because $d = g + t$, the exponent is $j - g + 1 \le 1$. Terms with $j > g$ vanish by the projection formula on the abelian base. After expansion of the reference bundle and division by the fixed top degree L_n^d, the quadratic core is therefore $O(n)$. This is the ambient analogue of the exact pullback calculation in [1, proof of Lemma 4.4, equations (4.5)–(4.6)].

For the compressed family, put $F_m = [m]^* O_{\overline{G}}(f)$ and $D_m = m^{-1}[m]^* D$. Applying the same projection-formula calculation to $[m]$, whose degree on the toric part is m^t, gives

$$F_m^2 \cdot \overline{G}(D_m)^{d-1-j} \cdot (\pi^* N)^j = O\big(m^{\,j-g+1}\big) = O(m)$$

for $j \le g$. Combining this with the n-division calculation yields the required $O(nm)$ bound for the compressed quadratic core. The higher-order terms are treated separately in Lemma 35; no higher-order conclusion is being hidden in the present quadratic calculation.

For integrable metrics, write the perturbation as a difference of semipositive model metrics, apply the preceding identities term by term, and pass to the limit by continuity of adelic intersections. The constants depend only on the fixed compactification, D, N, the reference models, and the chosen integrable norm of f, and are uniform for $m \le n$. □

Lemma 35 (Parameterized error bookkeeping)**.** *Let $O_{\overline{G}}(f)$ be an integrable adelic line bundle with trivial underlying algebraic class. Put*

$$(\overline{M}_{\star,n}, F_{\star,n}, \alpha_{\star,n}) = (\overline{L}_n^{\mathrm{can}}, \varphi_n^* O_{\overline{G}}(f), n)$$

in the canonical case, and

$$(\overline{M}_{\star,n}, F_{\star,n}, \alpha_{\star,n}) = (\overline{L}_{(m,n)}, \varphi_n^* O_{\overline{G}}(f \circ [m]_{\overline{G}}), nm)$$

in the compressed case, with $m \le n$. Then for every $2 \le r \le d+1$ and every $|\lambda| \le \alpha_{\star,n}^{-1}$,

$$|\lambda|^r \left| F_{\star,n}^r \cdot \overline{M}_{\star,n}^{d+1-r} \right| \le C_f L_n^d |\lambda|^2 \alpha_{\star,n},$$

where C_f is independent of n, of $m \le n$, of r, and of λ.

Proof. We first assume that f is a model function. For $r = 2$, the estimate is exactly the quadratic core estimate from Lemma 34, together with the arithmetic Hodge index estimate for two algebraically trivial metric factors as used in Kühne's first variation argument [1, Lemma 4.4]. Thus

$$\left| F_{\star,n}^2 \cdot \overline{M}_{\star,n}^{d-1} \right| \le C_f L_n^d \alpha_{\star,n}.$$

For $r > 2$, the same projection-formula computation used in Lemma 34 gives, for a monomial containing j factors pulled back from the abelian base, an n-exponent $r+j-g-1 \le r-1$. In the compressed case the independent calculation for $[m]$ gives the same bound for the m-exponent. Thus

$$\left| F_{\star,n}^r \cdot \overline{M}_{\star,n}^{d+1-r} \right| \le C_f L_n^d \alpha_{\star,n}^{r-1}.$$

Multiplying by $|\lambda|^r$, and using $|\lambda|\alpha_{\star,n} \leq 1$, gives

$$|\lambda|^r \left| F_{\star,n}^r \cdot \overline{M}_{\star,n}^{d+1-r} \right| \leq C_f L_n^d |\lambda|^2 \alpha_{\star,n}.$$

The number of possible r's is bounded by d, so the constant may be chosen uniformly in r.

For a general integrable test metric, approximate f from above and below by model functions in the integrable topology. Multilinearity and continuity of adelic intersections for integrable metrics allow passage to the limit, with the constant depending only on the chosen integrable norm of f. □

Definition 36 (Adelic perturbation size)**.** *Let $\overline{A}$ be fixed and semipositive and let $\overline{E} = O_Y(\phi)$ be integrable with trivial algebraic class. For a model function set*

$$\|\phi\|_{\mathrm{ad}} = \sum_v n_v \|\phi_v\|_{\sup}$$

and extend by completion. Define

$$\|\overline{E}\|_{\overline{A}} = \inf \max\{\|\phi^+\|_{\mathrm{ad}}, \|\phi^-\|_{\mathrm{ad}}\},$$

over decompositions $\overline{E} = O_Y(\phi^+) - O_Y(\phi^-)$ such that $\overline{A} + O_Y(\phi^\pm)$ are semipositive. Only finite-size perturbations occur below.

Proposition 37 (Kühne–Yuan–Ikoma quadratic arithmetic volume package)**.** *Let Y be a projective variety of dimension d over a number field, let $\overline{A}$ be a horizontally semipositive adelic line bundle on Y, and let $\overline{E}$ be an integrable adelic line bundle whose underlying algebraic line bundle is trivial. In the one-parameter range used below, and after writing the integrable perturbation as a difference of semipositive metrics, the arithmetic volume satisfies a quadratic comparison of the form*

$$\widehat{\mathrm{vol}}_\chi(\overline{A} + \overline{E}) - (\overline{A} + \overline{E})^{d+1} \geq -C_{\overline{A}} \|\overline{E}\|_{\overline{A}}^2,$$

where $\|\cdot\|_{\overline{A}}$ is the adelic size functional just defined. Thus for a perturbation $\overline{E} = \lambda \overline{F}$ the defect is $O(|\lambda|^2)$, with the constant controlled by the size of $\overline{F}$ and by the chosen semipositive reference bundle. More precisely, $C_{\overline{A}}$ depends only on d, the fixed numerical class and the chosen reference models. In the families below, C_f may depend on G, X_Σ, D, N, those models and the fixed integrable norm of f, but is independent of n, m, λ and K_n.

The general bigness inequality is [1, Lemma 2.6]. Its proof uses [4, Theorem 3.5.3 and Remark 3.5.4] together with Yuan's arithmetic bigness theorem [31]. Kühne's semiabelian implementation is [1, Lemma 4.6]: it applies Lemma 2.6 to a semipositive decomposition of the perturbation and then estimates the higher-order intersection terms. The constant-shift and continuity facts used to remove the auxiliary correction κ_n are [1, Lemma 2.5(a),(b)].

Remark 38. *Proposition 37 is the external Hilbert–Samuel type source package used in the proof, and it is not claimed as a new theorem of this paper. The lemmas below record exactly how it is used: they specify the perturbation $\overline{E}$, the horizontal semipositivity correction, and the n- and m-dependence of the constant. The dependence on n and m is not imported from the source package; it is verified here through Lemma 34 and Lemma 58.*

Raw versus normalized volume notation in the auxiliary tower. For an adelic line bundle over K_n, write

$$\widehat{\mathrm{vol}}_{\chi,K_n}^{\mathrm{raw}}(\overline{M}) := [K_n : \mathbb{Q}]\, \widehat{\mathrm{vol}}_{\chi,K_n}(\overline{M}), \qquad (\overline{M}^{d+1})_{K_n}^{\mathrm{raw}} := [K_n : \mathbb{Q}]\, (\overline{M}^{d+1})_{K_n}.$$

Every displayed defect carrying $[K_n : K]$ below is a raw defect and is marked as such. Unmarked heights, intersections, volumes and measures remain normalized.

Lemma 39 (Constant shifts in the volume defect)**.** *Let $\overline{B}_{m,n,\lambda}$ be any of the adelic line bundles occurring below, namely $\overline{A}_n+\varphi_n^*O_{\overline{G}}(\lambda f)$ or $\overline{A}_{m,n}+\varphi_n^*O_{\overline{G}}(\lambda f_m)$, with $|\lambda|\le n^{-1}$ in the canonical case and $|\lambda|\le (nm)^{-1}$ in the compressed case. If $0\le\kappa_n\le Cn^{-2}$, then*

$$\big|\widehat{\mathrm{vol}}^{\mathrm{raw}}_{\chi,K_n}(\overline{B}_{m,n,\lambda}\otimes\varphi_n^*O_{\overline{G}}(-\kappa_n))-\big((\overline{B}_{m,n,\lambda}\otimes\varphi_n^*O_{\overline{G}}(-\kappa_n))^{d+1}\big)^{\mathrm{raw}}_{K_n}$$
$$-\widehat{\mathrm{vol}}^{\mathrm{raw}}_{\chi,K_n}(\overline{B}_{m,n,\lambda})+(\overline{B}^{d+1}_{m,n,\lambda})^{\mathrm{raw}}_{K_n}\big|\le C_f[K_n:K]\kappa_n,$$

where C_f is independent of n, of $m\le n$, and of λ.

Proof. Tensoring by $O_{\overline{G}}(-\kappa_n)$ changes only the metric on the trivial line bundle. For the χ-arithmetic volume this is a Lipschitz perturbation: multiplying all sup norms by $e^{O(\kappa_n)}$ changes the raw asymptotic Euler characteristic by $O([K_n:K]\kappa_n)$. This is the uniform form of the constant-shift formula and continuity statement in [1, Lemma 6(a),(b)]. The same linear bound holds for the arithmetic self-intersection by multilinearity, because inserting one constant metrized trivial factor gives the algebraic degree, while terms with two or more such factors vanish. The degrees of the line bundles in the canonical and compressed families are uniformly controlled by the fixed data D, N, and the chosen test metric, after the normalizations used in this subsection. This gives the displayed uniform estimate. □

Lemma 40 (Field-extension normalization in the auxiliary tower)**.** *Fix a place v of K. In the auxiliary tower over K_n, the test metric at v is understood as the collection of the induced metrics at all places $v'\mid v$ of K_n. Write $N_{v'}^{K_n}$ for raw local degrees and $n_{v'}^{K_n}$ for normalized weights. Then*

$$\sum_{v'\mid v}N_{v'}^{K_n}=[K_n:K]N_v^K,\qquad \sum_{v'\mid v}n_{v'}^{K_n}=n_v^K.$$

Thus raw intersections and raw χ-volumes over K_n carry a common factor $[K_n:K]$, whereas normalized versions do not. Equivalently, after division by the global degree the linear local variation has coefficient n_v^K, and a raw volume-defect estimate of the form

$$\widehat{\mathrm{vol}}^{\mathrm{raw}}_{\chi,K_n}(\overline{M}_{\star,n,\lambda})-(\overline{M}^{d+1}_{\star,n,\lambda})^{\mathrm{raw}}_{K_n}\ge -C[K_n:K]E_{\star,n,\lambda}$$

contributes $O(E_{\star,n,\lambda})$ to the normalized Minkowski estimate before the usual division by $|\lambda|$.

Proof. The two identities are the raw and normalized forms of the standard decomposition formula. Raw arithmetic intersections and raw χ-volumes are additive over places, so installing the same Galois-invariant perturbation at all $v'\mid v$ multiplies the raw contribution by the raw local-degree sum. Global normalization cancels $[K_n:K]$, leaving coefficient n_v^K. The same cancellation applies to every quadratic or constant raw volume defect. Hence no normalized height in the sequel changes under extension of the ground field. □

Lemma 41 (Measure compression estimate)**.** *Let*

$$\mu_{n,v'}=\frac{c_1(\overline{L}^{\mathrm{can}}_{n,v'})^{g+t}}{(L_n)^{g+t}}.$$

Then for every continuous $\mathrm{Gal}(\mathbb{C}_v/K_v)$-invariant f,

$$\lim_{n\to\infty}\int_{\overline{G}^{an}_{n,\mathbb{C}_{v'}}}f_{n,v'}\,d\mu_{n,v'}=\int_{\overline{G}^{an}_{\mathbb{C}_v}}f\,d\mu_{\overline{L},v}.$$

Proof. Let $d=g+t$. Since $L_n=\overline{G}_n(D)\otimes\pi_n^*N$, the measure $\mu_{n,v'}$ is represented by

$$c_1(\overline{G}_n(\overline{D}^{\mathrm{can}})_{v'})^t\wedge c_1(\pi_n^*\overline{N}_{v'})^g$$

after division by the degree L_n^d. All other terms in the expansion of $c_1(\overline{L}^{\mathrm{can}}_{n,v'})^d$ are zero as measures. They are positive mixed measures because the two metrics are semipositive, and

their total masses are the corresponding top-degree intersection numbers. Those masses vanish by Lemma 32. This uses only its numerical statement, not a Chow-class identity.

The abelian part is unchanged by φ_n, while the toric part is pulled back by multiplication by n on T. The canonical toric metric is invariant under this multiplication after the normalization $[n]^*\overline{D}^{\mathrm{can}} = n\overline{D}^{\mathrm{can}}$. Therefore the projection formula gives

$$(\varphi_n)_*\left(\frac{c_1(\overline{G}_n(\overline{D}^{\mathrm{can}})_{v'})^t \wedge c_1(\pi_n^*\overline{N}_{v'})^g}{L_n^d}\right) \longrightarrow \frac{c_1(\overline{G}(\overline{D}^{\mathrm{can}})_v)^t \wedge c_1(\pi^*\overline{N}_v)^g}{L^d}.$$

The convergence is weak convergence of Chambert-Loir measures. Testing against the continuous function f and using $f_{n,v'} = f \circ \varphi_n^{an}$ gives the asserted limit. □

Lemma 42 (First variation of height)**.** *Assume $O_{\overline{G}}(f)$ is an integrable adelic line bundle. For*

$$\overline{L}_{n,\lambda}^{\mathrm{can}} = \overline{L}_n^{\mathrm{can}} + \varphi_n^* O_{\overline{G}}(\lambda f)$$

and $|\lambda| \leq n^{-1}$, one has

$$\left| h_{\overline{L}_{n,\lambda}^{\mathrm{can}}}(\overline{G}_n) - h_{\overline{L}_n^{\mathrm{can}}}(\overline{G}_n) - \lambda n_v \int_{\overline{G}_{n,\mathbb{C}_{v'}}^{an}} f_{n,v'}\, d\mu_{n,v'} \right| \leq C_f |\lambda|^2 n.$$

Proof. Put $d = g + t$ and $F = \varphi_n^* O_{\overline{G}}(f)$. By the multilinearity of adelic intersections,

$$h_{\overline{L}_n^{\mathrm{can}}+\lambda F}(\overline{G}_n) - h_{\overline{L}_n^{\mathrm{can}}}(\overline{G}_n) = \frac{1}{(d+1)L_n^d} \sum_{r=1}^{d+1} \binom{d+1}{r} \lambda^r \big(F^r \cdot (\overline{L}_n^{\mathrm{can}})^{d+1-r}\big).$$

The term with $r = 1$ equals

$$\lambda n_v \int_{\overline{G}_{n,\mathbb{C}_{v'}}^{an}} f_{n,v'}\, d\mu_{n,v'}.$$

This is the local intersection formula defining the Chambert-Loir measure, with the field-extension normalization explained in Lemma 40; in the semiabelian asymptotic argument it is the linear term isolated in [1, Lemma 4.4].

It remains to bound the terms with $r \geq 2$. Since $O(f)$ has trivial underlying algebraic line bundle, every such term is controlled by an intersection involving at least two metrically non-trivial but algebraically trivial factors. Approximate f by model functions from above and below. For model functions the estimate follows from the arithmetic Hodge index and the projection formula; the general integrable case follows by passage to the limit. The required scaling is the bridge from [1, Lemma 4.4] to the present toric compactification, and is supplied by Lemma 34. Hence, after normalization by L_n^d,

$$\left| \sum_{r=2}^{d+1} \binom{d+1}{r} \lambda^r \big(F^r \cdot (\overline{L}_n^{\mathrm{can}})^{d+1-r}\big) \right| \leq C_f L_n^d |\lambda|^2 n.$$

Dividing by $(d+1)L_n^d$ gives the stated bound. □

Lemma 43 (Arithmetic volume comparison)**.** *Assume that*

$$\overline{L}_n^{\mathrm{can}}(\kappa_n) = \overline{L}_n^{\mathrm{can}} \otimes \varphi_n^* O_{\overline{G}}(\kappa_n)$$

is horizontally semipositive, with $\kappa_n \ll n^{-2}$. Then for $|\lambda| \leq n^{-1}$,

$$\widehat{\mathrm{vol}}_{\chi,K_n}^{\mathrm{raw}}(\overline{L}_{n,\lambda}^{\mathrm{can}}) - \big((\overline{L}_{n,\lambda}^{\mathrm{can}})^{g+t+1}\big)_{K_n}^{\mathrm{raw}} \geq -C_f[K_n : K]\,\big(n|\lambda|^2(1+\kappa_n n) + \kappa_n\big).$$

Proof. Let

$$\overline{A}_n = \overline{L}_n^{\mathrm{can}}(\kappa_n)$$

and write

$$\overline{L}_{n,\lambda}^{\mathrm{can}} = \big(\overline{A}_n + \varphi_n^* O_{\overline{G}}(\lambda f)\big) \otimes \varphi_n^* O_{\overline{G}}(-\kappa_n).$$

By assumption $\overline{A}_n$ is horizontally semipositive. The arithmetic volume comparison used here is Proposition 37, namely the volume part of [1, Lemma 4.6] together with its source [1, Lemma 2.6]. Applying it to $\overline{A}_n$ and to $\varphi_n^* O_{\overline{G}}(\lambda f)$ gives a lower bound for

$$\widehat{\mathrm{vol}}_\chi(\overline{A}_n + \overline{E}) - (\overline{A}_n + \overline{E})^{d+1}$$

in terms of the square of the size of the integrable perturbation $\overline{E}$. Here $\overline{E} = \varphi_n^* O_{\overline{G}}(\lambda f)$.

The contribution linear in λf is already accounted for by the intersection term. The remaining error is quadratic in λf. Thus the only external volume theorem used at this point is the Kühne–Yuan–Ikoma package recorded in Proposition 37; the passage from Kühne's $(\mathbb{P}^1)^t$-compactification to the present X_Σ is supplied by Lemma 34. In particular, the pullback by the auxiliary n-isogeny contributes at most one n-factor in the non-linear error terms, while the κ_n-correction changes the bound by the factor $1 + \kappa_n n$, because the semipositive reference bundle is $\overline{A}_n = \overline{L}_n^{\mathrm{can}}(\kappa_n)$. Keeping the field-extension factor of Lemma 40 before normalization, one obtains

$$\widehat{\mathrm{vol}}^{\mathrm{raw}}_{\chi,K_n}(\overline{A}_n + \varphi_n^* O_{\overline{G}}(\lambda f)) - \big((\overline{A}_n + \varphi_n^* O_{\overline{G}}(\lambda f))^{d+1}\big)^{\mathrm{raw}}_{K_n} \geq -C_f [K_n : K]\, n|\lambda|^2 (1 + \kappa_n n).$$

Finally

$$\overline{L}^{\mathrm{can}}_{n,\lambda} = \big(\overline{A}_n + \varphi_n^* O_{\overline{G}}(\lambda f)\big) \otimes \varphi_n^* O_{\overline{G}}(-\kappa_n),$$

and Lemma 39 changes the volume defect by at most $C_f [K_n : K] \kappa_n$. Combining the two estimates gives

$$\widehat{\mathrm{vol}}^{\mathrm{raw}}_{\chi,K_n}(\overline{L}^{\mathrm{can}}_{n,\lambda}) - \big((\overline{L}^{\mathrm{can}}_{n,\lambda})^{d+1}\big)^{\mathrm{raw}}_{K_n} \geq -C_f [K_n : K] \big(n|\lambda|^2(1 + \kappa_n n) + \kappa_n\big).$$

The constant is independent of n, λ, and the place v' above v. □

5.4. **Canonical equidistribution.**

Theorem 44 (Canonical semiabelian equidistribution on X_Σ)**.** *Assume D is big, nef and T-effective, and all local toric metrics of $\overline{D}$ are canonical. Then $\overline{L}^{\mathrm{can}}$ has v-adic equidistribution at every place v, with measure*

$$\mu_{\overline{L},v} = \frac{c_1(\overline{G}(\overline{D}^{\mathrm{can}})_v)^t \wedge c_1(\overline{\pi}^* \overline{N}_v)^g}{G(D)^t (\pi^* N)^g}.$$

Proof. By approximation in the nef cone it is enough to treat D ample. Fix a continuous test function f, approximated by integrable test functions, and fix $v' \mid v$. Minkowski's theorem applied to $\overline{L}^{\mathrm{can}}_{n,\lambda}$ gives a non-zero section whose v'-norm is bounded by the normalized arithmetic volume. Using Lemma 43, then Lemma 42 and Proposition 29, one obtains

$$\limsup_i \left| \frac{1}{\#O_v(x_i)} \sum_{y \in O_v(x_i)} f(y) - \int_{\overline{G}^{an}_{n,\mathbb{C}_{v'}}} f_{n,v'}\, d\mu_{n,v'} \right| \leq C \left(n^{-2}|\lambda|^{-1} + |\lambda| n (1 + \kappa_n n)\right) + C|\lambda|^{-1}\epsilon.$$

Here Lemma 40 is used to pass from the $[K_n : K]$-weighted volume defect to the displayed normalized estimate. The additional term $\kappa_n |\lambda|^{-1}$ coming from Lemma 43 is absorbed into $n^{-2}|\lambda|^{-1}$, since $\kappa_n \ll n^{-2}$. Choose $\lambda = n^{-3/2}$ and $\epsilon = n^{-2}$. Since $\kappa_n \ll n^{-2}$, the right side is $O(n^{-1/2})$. Letting $n \to \infty$ and using Lemma 41 gives the result. □

6. Explicit compression for quasi-canonical metrics

6.1. **Compressed metrics.** Let $\overline{D}$ now be a quasi-canonical toric metrized divisor. For $m \geq 1$ put

$$\overline{D}_m = \frac{1}{m} [m]^* \overline{D},$$

where $[m]$ denotes multiplication on the toric part. Define

$$\overline{L}_{(m,n)} = \overline{G}_n(\overline{D}_m) \otimes \overline{\pi}_n^* \overline{N}_{K_n}.$$

The canonical endpoint is denoted

$$\overline{L}_{(\infty,n)} = \overline{L}_n^{\mathrm{can}}.$$

Throughout the compressed section, L_n denotes the common underlying algebraic line bundle of the adelic line bundles $\overline{L}_{(m,n)}$. Indeed, the toric divisor underlying $\overline{D}_m = m^{-1}[m]^*\overline{D}$ is again D, since the support function of a toric Cartier divisor is homogeneous. Thus

$$L_{(m,n)} = G_n(D) \otimes \pi_n^* N_{K_n} = L_n$$

as algebraic line bundles, although their metrics depend on m. In particular the normalizing degree in all height, measure, and intersection estimates below is L_n^d, $d = g + t$, uniformly in m.

Proposition 45 (Equivariance in the horizontal compression parameter)**.** *Fix $n \in \mathbb{N} \cup \{\infty\}$. For any two positive integers m, m', the v-adic equidistribution property for $\overline{L}_{(m,n)}$ is equivalent to the corresponding property for $\overline{L}_{(m',n)}$.*

Proof. It suffices to compare m with 1. The morphism $[m]_{\overline{G}_n}$ is generically finite and maps generic sequences to generic sequences. The height identity is

$$h_{\overline{L}_{(m,n)}}(x) = \frac{1}{m} h_{\overline{G}_n(\overline{D})}([m]x) + h_{\overline{\pi}_n^* \overline{N}}(x),$$

with the abelian part transforming quadratically under multiplication. For small sequences, the abelian canonical height tends to zero, so smallness is preserved in both directions after passing to preimages. The measure identity follows from the projection formula:

$$\int f \, d\mu_{\overline{L}_{(1,n)},v} = \int f \circ [m]_{\overline{G}_n} \, d\mu_{\overline{L}_{(m,n)},v}.$$

□

6.2. **Lower bound along a compression path.**

Proposition 46 (Quasi-canonical compression normal form)**.** *After changing $\overline{D}$ by an M_K-constant and normalizing $\mu^{\mathrm{abs}}_{\overline{D}}(T) = 0$, a quasi-canonical toric metric has critical vector $(u_v)_v$ and constants $(\gamma_v)_v$ with*

$$\sum_v n_v u_v = 0, \qquad \sum_v n_v \gamma_v = 0.$$

For $m \geq 1$, the compressed metric $\overline{D}_m = m^{-1}[m]^*\overline{D}$ has local support functions

$$\Psi_{\overline{D}_m,v}(u) = \Psi_D\left(u - \frac{u_v}{m}\right) - \frac{\gamma_v}{m}.$$

Moreover, on each fixed toric chart used in Lemma 27, the shift u_v/m changes the theta-factor on the abelian quotient by m^{-1} times a Picard-zero class depending only on the original quasi-canonical metric and on the chart; the constant terms γ_v/m cancel globally by the product formula.

Proof. By the BGPS normal form for quasi-canonical toric metrics, after adding an M_K-constant the local support function at v can be written as

$$\Psi_{\overline{D},v}(u) = \Psi_D(u - u_v) - \gamma_v.$$

The normalization $\mu^{\mathrm{abs}}_{\overline{D}}(T) = 0$, together with the product-formula condition for adelic toric translations, gives

$$\sum_v n_v u_v = 0, \qquad \sum_v n_v \gamma_v = 0.$$

The underlying toric Cartier divisor is unchanged by the normalized pullback: the support function of D is homogeneous for toric multiplication, so

$$\frac{1}{m}[m]^*D = D$$

as an $\mathbb{R}$-Cartier divisor. On support functions, multiplication by m sends u to mu. Hence

$$\Psi_{\overline{D}_m,v}(u) = \frac{1}{m}\Psi_{\overline{D},v}(mu) = \frac{1}{m}\Psi_D(mu - u_v) - \frac{\gamma_v}{m} = \Psi_D\left(u - \frac{u_v}{m}\right) - \frac{\gamma_v}{m},$$

where the last equality uses the homogeneity of the canonical support function Ψ_D. The constants have no global height contribution because their weighted sum is $m^{-1}\sum_v n_v\gamma_v = 0$.

It remains to record the effect on the semiabelian local trivialization. On a fixed chart the toric coordinate attached to a character is written as a semiabelian rational character divided by the theta representative of the corresponding character-pushout Picard-zero bundle, as in Theorem 4.1 and Proposition 4.25. Translating the tropical coordinate by u_v/m therefore changes only the automorphy factor in this character-pushout direction. Since the translation is linear in u_v/m, the resulting class in $\mathrm{Pic}^0(A)_{\mathbb{R}}$ is m^{-1} times the Picard-zero class attached to the original quasi-canonical translation on that chart. The finite chart system gives finitely many such classes, all depending only on $\overline{D}$ and on the chosen local trivialization data. This is the asserted theta-factor statement. □

Lemma 47 (Compressed chart lower bound)**.** *Assume that D is an integral ample T-effective toric divisor and that $\overline{D}$ is quasi-canonical, normalized by $\mu^{\mathrm{abs}}_{\overline{D}}(T) = 0$. Let $m_0, \dots, m_r$ be the lattice points chosen in Lemma 27. There are real Picard-zero classes $E_0, \dots, E_r \in \mathrm{Pic}^0(A)_{\mathbb{R}}$, depending only on $\overline{D}$ and on this finite chart system, such that for every $m, n \geq 1$ and every $x \in \overline{G}_n(\overline{K})$, one can choose j with*

$$h_{\overline{G}_n(\overline{D}_m)}(x) \geq -h_{\overline{\Theta}^{(n)}_{j,m}}(\pi_n(x)),$$

where

$$\Theta^{(n)}_{j,m} := H^{(n)}_{m_j} + \frac{1}{m}E^{(n)}_j \in \mathrm{Pic}^0(A)_{\mathbb{R}}.$$

Here $H^{(n)}_{m_j}$ is the Picard-zero class appearing in Lemma 27, and $E^{(n)}_j$ denotes the same class E_j transported through the n-division extension G_n.

Proof. The finite chart system is the same as in Lemma 27, because D_m has the same underlying algebraic toric divisor D. On such a chart, the canonical proof compares the local toric section with the theta factor $H^{(n)}_{m_j}$ on the abelian quotient. By Proposition 46, replacing Ψ_D by

$$\Psi_D\left(u - \frac{u_v}{m}\right) - \frac{\gamma_v}{m}$$

does not introduce a scalar global height contribution, since $\sum_v n_v\gamma_v = 0$. The remaining effect of the translated critical vector is linear in u_v/m. Through the local trivialization and the theta-function height identity of Proposition 4.25, this linear term is exactly m^{-1} times a Picard-zero height on A. Since only finitely many charts are used, these Picard-zero directions are a fixed finite family $E_0, \dots, E_r$. Thus, on the chart where the chosen toric section does not vanish,

$$h_{\overline{G}_n(\overline{D}_m)}(x) + h_{\overline{\Theta}^{(n)}_{j,m}}(\pi_n(x)) \geq 0,$$

which is the asserted inequality. □

Lemma 48 (Compressed Picard-zero translation bound)**.** *With the notation of Lemma 47, there is a constant $C > 0$, independent of m, n, j, and $y \in A(\overline{K})$, such that*

$$h_{\overline{N}}(y) - h_{\overline{\Theta}^{(n)}_{j,m}}(y) \geq -Cn^{-2}.$$

Proof. Fix j. Since $\phi_N : A \to A^\vee$ is an isogeny, after clearing denominators in the finite-dimensional real vector space generated by the classes H_{m_j} and E_j, there are a positive integer a and points $q_j, e_j \in A(\overline{K}) \otimes_{\mathbb{Z}} \mathbb{R}$ such that

$$\phi_N(q_j) = aH_{m_j}, \qquad \phi_N(e_j) = aE_j.$$

Choose $q_{j,n}$ and $e_{j,n}$ with

$$nq_{j,n} = q_j, \qquad ne_{j,n} = e_j.$$

Then

$$\phi_N\left(q_{j,n} + \frac{1}{m}e_{j,n}\right) = a\Theta_{j,m}^{(n)}.$$

By Corollary 16, we have

$$h_{\overline{\Theta}_{j,m}^{(n)}}(y) = \frac{2}{a}\left\langle y,\, q_{j,n} + \frac{1}{m}e_{j,n}\right\rangle_N.$$

Completing the square for the Neron–Tate quadratic form gives

$$h_{\overline{N}}(y) - h_{\overline{\Theta}_{j,m}^{(n)}}(y) \geq -a^{-2}h_{\overline{N}}\left(q_{j,n} + \frac{1}{m}e_{j,n}\right).$$

Since $m \geq 1$ and

$$h_{\overline{N}}\left(q_{j,n} + \frac{1}{m}e_{j,n}\right) = n^{-2}h_{\overline{N}}\left(q_j + \frac{1}{m}e_j\right) \leq C_j n^{-2},$$

and since j ranges over a finite set, the result follows after enlarging C. □

Lemma 49 (Uniformity in the compression parameter)**.** *In Lemmas 47 and 48, all constants can be chosen independently of the compression parameter $m \geq 1$. Consequently the lower-bound constants used below are uniform for every path $m = m(n)$.*

Proof. The chart system is fixed once D is fixed. The only new terms introduced by the quasi-canonical metric are the finitely many Picard-zero directions E_j, and the constants γ_v/m have zero global contribution by the normalization in Proposition 46. Thus the chart inequality itself has no constant which grows with m.

It remains to make explicit the uniformity in the Picard-zero estimate. With the notation used in the proof of Lemma 48, the possible translating points are

$$q_j + \frac{1}{m}e_j, \qquad m \geq 1.$$

They lie in the compact line segment

$$\{q_j + \theta e_j : 0 \leq \theta \leq 1\}$$

inside the finite-dimensional real vector space generated by the chosen Picard-zero data. Since the Neron–Tate height associated with N is a continuous quadratic form on this space, there is a constant C_j such that

$$h_{\overline{N}}\left(q_j + \frac{1}{m}e_j\right) \leq C_j \qquad (m \geq 1).$$

After passing to n-division points, this gives

$$h_{\overline{N}}\left(q_{j,n} + \frac{1}{m}e_{j,n}\right) = n^{-2}h_{\overline{N}}\left(q_j + \frac{1}{m}e_j\right) \leq C_j n^{-2}.$$

Taking the maximum over the finite set of charts gives a constant independent of m, n, j, and the point y. This is the asserted uniformity. □

Proposition 50 (Lower bound for compressed metrics)**.** *Assume D is ample and T-effective, and normalize $\mu^{\mathrm{abs}}_{\overline{D}}(T) = 0$. Let $m(n)$ be any increasing integer function with $m(n) \le n$. Then there is $C > 0$, depending only on the fixed data and not on the function $m(\cdot)$, such that*

$$\mu^{\mathrm{abs}}_{\overline{L}_{(m(n),n)}}(\overline{G}_n) \ge -Cn^{-2}, \qquad -Cn^{-2} \le h_{\overline{L}_{(m(n),n)}}(\overline{G}_n) \le 0.$$

Consequently there exists $0 \le \kappa_n \le Cn^{-2}$ such that

$$\overline{L}_{(m(n),n)} \otimes \varphi_n^* O_{\overline{G}}(\kappa_n)$$

is horizontally semipositive.

Proof. As in Proposition 29, it is enough first to treat integral ample T-effective divisors; the general ample $\mathbb{R}$-divisor case follows by linearity of support functions and heights.

Let $m = m(n)$. For any $x \in \overline{G}_n(\overline{K})$, choose j by Lemma 47. Combining that lemma with Lemmas 48 and 49, applied to $y = \pi_n(x)$, gives

$$h_{\overline{L}_{(m,n)}}(x) = h_{\overline{G}_n(\overline{D}_m)}(x) + h_{\overline{N}}(\pi_n(x)) \ge -Cn^{-2}$$

with C independent of $m \le n$. Hence

$$\mu^{\mathrm{abs}}_{\overline{L}_{(m(n),n)}}(\overline{G}_n) \ge -Cn^{-2}.$$

The quasi-canonical normalization and Proposition 46 imply $\mu^{\mathrm{abs}}_{\overline{D}_m}(T) = 0$ for every m: compression moves the local critical vector from u_v to u_v/m, while both the vector and constant parts have zero adelic sum. Taking torsion points on the abelian quotient and toric points of toric height 0, and using Proposition 4.27, gives

$$\mu^{\mathrm{ess}}_{\overline{L}_{(m(n),n)}}(\overline{G}_n) \le 0.$$

Theorem 25 therefore yields

$$-Cn^{-2} \le h_{\overline{L}_{(m(n),n)}}(\overline{G}_n) \le 0.$$

All constants in these two lower estimates are independent of the chosen compression path $m(n)$; the restriction $m(n) \le n$ is only needed later, when the moving test function $f \circ [m(n)]$ is paired with the n-division compression.

Finally apply Lemma 26 with $\delta_n = n^{-2}$. The preceding uniform estimates verify the hypotheses of that input with a constant independent of $m(n)$. After enlarging C, it gives $0 \le \kappa_n \le Cn^{-2}$ and the asserted horizontal semipositivity. □

6.3. **The three compressed estimates.**

Proposition 51 (Fixed algebraic data in the horizontal parameter)**.** *Put $d = g + t$. For every $m \ge 1$, the normalized pullback $D_m = m^{-1}[m]^*D$ has underlying algebraic divisor D. Consequently*

$$M_m := \overline{G}(D_m) = \overline{G}(D)$$

as an algebraic line bundle, although its metric depends on m. Moreover, for $i > g$,

$$\deg\big(c_1(\pi^*N)^i \cap (c_1(M_m)^{d-i} \cap [\overline{G}])\big) = 0,$$

whereas $M_m^t(\pi^*N)^g > 0$. These statements, and their intersection constants, are independent of m.

Proof. The first assertion follows from the homogeneity of the support function of D. The projection formula identifies the displayed degree with

$$\deg\big(c_1(N|_A)^i \cap (\pi|_{\overline{G}})_*(c_1(M_m|_{\overline{G}})^{d-i} \cap [\overline{G}])\big)\,,$$

which vanishes for $i > \dim A = g$. Positivity of the term with $i = g$ follows from relative ampleness of M_m and ampleness of N, exactly as in the proof of [1, Lemma 4.2]. □

Lemma 52 (Exact horizontal pushforward at the target level)**.** *For every positive integer m,*

$$([m]_{\overline{G}}^{an})_*\mu_{\overline{L}_{(m,1)},v} = \mu_{\overline{L},v}.$$

Equivalently, for every continuous f,

$$\int_{\overline{G}^{an}_{\mathbb{C}_v}} f \circ [m]_{\overline{G}}^{an}\, d\mu_{\overline{L}_{(m,1)},v} = \int_{\overline{G}^{an}_{\mathbb{C}_v}} f\, d\mu_{\overline{L},v}.$$

Proof. This is the projection-formula identity already established in Proposition 45. Notice that it is exact for each fixed m; no limiting comparison between different values of m is involved. □

Lemma 53 (Degree asymptotic with the horizontal parameter fixed)**.** *Let $m = m(n)$ be any positive integer-valued function and put $\delta_n = \deg(\varphi_n) = n^t$. Then*

$$\left| L_n^d - \delta_n n^{-d+g} \binom{d}{g} M_m^t (\pi^* N)^g \right| \ll_{G,D,N} \delta_n n^{-d+g-1}.$$

The implied constant is independent of m and n.

Proof. This is the degree calculation in [1, Lemmas 4.2 and 4.3], applied for each n with the same horizontal index $m = m(n)$ on source and target. The algebraic divisor underlying D_m is always D, so the projection formula gives the same expansion and the same constants for every m. Proposition 51 makes all terms with more than g factors from π^*N vanish. The term with exactly g such factors is the displayed leading term; all remaining terms have one additional power of n^{-1}. □

Lemma 54 (Integral asymptotic with the horizontal parameter fixed)**.** *Assume that $f \in C^0(\overline{G}^{an}_{\mathbb{C}_v})$ is $\mathrm{Gal}(\mathbb{C}_v/K_v)$-invariant. Put*

$$f_m = f \circ [m]_{\overline{G}}^{an}, \qquad f_{(m,n),v'} = f_m \circ \varphi_n^{an}.$$

For any positive integer-valued function $m = m(n)$,

$$\begin{aligned} &\Big| \int_{\overline{G}^{an}_{n,\mathbb{C}_{v'}}} f_{(m,n),v'}, c_1(\overline{L}_{(m,n),v'})^d \\ &\quad - \delta_n n^{-d+g} \binom{d}{g} \int_{\overline{G}^{an}_{\mathbb{C}_v}} f_m\, c_1(\overline{G}(\overline{D}_m)_v)^t \wedge c_1(\pi^*\overline{N}_v)^g \Big| \ll_{G,D,N,f} \delta_n n^{-d+g-1}. \end{aligned}$$

The implied constant is independent of m and n.

Proof. Repeat the proof of [1, Lemma 4.3]. The projection formula and Proposition 51 isolates the term with g abelian factors. Kühne's local intersection bounds [1, Lemma 2.2(a),(c)] then give the displayed estimate. The bound is uniform in m: the algebraic line bundle is fixed, all measures involved are positive, and $\|f_m\|_{\sup} = \|f\|_{\sup}$. Thus no comparison of local coordinates, no uniform-continuity argument, and no condition on $m(n)/n$ enters the proof. □

Proposition 55 (Normalized fixed-horizontal-index compression)**.** *Under the hypotheses of Lemma 54, for every positive integer-valued function $m = m(n)$,*

$$\int_{\overline{G}^{an}_{n,\mathbb{C}_{v'}}} f_{(m,n),v'}\, d\mu_{(m,n),v'} - \int_{\overline{G}^{an}_{\mathbb{C}_v}} f_m\, d\mu_{\overline{L}_{(m,1)},v} \longrightarrow 0.$$

Proof. Divide the integral asymptotic of Lemma 54 by the degree asymptotic of Lemma 53. The common leading intersection number is positive by Proposition 51; hence the quotient error is $O(n^{-1})$, uniformly in the chosen values of $m(n)$. □

Lemma 56 (Compressed measure estimate)**.** *Let $m(n)$ be any positive integer-valued function. For*

$$f_{(m(n),n),v'} = f \circ [m(n)]^{an}_{\overline{G}} \circ \varphi_n^{an},$$

one has

$$\lim_{n\to\infty} \int_{\overline{G}^{an}_{n,\mathbb{C}_{v'}}} f_{(m(n),n),v'}\, d\mu_{(m(n),n),v'} = \int_{\overline{G}^{an}_{\mathbb{C}_v}} f\, d\mu_{\overline{L},v}.$$

Proof. Proposition 55 compares $(m(n), n)$ directly with $(m(n), 1)$. The toric metric on both sides has the same horizontal index $m(n)$, so this step is exactly the compression path treated by Kühne's projection-formula argument. Lemma 52 then identifies the target integral with $\int f\, d\mu_{\overline{L},v}$ for every n. Combining the two statements proves the limit, with no assumption that $m(n)/n \to 0$. □

Remark 57 (Notation and the rank-one boundary calculation)**.** *Here $v' \mid v$ is a place of the auxiliary field above v, and the prime in $f_{(m(n),n),v'}$ is essential. The phrase "well-defined" refers to the independence of this pulled-back test function from the chosen identification $\mathbb{C}_{v'} \simeq \mathbb{C}_v$, ensured by the assumed Galois invariance of f; it does not concern the existence of φ_n.*

The boundary case requested by the referee is completely explicit. If $A = 0$, $G = \mathbb{G}_m$, and $X_\Sigma = \mathbb{P}^1$, then $G_n = G$ and $\varphi_n(z) = z^n$. For the canonical toric metric,

$$\frac{1}{n}\varphi_n^*\overline{D} = \overline{D}, \qquad (\varphi_n^{an})_*\mu_{\overline{D},v} = \mu_{\overline{D},v}.$$

At an Archimedean place this is invariance of Haar probability measure on the unit circle under $z \mapsto z^n$; at a non-Archimedean place it is the corresponding invariance of the canonical Gauss-point measure. Hence the projection identity used above is exact in this canonical rank-one case.

For a translated quasi-canonical metric one should distinguish the literal normalized pullback identity

$$\frac{1}{n}\varphi_n^*\overline{D}_m = \overline{D}_{mn}.$$

The proof of Lemma 56 does not replace $\overline{D}_m$ by this pullback and does not compare $\overline{D}_m$ with $\overline{D}_{mn}$ by uniform continuity. Instead, for each n it keeps the same horizontal index $m = m(n)$ on the two levels (m, n) and $(m, 1)$, and applies the degree and local-intersection asymptotics of Kühne's Lemma 4.3. This is why no hypothesis $m(n)/n \to 0$ is required.

Lemma 58 (Compressed quadratic intersection scaling)**.** *Let $O_{\overline{G}}(f)$ be an integrable test line bundle, put*

$$f_m = f \circ [m]_{\overline{G}}, \qquad F_{m,n} = \varphi_n^* O_{\overline{G}}(f_m),$$

and let $d = g + t$. There is a constant $C_f > 0$, independent of $m \le n$, n, and λ, such that for $|\lambda| \le (nm)^{-1}$,

$$\left| \sum_{r=2}^{d+1} \binom{d+1}{r} \lambda^r \big(F_{m,n}^r \cdot \overline{L}_{(m,n)}^{d+1-r}\big) \right| \le C_f L_n^d |\lambda|^2 nm.$$

Proof. Apply Lemma 35 in the compressed case, where

$$\alpha_{\star,n} = nm, \qquad F_{\star,n} = F_{m,n}.$$

For each $2 \le r \le d+1$, it gives

$$|\lambda|^r \left| F_{m,n}^r \cdot \overline{L}_{(m,n)}^{d+1-r} \right| \le C_f L_n^d |\lambda|^2 nm.$$

The binomial coefficients and the finitely many possible values of r are absorbed into the constant C_f. Summing the displayed inequalities gives the result. □

Lemma 59 (Compressed first variation)**.** *Assume $O_{\overline{G}}(f)$ is integrable. Let $m = m(n) \le n$, and put*

$$\overline{L}_{(m,n),\lambda} = \overline{L}_{(m,n)} + \varphi_n^* O_{\overline{G}}(\lambda f_m), \qquad f_m = f \circ [m]_{\overline{G}}.$$

For $|\lambda| \le (nm)^{-1}$,

$$\left| h_{\overline{L}_{(m,n),\lambda}}(\overline{G}_n) - h_{\overline{L}_{(m,n)}}(\overline{G}_n) - \lambda n_v \int f_{(m,n),v'}\, d\mu_{(m,n),v'} \right| \le C_f |\lambda|^2 nm.$$

Proof. Put $d = g + t$ and

$$F_{m,n} = \varphi_n^* O_{\overline{G}}(f_m), \qquad f_m = f \circ [m]_{\overline{G}}.$$

By multilinearity of adelic intersections,

$$h_{\overline{L}_{(m,n)}+\lambda F_{m,n}}(\overline{G}_n) - h_{\overline{L}_{(m,n)}}(\overline{G}_n) = \frac{1}{(d+1)L_n^d} \sum_{r=1}^{d+1} \binom{d+1}{r} \lambda^r \big(F_{m,n}^r \cdot \overline{L}_{(m,n)}^{d+1-r}\big).$$

The $r = 1$ term is the local variation formula for the Chambert-Loir measure, with the field-extension normalization of Lemma 40, as in the source estimate [1, Lemma 4.4]:

$$\lambda n_v \int_{\overline{G}_{n,\mathbb{C}_{v'}}^{an}} f_{(m,n),v'}\, d\mu_{(m,n),v'}.$$

The remaining terms are controlled by Lemma 58. Dividing the resulting bound by $(d+1)L_n^d$ gives the desired estimate. □

Lemma 60 (Compressed volume comparison)**.** *Assume that*

$$\overline{L}_{(m,n)}(\kappa_n) = \overline{L}_{(m,n)} \otimes \varphi_n^* O_{\overline{G}}(\kappa_n)$$

is horizontally semipositive, with $\kappa_n \ll n^{-2}$. If $|\lambda| \le (nm)^{-1}$, then

$$\widehat{\mathrm{vol}}^{\mathrm{raw}}_{\chi,K_n}(\overline{L}_{(m,n),\lambda}) - \big((\overline{L}_{(m,n),\lambda})^{g+t+1}\big)^{\mathrm{raw}}_{K_n} \ge -C_f[K_n : K]\,\big(nm|\lambda|^2(1+\kappa_n n) + \kappa_n\big).$$

Proof. Let

$$\overline{A}_{m,n} := \overline{L}_{(m,n)}(\kappa_n).$$

By assumption $\overline{A}_{m,n}$ is horizontally semipositive. We write

$$\overline{L}_{(m,n),\lambda} = \big(\overline{A}_{m,n} + \varphi_n^* O_{\overline{G}}(\lambda f_m)\big) \otimes \varphi_n^* O_{\overline{G}}(-\kappa_n).$$

Proposition 37, equivalently the volume part of [1, Lemma 4.6] after the toric-compactification bridge and its [1, Lemma 2.6] bigness input, applies to the semipositive adelic line bundle $\overline{A}_{m,n}$ and the integrable perturbation $\varphi_n^* O_{\overline{G}}(\lambda f_m)$. Its linear part is the intersection term associated with $\overline{A}_{m,n} + \varphi_n^* O_{\overline{G}}(\lambda f_m)$; the remainder is controlled by the square of the perturbation.

The nm-dependence is the only point not present in Kühne's published Lemma 4.6. It comes from the compressed perturbation $f_m = f \circ [m]_{\overline{G}}$, and the square of that perturbation is controlled by the compressed intersection scaling isolated in Lemma 58. The horizontally semipositive reference bundle is $\overline{A}_{m,n} = \overline{L}_{(m,n)}(\kappa_n)$, so the quadratic constant is changed by the factor $1 + \kappa_n n$. Keeping the field-extension factor of Lemma 40 before normalization gives

$$\begin{aligned} &\widehat{\mathrm{vol}}^{\mathrm{raw}}_{\chi,K_n}(\overline{A}_{m,n} + \varphi_n^* O_{\overline{G}}(\lambda f_m)) - \big((\overline{A}_{m,n} + \varphi_n^* O_{\overline{G}}(\lambda f_m))^{d+1}\big)^{\mathrm{raw}}_{K_n} \\ &\qquad \ge -C_f[K_n : K]\, nm|\lambda|^2(1+\kappa_n n). \end{aligned}$$

The final passage from $\overline{A}_{m,n} + \varphi_n^* O_{\overline{G}}(\lambda f_m)$ to $\overline{L}_{(m,n),\lambda}$ is the constant shift $\varphi_n^* O_{\overline{G}}(-\kappa_n)$. By Lemma 39, this costs at most $C_f[K_n : K]\kappa_n$. Hence

$$\widehat{\mathrm{vol}}^{\mathrm{raw}}_{\chi,K_n}(\overline{L}_{(m,n),\lambda}) - \big((\overline{L}_{(m,n),\lambda})^{d+1}\big)^{\mathrm{raw}}_{K_n} \ge -C_f[K_n : K]\,\big(nm|\lambda|^2(1+\kappa_n n) + \kappa_n\big).$$

The constant C_f is independent of n, $m \le n$, and λ. □

6.4. **Correct parameter choice.** A useful explicit choice is

$$m(n) = n^{1/2}, \qquad \lambda = n^{-7/4}, \qquad \epsilon = |\lambda| n^{-1/2},$$

where one may restrict to square n's if desired. This choice satisfies

$$|\lambda| = n^{-7/4} \leq n^{-3/2} = (nm(n))^{-1}.$$

The raw quadratic error terms in Lemmas 59 and 60 are of size

$$nm(n)|\lambda|^2(1+\kappa_n n) + \kappa_n.$$

In the Minkowski–variation argument this quantity is divided by $|\lambda|$. Hence the normalized error to be checked is

$$|\lambda|\, nm(n)(1+\kappa_n n) + \kappa_n |\lambda|^{-1}.$$

By Proposition 50 and Lemma 49, the estimate $\kappa_n \ll n^{-2}$ is uniform in the chosen compression path. Therefore the second summand is dominated by $n^{-2}|\lambda|^{-1}$. The computation gives

$$n^{-2}|\lambda|^{-1} = n^{-2} n^{7/4} = n^{-1/4},$$

$$|\lambda|\, n\, m(n)(1+\kappa_n n) \ll n^{-7/4} n n^{1/2} = n^{-1/4},$$

and

$$|\lambda|^{-1}\epsilon = n^{-1/2}.$$

Thus the total error is $O(n^{-1/4})$, not $O(n^{-1/2})$. This is enough.

More generally, choose

$$m(n) = \lfloor n^a \rfloor, \qquad 0 < a < 1,$$

and

$$\lambda = n^{-(3+a)/2}.$$

Then $|\lambda| \leq (nm(n))^{-1}$ for large n, and

$$n^{-2}|\lambda|^{-1} = O\big(n^{-(1-a)/2}\big),$$

$$|\lambda| nm(n)(1+\kappa_n n) = O\big(n^{-(1-a)/2}\big).$$

The remaining volume-shift term satisfies the same bound:

$$\kappa_n |\lambda|^{-1} \ll n^{-2}|\lambda|^{-1} = O\big(n^{-(1-a)/2}\big).$$

Taking $\epsilon = |\lambda| n^{-(1-a)/2}$, the final error is

$$O\big(n^{-(1-a)/2}\big).$$

The choice $a = 1/2$ gives $O(n^{-1/4})$. One may restrict to n a square or fourth power to avoid floor notation.

7. Quasi-canonical equidistribution by explicit compression

Theorem 61 (Quasi-canonical compression theorem)**.** *Assume that D is big, nef and T-effective, and that $\overline{D}$ is quasi-canonical. Then every generic $\overline{L}$-small sequence in $G(\overline{K})$, for*

$$\overline{L} = \overline{G}(\overline{D}) \otimes \overline{\pi}^* \overline{N},$$

has v-adic equidistribution at every place v, with measure

$$\mu_{\overline{L},v} = \frac{c_1(\overline{G}(\overline{D})_v)^t \wedge c_1(\overline{\pi}^* \overline{N}_v)^g}{G(D)^t (\pi^* N)^g}.$$

Proof. By the birational reduction for nef and big toric divisors, it is enough to treat D ample. Normalize the metric so that $\mu_{\overline{D}}^{\mathrm{abs}}(T) = 0$. Fix a generic $\overline{L}$-small sequence $(x_i) \subset G(\overline{K})$, a place v, and a continuous $\mathrm{Gal}(\mathbb{C}_v/K_v)$-invariant test function f, approximated by integrable test functions.

Let $m(n) = \lfloor n^a \rfloor$, $0 < a < 1$, and choose $\lambda = n^{-(3+a)/2}$. By Minkowski's theorem, applied to $\overline{L}_{(m(n),n),\lambda}$, there is an auxiliary non-zero section whose divisor avoids the inverse image of x_i for all large i. Combining Lemma 60, Lemma 59, and Proposition 50, and then averaging over the finite fiber of $[m(n)] \circ \varphi_n$, yields

$$\begin{aligned}\limsup_i \left| \frac{1}{\#O_v(x_i)} \sum_{y \in O_v(x_i)} f(y) - \int_{\overline{G}^{an}_{n,\mathbb{C}_{v'}}} f_{(m(n),n),v'}\, d\mu_{(m(n),n),v'} \right| \\ \leq C\left(n^{-2}|\lambda|^{-1} + |\lambda| n m(n)(1+\kappa_n n)\right) + C|\lambda|^{-1}\epsilon.\end{aligned}$$

By Lemma 40, the factor $[K_n : K]$ in this defect is exactly the factor which disappears in the normalized Minkowski–variation estimate. Here the term $|\lambda| n m(n)(1+\kappa_n n)$ is the normalized form of the quadratic error $n m(n)|\lambda|^2(1+\kappa_n n)$ from Lemmas 59 and 60; the normalization comes from the usual division by $|\lambda|$ in the variation argument. The additional term $\kappa_n|\lambda|^{-1}$ coming from Lemma 60 is absorbed into $n^{-2}|\lambda|^{-1}$, since Proposition 50 gives $\kappa_n \ll n^{-2}$ uniformly for the chosen compression path. With the parameter choice above, the right side is

$$O\big(n^{-(1-a)/2}\big).$$

Letting $n \to \infty$ and using Lemma 56 gives

$$\lim_i \frac{1}{\#O_v(x_i)} \sum_{y \in O_v(x_i)} f(y) = \int_{\overline{G}^{an}_{\mathbb{C}_v}} f\, d\mu_{\overline{L},v}.$$

This proves equidistribution. □

Remark 62 (Generic versus strict)**.** *Theorem 61 proves the generic equidistribution property. It is not, by itself, a strong equidistribution theorem for strict sequences. The standard passage from generic equidistribution to strong equidistribution uses a Bogomolov theorem: if a strict small sequence is not generic, one extracts a generic small sequence on a proper subvariety and then uses Bogomolov to force that subvariety into a proper algebraic subgroup. This is the argument used by Kühne after proving the Bogomolov theorem for semiabelian varieties. In the present paper the strict restricted statement is itself used in the proof of Bogomolov, so this upgrade cannot be imported from the present Bogomolov theorem without circularity.*

Recent general approximation results, such as those of Ballay–Sombra [8], give powerful generic equidistribution statements and recover the generic semiabelian equidistribution theorem. They do not remove the logical distinction above. Thus any use of strict quasi-canonical equidistribution below must either be cited as an independent source theorem or proved by a separate non-circular argument.

8. Monocritical case

Theorem 63 (Monocritical equidistribution)**.** *Suppose D is big and T-effective, and $\overline{D} = (D, \{\|\cdot\|_v\})$ is a semipositive, monocritical and arithmetically T-effective toric metrized $\mathbb{R}$-divisor in the sense of Definitions 3.12 and 4.20. Then*

$$\overline{L} = \overline{G}(\overline{D}) \otimes \overline{\pi}^* \overline{N}$$

satisfies v-adic equidistribution at every place v of K. In the quasi-canonical case the limiting measure is

$$\mu_{\overline{L},v} = \frac{\hat{c}_1(\overline{G}(\overline{D})_v)^t \wedge \hat{c}_1(\overline{\pi}^* \overline{N}_v)^g}{\overline{G}(D)^t (\overline{\pi}^* N)^g}.$$

Remark 64 (Source ledger for the monocritical reduction)**.** *The reduction in this section uses the toric measure theory of Burgos–Philippon–Rivera-Letelier–Sombra [16], especially their Kantorovich–Rubinstein topology, the functions Φ_v, and the rigidity of monocritical metrics. The new point here is not to reprove that toric package, but to connect it with the semiabelian height decomposition supplied by Section 4 and with the quasi-canonical compression theorem 61.*

Lemma 65 (Arithmetic T-effectivity fixes the origin)**.** *If $\overline{D}$ is arithmetically T-effective, then*

$$\upsilon_{\overline{D}}(0) = \mu^{\mathrm{ess}}_{\overline{D}}(X_\Sigma).$$

In particular $0 \in \Delta_{D,\max}$.

Proof. Since D is T-effective, $0 \in \Delta_D$ [17, Proposition 4.9]. By arithmetic T-effectivity, the distinguished invariant section is small with local constants l_v, so

$$\upsilon_{\overline{D},v}(0) = \inf_{u\in N_{\mathbb{R}}} \big(-\Psi_{\overline{D},v}(u)\big) \geq l_v$$

for every place v. Hence

$$\upsilon_{\overline{D}}(0) = \sum_v n_v \upsilon_{\overline{D},v}(0) \geq \sum_v n_v l_v = \mu^{\mathrm{abs}}_{\overline{D}}(T).$$

By the toric successive-minima formula [18, Theorem 3.9 and Corollary 3.10],

$$\mu^{\mathrm{abs}}_{\overline{D}}(T) = \mu^{\mathrm{ess}}_{\overline{D}}(X_\Sigma) = \max_{x\in\Delta_D} \upsilon_{\overline{D}}(x).$$

The reverse inequality is automatic because $0 \in \Delta_D$. Thus $\upsilon_{\overline{D}}(0) = \mu^{\mathrm{ess}}_{\overline{D}}(X_\Sigma)$. □

In the rest of this section D is big and T-effective and $\overline{D}$ is semipositive, monocritical and arithmetically T-effective. Let

$$\mathbf{u} = (u_v)_v \in \bigoplus_v N_{\mathbb{R}}$$

be the critical vector of $\overline{D}$.

Definition 66 (Adelic valuation measures)**.** *Let $\mathcal{E}$ be the space of Borel probability measures on $N_{\mathbb{R}}$ with finite first moment, endowed with the Kantorovich–Rubinstein topology. An adelic measure with center $c \in N_{\mathbb{R}}$ is a collection $\boldsymbol{\nu} = (\nu_v)_v$, with $\nu_v \in \mathcal{E}$ and $\nu_v = \delta_0$ for all but finitely many v, such that*

$$\sum_v n_v E[\nu_v] = c.$$

The set of such adelic measures is denoted by $\mathcal{H}^c_K$. For such $\boldsymbol{\nu}$, put

$$\eta_{\overline{G}(\overline{D})}(\boldsymbol{\nu}) = -\sum_v n_v \int_{N_{\mathbb{R}}} \Psi_{\overline{D},v}\, d\nu_v.$$

Lemma 67 (Point measures and the semiabelian height)**.** *For $x \in G(\overline{K})$, the local trivializations of Proposition 4.27 define an adelic valuation measure*

$$\boldsymbol{\nu}_x = (\nu_{x,v})_v, \qquad \nu_{x,v} := \frac{1}{\#O_v(x)} \sum_{y\in O_v(x)} \delta_{\mathrm{val}_v(\psi_{j(y),v}(y))},$$

where for each y one chooses any chart $j(y)$ with $\pi(y) \in U_{j(y),v}$. This definition is independent of all such choices, and

$$\sum_v n_v E[\nu_{x,v}] = \big(-h_{\overline{H}_1}(\pi(x)), \dots, -h_{\overline{H}_t}(\pi(x))\big).$$

Moreover

$$\eta_{\overline{G}(\overline{D})}(\boldsymbol{\nu}_x) = h_{\overline{G}(\overline{D})}(x).$$

Proof. This is Proposition 4.27, rewritten in the notation of adelic valuation measures. The center identity is the displayed center formula there, and the equality $\eta_{\overline{G}(\overline{D})}(\boldsymbol{\nu}_x) = h_{\overline{G}(\overline{D})}(x)$ is exactly its height identity. □

For $v \in \Sigma(K)$, set

$$g_{1,v} = \upsilon_{\overline{D},v}, \qquad g_{2,v} = \sum_{w \neq v} \frac{n_w}{n_v} \upsilon_{\overline{D},w}.$$

For $c \in N_{\mathbb{R}}$ and $\mu \in \mathcal{E}$, define

$$\Phi_{v,c}(\mu) = \int g_{1,v}^{\vee}\, d\mu + g_{2,v}^{\vee}\left(\frac{c}{n_v} - E[\mu]\right) + \max_{x \in \Delta_D} (g_{1,v} + g_{2,v})(x), \qquad \Phi_v := \Phi_{v,0}.$$

If $x \in \Delta_{D,\max}$, put

$$B_v = \partial g_{1,v}(x) \cap (-\partial g_{2,v}(x)),$$

and let F_v be the minimal face of $\partial g_{1,v}(x)$ containing B_v.

Proposition 68 (BGPS package and the general-center defect inequality)**.** *Let $\boldsymbol{\nu} = (\nu_v)_v \in \mathcal{H}_K^c$. Then*

$$\max_v\{-n_v \Phi_{v,c}(\nu_v)\} \leq \eta_{\overline{G}(\overline{D})}(\boldsymbol{\nu}) - \mu^{\mathrm{ess}}_{\overline{G}(\overline{D})}(\overline{G}) \leq \sum_v -n_v \Phi_{v,c}(\nu_v).$$

No additional affine term occurs in the middle expression: arithmetic T-effectivity permits the choice $x = 0 \in \Delta_{D,\max}$. Moreover, if

$$R_D := \sup_{x \in \Delta_D} \|x\|,$$

then

$$\sup_{\mu \in \mathcal{E}} \left|\Phi_{v,c}(\mu) - \Phi_v(\mu)\right| \leq \frac{R_D}{n_v}\|c\|.$$

For the centered function $\Phi_v = \Phi_{v,0}$, the following facts are the toric results of [16, Propositions 3.9, 3.11, 3.14 and 4.16, Lemma 4.5, and Theorem 4.19]: $\Phi_v \leq 0$, and $\Phi_v(\mu) = 0$ if and only if

$$\mathrm{supp}(\mu) \subseteq F_v, \qquad E[\mu] \in B_v.$$

If $\Phi_v(\mu_i) \to 0$, then every weak cluster point of (μ_i) belongs to $\mathcal{E}$ and satisfies these two conditions. Finally, if $\overline{D}$ is monocritical with critical vector $\mathbf{u} = (u_v)_v$, then $F_v = \{u_v\}$; in this case BGPS Theorem 4.19 gives adelic KR convergence to $\boldsymbol{\delta}_{\mathbf{u}}$ for toric small nets.

Proof. Put

$$\widehat{\phi}_{i,v}(z) = g_{i,v}^{\vee}(z) + g_{i,v}(0), \qquad i = 1, 2.$$

Because $0 \in \Delta_{D,\max}$, Fenchel duality gives $\widehat{\phi}_{i,v} \leq 0$, and

$$\eta_{\overline{G}(\overline{D})}(\boldsymbol{\nu}) - \mu^{\mathrm{ess}}_{\overline{G}(\overline{D})}(\overline{G}) = -\sum_w n_w \int \widehat{\phi}_{1,w}\, d\nu_w. \tag{7.1}$$

Also

$$\Phi_{v,c}(\nu_v) = \int \widehat{\phi}_{1,v}\, d\nu_v + \widehat{\phi}_{2,v}\left(\frac{c}{n_v} - E[\nu_v]\right). \tag{7.2}$$

The nonpositivity of the second term in (7.2) immediately gives the right-hand inequality.

For the left-hand inequality, fix v. The sup-convolution and right-scaling identities for concave duals [13, Propositions 2.3.1(1) and 2.3.3(3)] give

$$\widehat{\phi}_{2,v} = \boxplus_{w \neq v} \left(\widehat{\phi}_{1,w} \frac{n_w}{n_v}\right).$$

Since $\boldsymbol{\nu}$ has center c,

$$\frac{c}{n_v} - E[\nu_v] = \sum_{w \neq v} \frac{n_w}{n_v} E[\nu_w].$$

The definition of sup-convolution followed by Jensen's inequality therefore gives

$$\widehat{\phi}_{2,v}\left(\frac{c}{n_v}-E[\nu_v]\right)\geq\sum_{w\neq v}\frac{n_w}{n_v}\widehat{\phi}_{1,w}(E[\nu_w])\geq\sum_{w\neq v}\frac{n_w}{n_v}\int\widehat{\phi}_{1,w}\,d\nu_w.$$

Together with (7.1)–(7.2), this yields

$$-n_v\Phi_{v,c}(\nu_v)\leq\eta_{\overline{G}(\overline{D})}(\boldsymbol{\nu})-\mu^{\mathrm{ess}}_{\overline{G}(\overline{D})}(\overline{G}).$$

Taking the maximum over v proves the left-hand inequality.

Finally, the concave dual $g^{\vee}_{2,v}$ is R_D-Lipschitz because its primal domain is contained in Δ_D. Translating its argument by c/n_v proves the displayed uniform estimate. The remaining centered compactness, vanishing and rigidity assertions are exactly the cited BGPS results. □

Lemma 69 (KR upgrade from BGPS concentration)**.** *Let $(\boldsymbol{\nu}_i)_i$, $\boldsymbol{\nu}_i=(\nu_{i,v})_v$, be adelic valuation measures whose centers $c_i=\sum_v n_vE[\nu_{i,v}]$ tend to 0. Assume*

$$\eta_{\overline{G}(\overline{D})}(\boldsymbol{\nu}_i)\longrightarrow\mu^{\mathrm{ess}}_{\overline{G}(\overline{D})}(\overline{G})$$

and $\Phi_v(\nu_{i,v})\to 0$ for every place v. If $\overline{D}$ is monocritical with critical vector $\mathbf{u}=(u_v)_v$, then

$$\boldsymbol{\nu}_i\longrightarrow\boldsymbol{\delta}_{\mathbf{u}}$$

in the adelic Kantorovich–Rubinstein topology.

Proof. The cluster statement in Proposition 68, together with monocriticality and BGPS Proposition 4.16, gives $F_v=\{u_v\}$. Hence every weak cluster point of $(\nu_{i,v})_i$ is δ_{u_v}, and the compactness part of BGPS Proposition 3.11 gives weak convergence

$$\nu_{i,v}\longrightarrow\delta_{u_v}$$

for every v.

It remains to upgrade weak convergence to KR convergence. Put

$$\eta_{\mathbf{u}}(\boldsymbol{\nu})=-\sum_v n_v\int_{N_{\mathbb{R}}}\Psi_D(z-u_v)\,d\nu_v(z).$$

For all but finitely many places, $u_v=0$ and $\Psi_{\overline{D},v}=\Psi_D$. Thus

$$f_v(z)=\Psi_{\overline{D},v}(z)-\Psi_D(z-u_v)$$

is a finite adelic family of bounded continuous functions. By the weak convergence just proved,

$$\eta_{\mathbf{u}}(\boldsymbol{\nu}_i)=\eta_{\overline{G}(\overline{D})}(\boldsymbol{\nu}_i)+\sum_v n_v\int f_v\,d\nu_{i,v}\longrightarrow\mu^{\mathrm{ess}}_{\overline{G}(\overline{D})}(\overline{G})+\sum_v n_vf_v(u_v).$$

The identity

$$\mu^{\mathrm{ess}}_{\overline{G}(\overline{D})}(\overline{G})+\sum_v n_vf_v(u_v)=0$$

is precisely the critical-vector normalization used in the proof of [16, Theorem 4.19]. Therefore $\eta_{\mathbf{u}}(\boldsymbol{\nu}_i)\to 0$.

Choose $x_0\in\operatorname{int}(\Delta_D)$. Since Δ_D is full dimensional, there is $C>0$ such that

$$\|z\|\leq -C\big(\Psi_D(z)-\langle x_0,z\rangle\big),\qquad z\in N_{\mathbb{R}}.$$

Using the elementary bound

$$W(\nu_{i,v},\delta_{u_v})\leq\int\|z-u_v\|\,d\nu_{i,v}(z),$$

we obtain

$$W_K(\boldsymbol{\nu}_i, \boldsymbol{\delta}_{\mathbf{u}}) \le \sum_v n_v \int \|z - u_v\| \, d\nu_{i,v}(z)$$
$$\le C\, \eta_{\mathbf{u}}(\boldsymbol{\nu}_i) + C\, \big|\langle x_0, c_i - \sum_v n_v u_v \rangle\big|.$$

The critical vector is centered, $\sum_v n_v u_v = 0$, and $c_i \to 0$. Together with $\eta_{\mathbf{u}}(\boldsymbol{\nu}_i) \to 0$, this proves adelic KR convergence. □

Proposition 70 (Defect vanishing for small sequences)**.** *Let (x_i) be a $\overline{L}$-small sequence in $G(\overline{K})$. Then for every place v,*

$$\Phi_v(\nu_{x_i,v}) \longrightarrow 0.$$

The centers of the adelic measures $\boldsymbol{\nu}_{x_i}$ tend to 0.

Proof. By Proposition 4.28 and the height decomposition

$$h_{\overline{L}}(x_i) = h_{\overline{G}(\overline{D})}(x_i) + \hat{h}_{\overline{N}}(\pi(x_i)),$$

smallness for $\overline{L}$ implies

$$h_{\overline{G}(\overline{D})}(x_i) \to \mu^{\mathrm{ess}}_{\overline{D}}(X_\Sigma), \qquad \hat{h}_{\overline{N}}(\pi(x_i)) \to 0.$$

Since the canonical heights $h_{\overline{H}_j}$ are quadratic forms controlled by $\hat{h}_{\overline{N}}$, we have

$$h_{\overline{H}_j}(\pi(x_i)) \to 0, \qquad 1 \le j \le t.$$

By Lemma 67, the centers therefore tend to 0, and

$$\eta_{\overline{G}(\overline{D})}(\boldsymbol{\nu}_{x_i}) = h_{\overline{G}(\overline{D})}(x_i) \to \mu^{\mathrm{ess}}_{\overline{G}(\overline{D})}(\overline{G}).$$

Write

$$c_i = \sum_w n_w E[\nu_{x_i,w}].$$

The general-center defect inequality in Proposition 68 gives, for each v,

$$0 \le -n_v \Phi_{v,c_i}(\nu_{x_i,v}) \le \eta_{\overline{G}(\overline{D})}(\boldsymbol{\nu}_{x_i}) - \mu^{\mathrm{ess}}_{\overline{G}(\overline{D})}(\overline{G}).$$

Thus $\Phi_{v,c_i}(\nu_{x_i,v}) \to 0$. Since $c_i \to 0$, the uniform estimate in Proposition 68 yields

$$\big|\Phi_{v,c_i}(\nu_{x_i,v}) - \Phi_v(\nu_{x_i,v})\big| \le \frac{R_D}{n_v}\|c_i\| \longrightarrow 0.$$

Consequently $\Phi_v(\nu_{x_i,v}) \to 0$, as required. □

Proposition 71 (KR convergence to the critical vector)**.** *Let (x_i) be a $\overline{L}$-small sequence in $G(\overline{K})$. Then*

$$\boldsymbol{\nu}_{x_i} \longrightarrow \boldsymbol{\delta}_{\mathbf{u}}$$

in the adelic Kantorovich–Rubinstein topology. Equivalently, for every place v,

$$\nu_{x_i,v} \longrightarrow \delta_{u_v}$$

in the local Kantorovich–Rubinstein topology.

Proof. By Proposition 70, the BGPS defects $\Phi_v(\nu_{x_i,v})$ tend to 0, and the centers of $\boldsymbol{\nu}_{x_i}$ tend to 0. The equality

$$\eta_{\overline{G}(\overline{D})}(\boldsymbol{\nu}_{x_i}) = h_{\overline{G}(\overline{D})}(x_i)$$

and the proof of Proposition 70 show that this quantity tends to the essential minimum. Lemma 69 therefore gives adelic KR convergence to $\boldsymbol{\delta}_{\mathbf{u}}$, and hence local KR convergence at every place. □

Proposition 72 (Quasi-canonical replacement)**.** *After changing $\overline{D}$ by an M_K-constant, assume $\mu^{\mathrm{abs}}_{\overline{D}}(T) = 0$. Define a quasi-canonical toric metric $\overline{D}'$ by*

$$\Psi_{\overline{D}',v}(z) = \Psi_D(z - u_v).$$

Let

$$\overline{L}' = \overline{G}(\overline{D}') \otimes \overline{\pi}^* \overline{N}.$$

Then $\mu^{\mathrm{abs}}_{\overline{D}'}(T) = \mu^{\mathrm{abs}}_{\overline{D}}(T) = 0$. Moreover every $\overline{L}$-small sequence in $G(\overline{K})$ is $\overline{L}'$-small.

Proof. The metric $\overline{D}'$ is quasi-canonical by construction: at the place v it is the canonical toric metric translated by the vector u_v. The product formula for the critical vector, $\sum_v n_v u_v = 0$, implies that the global absolute minimum is unchanged.

Let (x_i) be a $\overline{L}$-small sequence. By the proof of Proposition 70,

$$\hat{h}_{\overline{N}}(\pi(x_i)) \to 0.$$

By Proposition 71,

$$\boldsymbol{\nu}_{x_i} \to \boldsymbol{\delta}_{\mathbf{u}}$$

in the adelic KR topology. The functions $\Psi_{\overline{D}',v}$ are Lipschitz on $N_{\mathbb{R}}$, with constants controlled by the fixed polytope Δ_D. Hence

$$\eta_{\overline{G}(\overline{D}')}(\boldsymbol{\nu}_{x_i}) \to \eta_{\overline{G}(\overline{D}')}(\boldsymbol{\delta}_{\mathbf{u}}) = 0 = \mu^{\mathrm{ess}}_{\overline{D}'}(X_\Sigma).$$

Using Lemma 67 for $\overline{D}'$, this says

$$h_{\overline{G}(\overline{D}')}(x_i) \to 0.$$

Together with $\hat{h}_{\overline{N}}(\pi(x_i)) \to 0$, we obtain

$$h_{\overline{L}'}(x_i) \to \mu^{\mathrm{ess}}_{\overline{L}'}(\overline{G}),$$

so (x_i) is $\overline{L}'$-small. □

Proof of Theorem 63. Let (x_i) be a generic $\overline{L}$-small sequence. By Proposition 72, after the harmless M_K-constant normalization, the sequence is also small for the quasi-canonical adelic line bundle $\overline{L}'$. The quasi-canonical compression theorem, Theorem 61, applies to $\overline{L}'$, hence the Galois orbit measures of (x_i) equidistribute for $\overline{L}'$.

Proposition 71 identifies the tropical part of the limit: at the place v, the valuation measure is the Dirac mass δ_{u_v}. Thus the quasi-canonical limiting measure is exactly the measure obtained from the original monocritical metric. This proves v-adic equidistribution for $\overline{L}$. In the quasi-canonical case, the standard Chambert-Loir description gives the displayed formula for $\mu_{\overline{L},v}$. □

Remark 73. *The T-effectivity of D is used twice. First, together with arithmetic T-effectivity, it places the origin in $\Delta_{D,\max}$ by Lemma 65. Second, it makes the Deligne-pairing term $\langle \overline{G}(D), \dots, \overline{G}(D) \rangle$ vanish in the direction needed for equidistribution over the abelian quotient. Without this condition, the reduction to the quasi-canonical metric need not preserve the equidistribution property.*

9. Applications: Bogomolov theorem

9.1. **Set-up.** Let

$$1 \longrightarrow T \longrightarrow G \xrightarrow{\pi} A \longrightarrow 0$$

be a semiabelian variety over K, with $T \simeq \mathbb{G}_m^t$. Let

$$\overline{L} = \overline{G}(\overline{D}) \otimes \pi^* \overline{N}$$

be as in the main theorem: D is ample and T-effective, $\overline{D}$ is semipositive, monocritical and arithmetically T-effective, and $\overline{N}$ is symmetric ample with its canonical metric. Write

$$\mu_G := \mu_{\overline{L}}^{\mathrm{ess}}(\overline{G}) = \mu_{\overline{L}}^{\mathrm{abs}}(G).$$

Let

$$\mathbf{u} = (u_v)_v \in \bigoplus_v N_{\mathbb{R}}$$

be the critical vector of $\overline{D}$, as in Proposition 71.

9.2. **Special subvarieties and essential minima.**

Definition 74 (Special and $\overline{L}$-special subvarieties)**.** *An irreducible closed subvariety $X \subseteq G_{\overline{K}}$ is called* special *if*

$$X = a + B$$

for a connected algebraic subgroup $B \subseteq G_{\overline{K}}$ and a point $a \in G(\overline{K})$.

A point $p \in G(\overline{K})$ is called $\overline{L}$-special if

$$h_{\overline{L}}(p) = \mu_G.$$

An irreducible closed subvariety $X \subseteq G_{\overline{K}}$ is called $\overline{L}$-special if

$$X = p + B$$

for a connected algebraic subgroup $B \subseteq G_{\overline{K}}$ and an $\overline{L}$-special point $p \in G(\overline{K})$.

Remark 75. *The level in the definition of a special point is the ambient minimum μ_G, not $\mu_{\overline{L}}^{\mathrm{ess}}(X)$. The equality $\mu_{\overline{L}}^{\mathrm{ess}}(X) = \mu_G$ is a conclusion to be proved.*

Definition 76 (Strict sequence)**.** *A sequence (x_i) in $G(\overline{K})$ is strict if, for every proper translate $U \subsetneq G_{\overline{K}}$ of a connected algebraic subgroup, all but finitely many x_i lie outside U.*

9.3. **Quasi-canonical Bogomolov and Strict Upgrade.**

Theorem 77 (Restricted quasi-canonical Bogomolov property)**.** *Let $\overline{M}$ be an adelic line bundle on a semiabelian variety H of the form*

$$\overline{M} = \overline{H}(\overline{E}) \otimes \rho^* \overline{N}_H,$$

where $\overline{E}$ is the quasi-canonical toric metric attached to an ample T_H-effective divisor and $\overline{N}_H$ is canonical on the abelian quotient. Assume at every archimedean place either the standard $(\mathbb{P}^1)^t$-compactification, or the independently verified general-fan curvature–Haar slice input 9.5, with lattice-index normalization and no additional boundary mass, for all subvarieties used below. If an irreducible proper subvariety $Z \subsetneq H_{\overline{K}}$ carries a Z-generic sequence (z_i) with

$$h_{\overline{M}}(z_i) \longrightarrow \mu_{\overline{M}}^{\mathrm{ess}}(H),$$

then Z is contained in a proper translate of a connected algebraic subgroup of $H_{\overline{K}}$.

Proposition 78 (Dominant quasi-canonical semiabelian bridge)**.** *In the setting of Theorem 77, assume after a torsion translation and after replacing H by the inverse image of an abelian subvariety of its quotient that*

$$\rho(Z) = A_H.$$

If $Z \subsetneq H_{\overline{K}}$ carries a Z-generic sequence (z_i) with

$$h_{\overline{M}}(z_i) \longrightarrow \mu_{\overline{M}}^{\mathrm{ess}}(H),$$

then Z is contained in a proper translate of a connected algebraic subgroup of $H_{\overline{K}}$.

9.4. **Kühne-operation route for the dominant bridge.** The dominant bridge above is the point at which the proof can most naturally return to Kühne's original argument. The toric part in Proposition 78 is quasi-canonical; at an archimedean place v, its local support function has the form

$$\Psi_{\overline{E},v}(u) = \Psi_E(u - u_v) - \gamma_v.$$

Thus the limiting toric measure is not a new object: it is the Haar measure which occurs in Kühne's canonical case, translated from the maximal compact torus to the compact torus with tropical value u_v. We therefore transport Kühne's local-trivialization analysis [1, Lemma 3.3, Proposition 4.1, Lemmas 5.1–5.3, Proposition 6.1, Lemma 6.2, and equation (6.2)] from the standard $(\mathbb{P}^1)^t$ compactification to the present toric compactification.

Lemma 79 (Quasi-canonical Haar shift)**.** *Let v be an archimedean place, and let $\overline{E}$ be a quasi-canonical toric metric on the toric part T_H of H. Write its local support function in BGPS normal form*

$$\Psi_{\overline{E},v}(u) = \Psi_E(u - u_v) - \gamma_v, \qquad u \in N_{H,\mathbb{R}}.$$

Let $S_v \subset T_H(\mathbb{C}_v)$ be the maximal compact subgroup, and put

$$S_{u_v} := \mathrm{val}_v^{-1}(u_v) = S_v \cdot e^{-u_v}$$

in the logarithmic coordinates used in the archimedean toric uniformization. Then the canonical toric measure attached to $\overline{E}_v$ is the translate by e^{-u_v} of the canonical toric measure attached to E. In particular its toric support is S_{u_v}, and the normalized Haar volume form on the compact toric factor is unchanged.

Moreover this translation is functorial for products, inverses and difference morphisms: the center is carried by the corresponding linear map on cocharacter spaces. Thus, for the Kühne difference map

$$(x_1, \dots, x_m) \longmapsto (x_1 x_2^{-1}, \dots, x_{m-1} x_m^{-1}),$$

the common quasi-canonical center $(u_v, \dots, u_v)$ is sent to $(0, \dots, 0)$.

Proof. The displayed normal form says exactly that the local potential of $\overline{E}_v$ on the analytic torus is the canonical toric potential composed with the translation $u \mapsto u - u_v$, up to the additive constant γ_v. The constant does not affect curvature currents. Hence the Chern current, and therefore its top-degree toric Monge–Ampere measure, is the translate by e^{-u_v} of the canonical one. In the canonical archimedean toric case, this measure is the normalized Haar measure on S_v; equivalently this is the toric part of Kühne's canonical measure description and of the standard canonical toric metric description. Translating gives Haar measure on $S_v \cdot e^{-u_v} = \mathrm{val}_v^{-1}(u_v)$.

The last assertion follows from the group law in tropical coordinates: $\mathrm{val}_v(xy) = \mathrm{val}_v(x) + \mathrm{val}_v(y)$ and $\mathrm{val}_v(x^{-1}) = -\mathrm{val}_v(x)$. Consequently products, inverses and difference maps transport the translated compact toric factors by the same linear maps as in the canonical case. Haar volume is invariant under translation, so no Jacobian or normalization factor is introduced. □

Lemma 80 (Fan-chart compact-torus coordinates)**.** *Let v be an archimedean place. Use Kühne's archimedean local trivialization, in the form of [1, Section 3.5, Lemma 3.3], and compose the resulting toric coordinate with the affine toric charts of the chosen fan Σ. Let*

$$\Psi_j : \overline{H}^{an}_{\mathbb{C}_v}|_{U_j} \simeq U_j \times X^{an}_{\Sigma,\mathbb{C}_v}$$

be one of the local trivializations, and let

$$\psi_j : H^{an}_{\mathbb{C}_v}|_{U_j} \longrightarrow T_H^{an}$$

be its toric coordinate on the open semiabelian part. For $m \in M_H$, put

$$\psi_{j,m} := \chi^m \circ \psi_j.$$

Then:

(a) *If $\sigma \in \Sigma$ and $m \in \sigma^\vee \cap M_H$, then $\psi_{j,m}$ extends to an analytic function on the affine chart*
$$U_j \times X^{an}_{\sigma,\mathbb{C}_v} \subset \overline{H}^{an}_{\mathbb{C}_v}|_{U_j}.$$
(b) *For $u \in N_{H,\mathbb{R}}$, the translated compact toric factor over the chart is described by*
$$\mathrm{val}_v(\psi_j(y)) = u \quad \Longleftrightarrow \quad |\psi_{j,m}(y)| = \exp(-\langle m, u\rangle) \quad \text{for all } m \in M_H.$$
(c) *If $m_1, \dots, m_r \in M_H$ are $\mathbb{Z}$-linearly independent, then on each toric fiber the phase map*
$$y \longmapsto \left(e^{\langle m_\ell, u\rangle}\psi_{j,m_\ell}(y)\right)_{\ell=1}^{r}$$
restricts on $\mathrm{val}_v(\psi_j(y)) = u$ to a real-analytic submersion onto $(S^1)^r$. If $m_1, \dots, m_t$ is a $\mathbb{Z}$-basis of M_H, this is a real-analytic isomorphism with $(S^1)^t$.

Proof. Part (a) is the affine-chart part of the archimedean local trivialization, not an application of Theorem 4.1, which was stated for non-archimedean places. Kühne's Lemma 3.3 gives local toric coordinates on the semiabelian torsor with unitary transition functions. Once these coordinates are fixed, the passage from the open torus to $X_\sigma = \operatorname{Spec} \mathbb{C}_v[\sigma^\vee \cap M_H]$ is purely toric: the characters χ^m with $m \in \sigma^\vee \cap M_H$ are regular, and their pullbacks through the toric coordinate ψ_j are the analytic monomials used to extend the local trivialization across the toric boundary. This is the same fan-chart argument used in the proof of Theorem 4.1, but with Kühne's archimedean local trivialization as the analytic input.

For (b), on the principal torus the tropicalization is a group homomorphism and satisfies
$$-\log|\chi^m(t)| = \langle m, \mathrm{val}_v(t)\rangle.$$
Taking $t = \psi_j(y)$ gives the displayed modulus condition. Conversely, the characters separate points of $N_{H,\mathbb{R}}$, so the displayed equalities for all $m \in M_H$ recover $\mathrm{val}_v(\psi_j(y)) = u$.

For (c), after fixing u, the fiber $\mathrm{val}_v^{-1}(u) \subset T_H(\mathbb{C}_v)$ is the translate $e^{-u}S_v$ of the maximal compact torus. Multiplication by $e^{\langle m_\ell, u\rangle}$ normalizes χ^{m_ℓ} to a unit-modulus character on this compact torus. Independent characters give a submersion of compact real tori, and a lattice basis gives an isomorphism. Transporting this description through ψ_j proves the claim. □

Definition 81 (Rank-active character frames)**.** *Let $X \subseteq H_{\overline{K}}$ be irreducible of positive dimension, put*
$$d = \dim X, \qquad d_0 = \dim \rho(X), \qquad r = d - d_0,$$
and fix an archimedean place v. For a tuple $\mathbf{m} = (m_1, \dots, m_r) \in M_H^r$ and a local trivialization U_j, write
$$\Phi_{j,\mathbf{m}} = (\psi_{j,m_1}, \dots, \psi_{j,m_r}) : H^{an}_{\mathbb{C}_v}|_{U_j} \longrightarrow (\mathbb{G}_m^r)^{an}_{\mathbb{C}_v}.$$
We say that $\mathbf{m}$ is rank-active for X if the characters $m_1, \dots, m_r$ are $\mathbb{Z}$-linearly independent and the locus
$$\ker d(\rho^{an}, \Phi_{j,\mathbf{m}}) \cap T^{1,0}_{\mathbb{C},y} X^{an}_{\mathbb{C}_v} \neq 0$$
does not contain the smooth locus of
$$X^{an}_{\mathbb{C}_v} \cap H^{an}_{\mathbb{C}_v} \cap \rho^{-1}(\rho(X)^{sm}).$$
The condition is independent of the chosen local trivialization because, on overlaps, the toric coordinates differ by compact-torus units and hence by invertible analytic changes of the target torus.

Lemma 82 (General fan rank slices)**.** *Let $\mathbf{m} = (m_1, \dots, m_r)$ be rank-active for X, and let $u_v \in N_{H,\mathbb{R}}$ be the quasi-canonical center at the archimedean place v. Define*
$$X_{\mathbf{m},u_v} = \bigcup_{j \in J} \left\{ y \in X^{an}_{\mathbb{C}_v} \cap \rho^{-1}(U_j) : |\psi_{j,m_\ell}(y)| = \exp(-\langle m_\ell, u_v\rangle) \text{ for } 1 \le \ell \le r \right\}.$$
Then there is a closed complex-analytic subset
$$E_{\mathbf{m}} \subset X^{an}_{\mathbb{C}_v}$$

of dimension $< d$ such that:

(a) *on*
$$\big(X^{an}_{\mathbb{C}_v} \cap \rho^{-1}(U_j)\big) \setminus E_{\mathbf{m}},$$
the map
$$(\rho^{an}, \Phi_{j,\mathbf{m}}) : X^{an}_{\mathbb{C}_v} \longrightarrow \rho(X)^{sm,an}_{\mathbb{C}_v} \times (\mathbb{G}^r_m)^{an}_{\mathbb{C}_v}$$
is a local biholomorphism;

(b) $X_{\mathbf{m},u_v} \setminus E_{\mathbf{m}}$ *is a finite union of embedded real-analytic submanifolds, each of real dimension* $d + d_0$*;*

(c) *on*
$$\big(X_{\mathbf{m},u_v} \cap \rho^{-1}(U_j)\big) \setminus E_{\mathbf{m}},$$
the map
$$y \longmapsto \Big(\rho(y), \Big(e^{\langle m_\ell, u_v\rangle}\psi_{j,m_\ell}(y)\Big)^r_{\ell=1}\Big)$$
is a real-analytic local isomorphism with target
$$\rho(X)^{sm,an}_{\mathbb{C}_v} \times (S^1)^r.$$

Proof. Let $R_{\mathbf{m},j}$ be the rank-failure locus of $(\rho^{an}, \Phi_{j,\mathbf{m}})$ on
$$X^{sm,an}_{\mathbb{C}_v} \cap H^{an}_{\mathbb{C}_v} \cap \rho^{-1}(\rho(X)^{sm,an}_{\mathbb{C}_v}) \cap \rho^{-1}(U_j).$$
It is cut out by the maximal minors of the analytic Jacobian, hence is complex-analytic. On overlaps these loci agree, because the transition between the two toric coordinate systems is an invertible analytic change in the toric variables, together with multiplication by compact-torus units. The rank-active hypothesis says that their union is not the whole smooth analytic space. Since $X^{an}_{\mathbb{C}_v}$ is irreducible, the closure of this union has dimension $< d$. Enlarging it by
$$(X \setminus X^{sm})^{an}_{\mathbb{C}_v}, \qquad \rho^{-1}(\rho(X) \setminus \rho(X)^{sm})^{an}_{\mathbb{C}_v}, \qquad (\overline{X} \setminus X)^{an}_{\mathbb{C}_v}$$
gives the required closed complex-analytic set $E_{\mathbf{m}}$.

Outside $E_{\mathbf{m}}$, the differential of $(\rho^{an}, \Phi_{j,\mathbf{m}})$ on the d-dimensional complex tangent space of X has trivial kernel. The target has complex dimension
$$d_0 + r = d,$$
so the analytic inverse function theorem gives the local biholomorphism in (a).

The equations defining $X_{\mathbf{m},u_v}$ are, after the local biholomorphism in (a),
$$|z_\ell| = \exp(-\langle m_\ell, u_v\rangle), \qquad 1 \le \ell \le r,$$
inside $(\mathbb{G}^r_m)^{an}_{\mathbb{C}_v}$. These are r independent real-analytic equations. Therefore the slice has real dimension
$$2d - r = 2d - (d - d_0) = d + d_0.$$
Compactness of the analytic closure and finiteness of the cover $\{U_j\}_{j\in J}$ give finitely many embedded real-analytic pieces. Finally Lemma 80 identifies the normalized coordinates
$$e^{\langle m_\ell, u_v\rangle}\psi_{j,m_\ell}$$
with phase coordinates on $(S^1)^r$, proving (c). □

Lemma 83 (One-wall toric corner current)**.** *Let v be archimedean, and let U_j be one of the local trivialization charts of Lemma 80. Assume that, on a relatively compact open subset of the principal semiabelian orbit, the canonical toric metric of E has two adjacent affine toric potentials*
$$\ell_0(u) = \langle m_0, u - u_v\rangle + c_0, \qquad \ell_1(u) = \langle m_1, u - u_v\rangle + c_1$$
and is locally equal, up to addition of a pluriharmonic affine term, to
$$\max\{\ell_0(u), \ell_1(u)\}.$$

Suppose their common wall is given by

$$\langle m, u - u_v \rangle = 0, \qquad m = m_1 - m_0 \neq 0.$$

Writing

$$z_m = e^{\langle m, u_v \rangle} \psi_{j,m},$$

the curvature current contributed by this wall is a positive constant multiple of

$$dd^c \big| \log |z_m| \big|.$$

Consequently, after testing against a compactly supported function and using Kühne's normalization of dd^c,

$$\int h \, dd^c \big| \log |z_m| \big| = \int_{|z_m|=1} h \, d\mu_m^{\mathrm{Haar}},$$

where $d\mu_m^{\mathrm{Haar}}$ denotes the normalized Haar measure in the m-phase direction, up to Kühne's harmless factor 2.

Proof. The affine terms ℓ_0 and ℓ_1 define pluriharmonic functions on the analytic torus, so their dd^c-currents vanish on the principal orbit. After subtracting the affine function ℓ_0, the local potential becomes

$$\max\{0, \ell_1 - \ell_0\} = \max\{0, \langle m, u - u_v \rangle\}.$$

In the local monomial coordinate $\psi_{j,m}$, the valuation convention gives

$$\log |z_m| = \log |\psi_{j,m}| + \langle m, u_v \rangle = -\langle m, u - u_v \rangle$$

up to the fixed sign convention for val_v. Since

$$\max\{0, s\} = \frac{s + |s|}{2}$$

and $dd^c s = 0$ for an affine logarithmic coordinate, we obtain

$$dd^c \max\{0, \langle m, u - u_v \rangle\} = \frac{1}{2} \, dd^c \big| \log |z_m| \big|.$$

The factor $1/2$, together with the lattice length of the jump of the support function across the wall, is absorbed into the positive wall weight. The final identity is exactly Kühne's elementary computation [1, Appendix B] after the substitution $z = z_m$. □

Proposition 84 (Product-corner normal form)**.** *Let τ be a codimension-r cell of the corner complex of the canonical toric potential. Assume that its dual cell is locally a Minkowski sum of r independent segments (the product-corner condition). In a neighborhood of the relative interior of τ, the canonical potential is, modulo an affine pluriharmonic function, a finite positive linear combination of simple one-wall potentials*

$$\max\{0, \langle m_{\tau,\ell}, u - u_v \rangle\}, \qquad 1 \leq \ell \leq r,$$

with the $m_{\tau,\ell}$ linearly independent and rank-active on X after discarding a lower-dimensional analytic exceptional set. The Bedford–Taylor mixed product along τ is the corresponding positive mixed toric intersection weight times the product of the r one-wall currents supplied by Lemma 83.

Proof. The assertion is local on the real polyhedral space $N_{H,\mathbb{R}}$ near the relative interior of τ. The product-corner hypothesis gives primitive integral normal characters

$$m_{\tau,1}, \ldots, m_{\tau,r} \in M_H$$

which are linearly independent. After a common regular refinement, the remaining strata of the corner complex inside the same neighborhood have smaller dimension; their inverse images in $X_{\mathbb{C}_v}^{an}$ are absorbed into the exceptional analytic set already allowed in Lemma 82.

On such a simple star the canonical toric weight is piecewise affine and concave. Choose one affine branch ℓ_0 as a base branch. Subtracting ℓ_0, which gives a pluriharmonic factor on

the analytic torus, leaves a piecewise affine concave function whose bending locus is exactly the union of the r coordinate walls in the normal directions to τ. The jump of the affine slope across the ℓ-th wall is a positive multiple of $m_{\tau,\ell}$, because the metric is the canonical semipositive toric metric attached to the ample divisor E. Hence, after possibly shrinking the simple star,

$$\Psi_E(u) - \ell_0(u) = \sum_{\ell=1}^{r} b_{\tau,\ell} \max\{0, \langle m_{\tau,\ell}, u - u_v \rangle\}, \qquad b_{\tau,\ell} > 0,$$

up to an additive constant and with the harmless sign fixed by the valuation convention in Lemma 80. Refining once more if several simple stars meet along the same τ gives the finite positive linear combination appearing in the statement.

It remains to identify the current. Bedford–Taylor products for locally bounded plurisubharmonic toric potentials are local and multilinear, and affine logarithmic terms have zero dd^c on the principal torus. Therefore the contribution of the cell τ is the mixed coefficient obtained by polarizing the positive jumps $b_{\tau,\ell}$, multiplied by the wedge product of the one-wall currents. The positivity and the agreement of this coefficient with the mixed toric intersection weight are precisely the BPS polyhedral Monge–Ampere description [13, Proposition 2.7.4, Definition 2.7.12 and Proposition 2.7.13], together with the toric Chern-current comparison [13, Theorem 4.8.11 and Corollary 4.8.12]. Finally, if the resulting normal frame fails to be rank-active on a component of X, that component is part of the rank-failure analytic set in Lemma 82. This proves the stated local normal form and mixed-product description. □

Remark 85 (BPS source of the mixed corner input)**.** *Proposition 84 is the exact place where the standard toric metric-current theory should be cited or unpacked. In BPS [13, §2.7], the real Monge–Ampere measure of a piecewise affine concave function is expressed by the dual polyhedral complex [13, Proposition 2.7.4], and the mixed Monge–Ampere operator is defined and shown continuous in [13, Definition 2.7.12 and Proposition 2.7.13]. The passage from toric metrized line bundles to these real Monge–Ampere measures is [13, Theorem 4.8.11 and Corollary 4.8.12]. What remains in the present notation is therefore not a new equidistribution theorem: it is the local translation from the BPS polyhedral mixed Monge–Ampere weights to the monomial wall coordinates $\psi_{j,m_{\tau,\ell}}$ supplied by Lemma 80.*

Lemma 86 (Torus-invariant lift of atomic tropical measures)**.** *Let $T = (\mathbb{C}^\times)^r$, let*

$$\mathrm{val} : T \longrightarrow \mathbb{R}^r$$

be the logarithmic absolute-value map, and let μ be a finite positive Radon measure on T which is invariant under the compact torus $(S^1)^r$. If

$$\mathrm{val}_*\,\mu = \sum_{\nu \in F} a_\nu\, \delta_{u_\nu}$$

for a finite set F, with $a_\nu > 0$, then

$$\mu = \sum_{\nu \in F} a_\nu\, \mu_{u_\nu}^{\mathrm{Haar}},$$

where $\mu_{u_\nu}^{\mathrm{Haar}}$ is the normalized Haar measure on the compact real torus

$$\mathrm{val}^{-1}(u_\nu) = \{|z_1| = e^{-u_{\nu,1}}, \dots, |z_r| = e^{-u_{\nu,r}}\}.$$

Proof. Disintegrate μ with respect to $\mathrm{val}_*\,\mu$. For $\mathrm{val}_*\,\mu$-almost every u, the conditional measure on the fiber $\mathrm{val}^{-1}(u)$ is invariant under the compact torus $(S^1)^r$. This action is transitive on the fiber, hence the conditional probability measure is the normalized Haar measure. Since the base measure is the finite atomic measure displayed above, the asserted formula follows. The same argument is equivalently obtained by testing against a compactly supported continuous function and first averaging it over the compact torus fibers. □

Definition 87 (Corner-active character frames)**.** *After replacing the toric fan by a finite regular subdivision, let τ be a codimension-r cell in the corner complex of the canonical toric potential which meets the rank-active locus of X. A tuple*

$$\mathbf{m}_\tau = (m_{\tau,1}, \dots, m_{\tau,r}) \in M_H^r$$

is called corner-active for X along τ if the characters $m_{\tau,\ell}$ are linearly independent, span the normal character space of τ in the local star of the corner complex, and are rank-active for X in the sense of Definition 81, after removing the lower-dimensional analytic exceptional set.

Proposition 9.1 (Corner-active polyhedral quotient factorization)**.** *Let τ be a codimension-r corner cell as in Definition 87, and let $\mathbf{m}_\tau = (m_{\tau,1}, \dots, m_{\tau,r})$ be a corner-active frame. For every local trivialization chart, the local map*

$$q_{j,\mathbf{m}_\tau} := (\rho^{an}, \Phi_{j,\mathbf{m}_\tau})$$

has the following polyhedral factorization property. On the relevant corner-active cone or cell of the chosen regular subdivision, the local toric weight of $\overline{H}(\overline{E})_v$ can be written as

$$\varphi_\tau(\mathrm{val}(z_1, \dots, z_r)) + \ell_\tau(u) + c_\tau,$$

where φ_τ is the concave BPS toric weight on the r-dimensional quotient torus with coordinates

$$z_\ell = e^{\langle m_{\tau,\ell}, u_v\rangle}\psi_{j,m_{\tau,\ell}}.$$

Here ℓ_τ is an integral affine function in the inactive toric directions and c_τ is a constant. The BPS tropicalization coordinate on this quotient is the logarithmic absolute-value coordinate of the z_ℓ's, with the valuation sign convention used in Lemma 80.

Proof. Work in the star of the cell τ. Since the canonical toric potential is piecewise affine after the chosen regular subdivision, it is a finite polyhedral envelope, with the sign convention fixed by the metric convention, of affine functions

$$u \longmapsto \langle a_\alpha, u - u_v\rangle + c_\alpha$$

on this star. Choose one affine branch $\ell_\tau(u) + c_\tau$ as a base branch. If two affine branches meet along τ, their difference vanishes on the tangent space of τ. Hence every difference

$$\langle a_\alpha - a_0, u - u_v\rangle + (c_\alpha - c_0)$$

belongs to the real span of the normal characters $m_{\tau,1}, \dots, m_{\tau,r}$. Therefore, after subtracting the base affine branch, the remaining piecewise affine function depends only on the r normal coordinates

$$\langle m_{\tau,\ell}, u - u_v\rangle, \qquad 1 \le \ell \le r.$$

This defines the quotient piecewise affine concave function φ_τ. In the analytic local trivialization,

$$\log|z_\ell| = \log|\psi_{j,m_{\tau,\ell}}| + \langle m_{\tau,\ell}, u_v\rangle = -\langle m_{\tau,\ell}, u - u_v\rangle,$$

with the sign convention fixed in Lemma 80. Thus the same piecewise affine function is exactly the pullback of the BPS quotient toric weight through the displayed quotient coordinates. The base affine branch is the inactive affine term $\ell_\tau(u) + c_\tau$. This proves the factorization. □

Lemma 88 (Toric quotient descent criterion)**.** *Let $\mathbf{m}_\tau$ be a corner-active frame. Then, on the smooth rank-active locus associated with $\mathbf{m}_\tau$, the toric metrized line bundle $\overline{H}(\overline{E})_v$ is the pullback, up to a pluriharmonic local factor, of the semipositive BPS toric metrized line bundle on the $\mathbf{m}_\tau$-quotient.*

Proof. The characters $m_{\tau,1}, \dots, m_{\tau,r}$ define a quotient torus, and the coordinates

$$z_\ell = e^{\langle m_{\tau,\ell}, u_v\rangle}\psi_{j,m_{\tau,\ell}}$$

are exactly the analytic pullbacks of its standard characters in the local trivialization of Lemma 80. The term $\varphi_\tau(\mathrm{val}(z_1, \dots, z_r))$ is therefore the pullback of the quotient BPS toric

weight by Proposition 9.1. The remaining affine term $\ell_\tau(u) + c_\tau$ is the logarithm of the norm of a monomial times a constant on the torus chart. Hence it is pluriharmonic on the smooth torus-bundle locus and has zero dd^c. Thus the two metrics have the same curvature currents in the rank-active directions, which is the claimed descent up to a pluriharmonic factor. □

Proposition 9.2 (Corner-active BPS Chern-current compatibility)**.** *Let $\mathbf{m}_\tau$ be a corner-active frame. Then the local map $q_{j,\mathbf{m}_\tau}$ identifies the toric part of*

$$c_1(\overline{H}(\overline{E})_v|_{\overline{X}})^r$$

on the rank-active locus of X with the pullback of the semipositive toric Chern measure on the corresponding r-dimensional BPS quotient.

Proof. Work on a smooth rank-active chart outside the analytic exceptional set in Lemma 82. There the functions $\psi_{j,m_{\tau,1}}, \dots, \psi_{j,m_{\tau,r}}$ give holomorphic toric coordinates in the active directions, and $q_{j,\mathbf{m}_\tau}$ is a local holomorphic submersion onto its image. By Lemma 88, the local weight of $\overline{H}(\overline{E})_v$ differs from the pullback quotient weight by a pluriharmonic function. Hence the Bedford–Taylor product is local for locally bounded plurisubharmonic weights and the pluriharmonic discrepancy has zero dd^c. Therefore

$$c_1(\overline{H}(\overline{E})_v|_{\overline{X}})^r = q_{j,\mathbf{m}_\tau}^* c_1(\overline{L}_{\mathbf{m}_\tau,v})^r$$

on the rank-active locus, where $\overline{L}_{\mathbf{m}_\tau}$ denotes the quotient toric metrized line bundle with BPS weight φ_τ. The sign and shift in the tropical coordinate are exactly those fixed in Lemma 80, because

$$z_\ell = e^{\langle m_{\tau,\ell}, u_v \rangle} \psi_{j,m_{\tau,\ell}}.$$

This proves the claimed compatibility. □

Proposition 9.3 (BPS atomic tropical pushforward on corner-active quotients)**.** *For every corner-active local toric quotient produced by Definition 87 and Lemma 82, the toric part of the measure*

$$c_1(\overline{H}(\overline{E})_v|_{\overline{X}})^r$$

has finite atomic tropical pushforward

$$\mathrm{val}_*\big(c_1(\overline{H}(\overline{E})_v|_{\overline{X}})^r\big) = \sum_\nu a_\nu \, \delta_{u_\nu}$$

on the rank-active toric directions, with $a_\nu > 0$. The corresponding archimedean measure is invariant under the compact phase torus.

Proof. By Proposition 9.2, the toric part of the current on a rank-active chart is the pullback of a semipositive toric Chern measure on an r-dimensional toric quotient. BPS [13, Theorem 4.8.11] identifies the tropical pushforward of the top toric Chern measure with $r!$ times the real Monge–Ampere measure of the corresponding concave metric function, and characterizes the archimedean measure by this pushforward together with toric invariance. In the mixed case, [13, Corollary 4.8.12] gives the same statement for mixed Chern measures, by multilinearity.

The canonical toric metric function is piecewise affine after the chosen regular subdivision. For a piecewise affine concave function, the real Monge–Ampere measure is supported on the finite polyhedral cell complex and has positive dual-cell weights by [13, Proposition 2.7.4]. The mixed operator is obtained by polarization and is symmetric and multilinear by [13, Definition 2.7.12 and Proposition 2.7.13]. Hence the tropical pushforward is a finite positive atomic measure. The compact phase invariance is the archimedean toric invariance in BPS Theorem 4.8.11, pulled back through the local quotient coordinates supplied by Proposition 9.1. □

Lemma 89 (Monomial functoriality of compact Haar fibers)**.** *Let*

$$\varphi : (\mathbb{C}^\times)^r \longrightarrow (\mathbb{C}^\times)^s$$

be a homomorphism given by monomials, and let

$$A : \mathbb{R}^r \longrightarrow \mathbb{R}^s$$

be the induced homomorphism on logarithmic valuation spaces. For every $u \in \mathbb{R}^r$, the pushforward of the normalized Haar probability measure on the compact fiber $\mathrm{val}^{-1}(u)$ is the normalized Haar probability measure on the compact subtorus

$$\varphi(\mathrm{val}^{-1}(u)) \subseteq \mathrm{val}^{-1}(Au).$$

Consequently, finite positive sums of compact phase-fiber Haar measures are sent by φ to finite positive sums of compact phase-fiber Haar measures, with the atoms on valuation space pushed forward by A.

Proof. The restriction of φ to the compact torus $\mathrm{val}^{-1}(u)$ is a continuous homomorphism from a compact abelian Lie group to the compact torus $\mathrm{val}^{-1}(Au)$, up to translation by the fixed moduli u. The pushforward of normalized Haar measure under a continuous homomorphism of compact groups is normalized Haar measure on the image compact subgroup. The final assertion follows by linearity and from the identity

$$\mathrm{val} \circ \varphi = A \circ \mathrm{val}\,.$$

□

Proposition 9.4 (Local corner currents from simple mixed corners)**.** *Let τ be a codimension-r cell of the canonical corner complex meeting the rank-active locus of X, and let*

$$\mathbf{m}_\tau = (m_{\tau,1}, \dots, m_{\tau,r})$$

be the frame supplied by Proposition 84. Then, on each local trivialization chart U_j and away from the lower-dimensional analytic exceptional set, the Bedford–Taylor contribution of τ to

$$c_1(\overline{H}(\overline{E})_v|_{\overline{X}})^r$$

is a positive mixed toric intersection weight $a_\tau > 0$ times

$$\bigwedge_{\ell=1}^{r} dd^c \left|\log|\psi_{j,m_{\tau,\ell}}| + \langle m_{\tau,\ell}, u_v\rangle\right|.$$

Proof. Apply Lemma 83 to each one-wall factor in the simple normal form. The Bedford–Taylor product is local on the complement of pluripolar sets and is multilinear for locally bounded plurisubharmonic potentials. Therefore the mixed product attached to a codimension-r cell τ is a positive mixed toric intersection weight times

$$\bigwedge_{\ell=1}^{r} dd^c \left|\log|\psi_{j,m_{\tau,\ell}}| + \langle m_{\tau,\ell}, u_v\rangle\right|.$$

The rank-active condition and the analytic exceptional set are exactly those handled in Lemma 82. Monomial changes of toric charts preserve the wall characters and multiply phase coordinates by compact-torus units; hence the local descriptions agree on overlaps. □

9.5. **Geometric role of the three general-fan hypotheses.** The three hypotheses used below are not inserted merely to make a formal argument go through. They separate three genuinely different operations in the passage from a tropical corner measure to the archimedean measure used in the difference-morphism argument: separation of the corner variables, integral normalization of the phase map, and exclusion of mass on the compactification boundary. They are natural sufficient hypotheses for the present proof. We do not claim that they are necessary or minimal; removing or replacing them is the subject of the general-fan problem left open here.

Product-corner normal form. Fix a codimension-r corner τ meeting the rank-active locus. The product-corner condition is the concrete polyhedral requirement that, locally at τ, its dual cell be a Minkowski sum of r independent lattice segments. If their directions are $m_{\tau,1},\dots,m_{\tau,r}$, this is exactly what allows the local canonical potential, after subtraction of one affine branch, to be written as a positive sum of the independent hinges

$$\max\{0,\langle m_{\tau,\ell},u-u_v\rangle\},\qquad 1\le\ell\le r.$$

Thus the polyhedral hypothesis supplies the analytic separability used in the proof: the Bedford–Taylor mixed product becomes the wedge of the r one-wall currents, and Fubini gives the corresponding Haar-slice expression. Without the Minkowski-sum description, the mixed term need not split into these one-dimensional factors, so the one-wall calculation by itself does not prove the slice formula. This identifies the precise missing implication; it does not assert that every non-product corner is a counterexample.

Primitive frame in the saturated active lattice. For the directions above set

$$L_\tau=\sum_{\ell=1}^{r}\mathbb{Z}m_{\tau,\ell},\qquad L_\tau^{\mathrm{sat}}=(L_\tau\otimes_{\mathbb{Z}}\mathbb{Q})\cap M_H.$$

The hypothesis is $L_\tau=L_\tau^{\mathrm{sat}}$, not merely that each $m_{\tau,\ell}$ is primitive. It says that the monomial phase map defined by this frame has no finite residual kernel. If saturation fails, its degree is the index $[L_\tau^{\mathrm{sat}}:L_\tau]$; normalized Haar probability still pushes forward to Haar probability, whereas the pullback of the normalized top Haar form acquires precisely this index. The saturated-frame hypothesis therefore prevents a lattice factor from being counted twice when the Haar form is paired with the mixed-intersection coefficient. A more general statement would retain the displayed index explicitly; that indexed formula is not asserted here.

No additional boundary mass. Let $\nu:\overline{X}^{\nu}\to\overline{X}$ be the normalization and let μ_X denote the full top-degree restricted Bedford–Taylor measure, including the smooth factors pulled back from the abelian quotient. The exact hypothesis is

$$\mu_X\big(\nu^{-1}(\overline{X}\setminus X)\big)=0.$$

Only under this equality does the current formula proved on the principal semiabelian orbit determine the global restricted measure used by the difference morphism. Otherwise one must add the boundary functional

$$B_\partial(f)=\int_{\nu^{-1}(\overline{X}\setminus X)}f\,d\mu_X$$

and control its pullback and pushforward under every product and difference map. For a continuous semipositive metric on the compactification, the expected vanishing mechanism is analytic: its pullback to $\overline{X}^{\nu}$ has locally bounded psh weights, and the associated Bedford–Taylor measure does not charge the proper analytic, hence pluripolar, boundary [1, Subsection 2.3]. Proper intersection with toric boundary strata is useful for the algebraic stable-intersection calculation but does not by itself imply this measure-zero statement. This distinction is why boundary no-mass remains an explicit input in the general-fan formulation.

In Kühne's standard $(\mathbb{P}^1)^t$-compactification the coordinate walls provide the product decomposition, the coordinate characters form a primitive saturated basis, and the explicit one-variable current formula has no boundary contribution. This explains why the three issues are invisible in that model. For a general fan they must be verified separately (or replaced by a theorem with lattice-index and boundary correction terms).

Proposition 90 (Conditional toric corner-current decomposition)**.** *Let E be the ample toric divisor on X_Σ, endowed at the archimedean place v with the canonical toric metric, and let $\overline{E}$ be its quasi-canonical translate with center u_v. Let $X\subseteq H_{\overline{K}}$ be irreducible, and put*

$$d=\dim X,\qquad d_0=\dim\rho(X),\qquad r=d-d_0.$$

Assume that every codimension-r corner meeting the rank-active locus admits, after subdivision, the product-corner normal form of Proposition 84, with the character frame primitive in its saturated lattice, and assume that the compactification boundary contributes no additional mass to the restricted Bedford–Taylor product. After replacing Σ by a finite regular toric subdivision and pulling back E, the Bedford–Taylor current

$$c_1(\overline{H}(\overline{E})_v|_{\overline{X}})^r$$

restricted to the principal semiabelian orbit has the following finite polyhedral description. There is a finite set $\mathscr{C}_X(E)$ of codimension-r cells in the corner complex of the canonical toric potential, and for each $\tau \in \mathscr{C}_X(E)$ there are

(i) *a tuple of characters*

$$\mathbf{m}_\tau = (m_{\tau,1}, \dots, m_{\tau,r}) \in M_H^r$$

which is rank-active for X;

(ii) *a positive mixed toric intersection weight $a_\tau > 0$;*

(iii) *a closed complex-analytic exceptional set*

$$E_\tau \subset X^{an}_{\mathbb{C}_v}, \qquad \dim E_\tau < d;$$

such that, on each local trivialization chart U_j and away from the union of the E_τ, the current is the finite positive sum of the local corner currents

$$a_\tau \bigwedge_{\ell=1}^{r} dd^c \left|\log|\psi_{j,m_{\tau,\ell}}| + \langle m_{\tau,\ell}, u_v\rangle\right|.$$

This identity is meant after testing against compactly supported continuous functions and wedging with smooth forms pulled back from the abelian quotient. It is invariant under monomial changes of toric charts, and it descends from the subdivision back to X_Σ by the projection formula.

Proof. This is the global form of the preceding local calculation. After the finite regular subdivision, the corner complex of the canonical toric potential has only finitely many cells meeting the tropical image of X. For each codimension-r cell τ, Proposition 84 gives a simple local normal form away from the lower-dimensional analytic exceptional set. Proposition 9.4 then identifies the Bedford–Taylor contribution of τ with the displayed wedge product of one-wall currents and the positive mixed toric intersection weight a_τ.

The sum over τ is finite. On overlaps of toric charts the coordinates $\psi_{j,m_{\tau,\ell}}$ differ by monomial changes and compact-torus units, so Lemma 89 and the one-wall current formula show that the local current descriptions define the same distribution after testing against compactly supported functions. The exceptional sets are proper complex-analytic subsets, hence pluripolar, and the locally bounded Bedford–Taylor products do not charge them. Finally the subdivision map is proper, toric and birational, and is an isomorphism over the principal orbit; the projection formula pushes the current identity down to the original compactification X_Σ. This proves the stated decomposition. □

Proposition 9.5 (Conditional general-fan curvature–Haar slice formula)**.** *Keep the notation of Lemma 82. Assume the product-corner, saturated-lattice, and boundary no-mass hypotheses of Proposition 90. There is a finite set $\mathcal{A}_X$ of corner-active, hence rank-active, character frames*

$$\mathbf{m} = (m_1, \dots, m_r)$$

and positive constants $a_{\mathbf{m}}$, depending only on the ample toric divisor E and on X, such that the following holds. For each $\mathbf{m} \in \mathcal{A}_X$, the pullbacks of the normalized Haar form on $(S^1)^r$ through the local phase maps of Lemma 82 glue to a smooth positive r-form $\omega_{\mathbf{m},u_v}$ on

$$X_{\mathbf{m},u_v} \setminus E_{\mathbf{m}}.$$

For every compactly supported continuous function f on $X^{an}_{\mathbb{C}_v}$,

$$\int_{X^{an}_{\mathbb{C}_v}} f\, c_1(\overline{H}(\overline{E})_v|_{\overline{X}})^r \wedge c_1(\rho^*\overline{N}_{H,v}|_{\overline{X}})^{d_0} = \sum_{\mathbf{m}\in\mathcal{A}_X} a_{\mathbf{m}} \int_{X_{\mathbf{m},u_v}\setminus E_{\mathbf{m}}} f\,\omega_{\mathbf{m},u_v} \wedge c_1(\rho^*\overline{N}_{H,v})^{d_0}.$$

Equivalently, the Kühne Proposition 4.1 measure is a finite positive sum of Haar-slice measures on the real-analytic slices supplied by Lemma 82.

Proof. First use Proposition 9.3. On each corner-active local toric quotient, BPS Theorem 4.8.11 and Corollary 4.8.12 identify the tropical pushforward of the relevant toric Chern measure with the mixed real Monge–Ampere measure. Since the canonical toric potential is piecewise affine after the chosen regular subdivision, Proposition 2.7.4 and the mixed Monge–Ampere polarization give a finite atomic measure on the tropical side. The archimedean toric Chern measure is toric, hence invariant under the compact real torus in the phase directions. Lemma 86 therefore lifts every tropical atom to the normalized Haar measure on the corresponding compact phase fiber.

In the coordinates of Lemma 80, these fibers are exactly the real-analytic slices

$$|z_\ell| = 1, \qquad z_\ell = e^{\langle m_{\nu,\ell}, u_v\rangle}\psi_{j,m_{\nu,\ell}},$$

or equivalently

$$|\psi_{j,m_{\nu,\ell}}| = \exp(-\langle m_{\nu,\ell}, u_v\rangle).$$

Lemma 82 supplies the underlying rank-active charts, and Definition 87 selects the relevant corner-active frames after deleting the lower-dimensional analytic exceptional sets. Grouping the finitely many tropical atoms with the same character frame gives the finite family $\mathcal{A}_X$, the weights $a_{\mathbf{m}}$, and the glued Haar forms $\omega_{\mathbf{m},u_v}$. Wedging with the smooth form pulled back from the abelian quotient gives the displayed formula in this case.

For comparison, one can also unpack BPS by local Bedford–Taylor calculus using the corner-current decomposition of Proposition 90. By Proposition 90, after a finite toric subdivision and after deleting the pluripolar exceptional sets, the relevant toric curvature product is a finite positive sum of wedge products of one-dimensional corner currents. The subdivision is harmless on the principal semiabelian orbit, and the projection formula pushes the resulting measure back to the original compactification.

Fix one corner cell τ and one local trivialization U_j. Put

$$z_\ell = e^{\langle m_{\tau,\ell}, u_v\rangle}\psi_{j,m_{\tau,\ell}}, \qquad 1 \le \ell \le r.$$

On the rank-active locus, Lemma 82 says that $(\rho^{an}, z_1, \dots, z_r)$ is a local biholomorphic coordinate system on the smooth part of X. In these coordinates, the local summand supplied by Proposition 90 is

$$a_\tau \bigwedge_{\ell=1}^{r} dd^c\big|\log|z_\ell|\big|.$$

Kühne's elementary computation [1, Appendix B] gives, for one coordinate,

$$\int_{\mathbb{C}^\times} h(z)\, dd^c\big|\log|z|\big| = \int_0^{2\pi} h(e^{i\theta})\,\frac{d\theta}{\pi},$$

with Kühne's normalization of dd^c. Iterating this identity by Fubini, and wedging with the smooth base form $c_1(\rho^*\overline{N}_{H,v})^{d_0}$, converts the local corner-current summand into the Haar measure on the real-analytic slice

$$|z_\ell| = 1 \quad (1 \le \ell \le r),$$

that is,

$$|\psi_{j,m_{\tau,\ell}}| = \exp(-\langle m_{\tau,\ell}, u_v\rangle).$$

These are exactly the slices $X_{\mathbf{m}_\tau,u_v}$ of Lemma 82.

The monomial transition functions between local trivializations are compact torus units on the slices, so the local Haar forms glue. Grouping cells with the same character frame gives the finite set $\mathcal{A}_X$ and the constants $a_{\mathbf{m}}$. Since the exceptional sets are complex analytic of dimension $< d$, the Bedford–Taylor measures under consideration do not charge them. Summing over the finitely many cells proves the formula. □

Remark 91. *Proposition 9.5 is a conditional general-fan replacement for the computational heart of Kühne's Lemma 5.2, with the standard toric corner-current decomposition supplied by Proposition 90. In the standard $(\mathbb{P}^1)^t$ compactification, the corner cells are the coordinate walls and the formula follows from Kühne's identity for $dd^c\big|\log|z|\big|$, not from $dd^c\log|z|$. For an arbitrary ample toric divisor E, the product-corner reduction, saturated-lattice factors, and absence of compactification boundary mass remain additional inputs; they do not follow from the local one-wall calculation alone. Accordingly no unconditional arbitrary-fan Bogomolov statement is deduced here.*

Proposition 9.6 (Finite Haar-slice functoriality for monomial semiabelian maps)**.** *Let $\beta : H_1 \to H_2$ be a semiabelian homomorphism such that, on the toric parts in local trivialization coordinates, β is given by monomials. Assume that the source measure on an irreducible $X \subset H_1$ has the finite Haar-slice expression of Proposition 9.5. Then, on every smooth rank-active chart on which $\beta|_X$ has constant rank, the pushforward of each source Haar-slice summand is a finite positive Haar-slice measure on the image. In particular the class of finite Haar-slice measures is stable under products, torus inversion, and the Kühne difference morphisms*

$$(x_1,\dots,x_m) \longmapsto (x_1x_2^{-1},\dots,x_{m-1}x_m^{-1}).$$

If, in addition, the limiting measures satisfy the pushforward identity, then the finite Haar-slice descriptions are compatible with β_*. In the application below this pushforward identity is obtained directly from equidistribution of Galois orbit measures on the source and on the target, rather than from a separate local Chern-measure computation.

Proof. On a local trivialization chart, the toric part of β has the form

$$(z_1,\dots,z_r) \longmapsto \left(\prod_i z_i^{a_{1i}},\dots,\prod_i z_i^{a_{si}}\right)$$

for an integral matrix $A = (a_{ji})$. The induced map on logarithmic valuation spaces is the linear map $u \mapsto Au$. Hence Lemma 89 sends the normalized Haar measure on each compact phase fiber over u to the normalized Haar measure on the compact image phase fiber over Au, possibly inside a lower dimensional compact subtorus if the rank drops. After partitioning the rank-active locus into finitely many constant-rank real-analytic pieces, these lower-rank images are again among the allowed finite real-analytic Haar-slice pieces on the image.

Products correspond to block diagonal monomial maps, inversion corresponds to the matrix $-I$, and the Kühne difference morphism corresponds on the toric coordinates of H^m to the block matrix

$$(z^{(1)},\dots,z^{(m)}) \longmapsto \big(z^{(1)}(z^{(2)})^{-1},\dots,z^{(m-1)}(z^{(m)})^{-1}\big).$$

Thus all three operations preserve the finite Haar-slice class. The final assertion is the functorial statement used below after the relevant pushforward identity has been supplied by equidistribution. □

Lemma 92 (Toric quasi-canonical normalization under homomorphisms)**.** *Let $\beta : T_1 \to T_2$ be a homomorphism of algebraic tori, and let*

$$A_\beta : N_{1,\mathbb{R}} \longrightarrow N_{2,\mathbb{R}}$$

be the induced linear map on cocharacter spaces. Let $\overline{E}_2$ be a quasi-canonical toric metrized divisor on T_2, with archimedean local weight

$$\Psi_{\overline{E}_2,v}(w) = \Psi_{E_2}(w - u_{2,v}) - \gamma_{2,v}.$$

Equip $\beta^*\overline{E}_2$ with the pulled-back metric. Then its toric local weight on T_1 is

$$u \longmapsto \Psi_{E_2}(A_\beta u - u_{2,v}) - \gamma_{2,v}.$$

In particular, if $u_{2,v} = A_\beta u_{1,v}$, this is the quasi-canonical metric on the pulled-back toric divisor with center $u_{1,v}$. Products, torus inversion, quotients of tori, and the toric part of the Kühne difference morphisms satisfy this condition with the centers transported by the same integral linear maps as in Lemma 79.

Proof. On characters, the pullback by β is the lattice homomorphism dual to A_β. Therefore the tropicalization satisfies

$$\mathrm{val}_v(\beta(t)) = A_\beta \,\mathrm{val}_v(t).$$

Pulling back the toric metric simply composes its local weight with this linear map, giving the displayed formula. If $u_{2,v} = A_\beta u_{1,v}$, then

$$\Psi_{E_2}(A_\beta u - u_{2,v}) = \Psi_{E_2}(A_\beta(u - u_{1,v})),$$

which is the support function of the pulled-back toric divisor evaluated at $u - u_{1,v}$. This is precisely the quasi-canonical normal form on the source. The product case corresponds to block diagonal A_β, inversion to $-I$, quotient maps to the induced quotient linear map, and the Kühne difference morphism to the matrix

$$(u_1, \dots, u_m) \longmapsto (u_1 - u_2, \dots, u_{m-1} - u_m).$$

For the common center $(u_v, \dots, u_v)$, this last matrix gives $(0, \dots, 0)$, as stated in Lemma 79. □

Lemma 93 (Picard-zero character pushouts under operations)**.** *Let G be a semiabelian variety with split torus T_G, abelian quotient $\rho_G : G \to A_G$, and character lattice $M_G = X^*(T_G)$. For $m \in M_G$, let*

$$\mathcal{H}_{G,m} \in \mathrm{Pic}^0(A_G)$$

denote the rigidified line bundle whose complement of the zero section is the $\mathbb{G}_m$-torsor obtained by pushing out

$$0 \longrightarrow T_G \longrightarrow G \longrightarrow A_G \longrightarrow 0$$

through $\chi^m : T_G \to \mathbb{G}_m$. Then:

(a) *The assignment $m \mapsto \mathcal{H}_{G,m}$ is additive:*

$$\mathcal{H}_{G,m_1+m_2} \simeq \mathcal{H}_{G,m_1} \otimes \mathcal{H}_{G,m_2}, \qquad \mathcal{H}_{G,-m} \simeq \mathcal{H}_{G,m}^\vee, \qquad \mathcal{H}_{G,0} \simeq \mathcal{O}_{A_G}.$$

(b) *If $\beta : G_1 \to G_2$ is a semiabelian homomorphism, with abelian part $\beta_A : A_{G_1} \to A_{G_2}$ and character pullback $\beta_M : M_{G_2} \to M_{G_1}$, then there is a canonical rigidified isomorphism*

$$\mathcal{H}_{G_1,\beta_M(m)} \simeq \beta_A^* \mathcal{H}_{G_2,m}.$$

(c) *For products,*

$$\mathcal{H}_{G_1\times G_2,(m_1,m_2)} \simeq p_1^*\mathcal{H}_{G_1,m_1} \otimes p_2^*\mathcal{H}_{G_2,m_2}.$$

(d) *For the Kühne difference morphism*

$$\alpha_r : G^r \longrightarrow G^{r-1}, \qquad (x_1, \dots, x_r) \longmapsto (x_1x_2^{-1}, \dots, x_{r-1}x_r^{-1}),$$

and a character tuple $\mathbf{m} = (m_1, \dots, m_{r-1}) \in M_G^{r-1}$, the pullback character on T_G^r is

$$\alpha_{r,M}(\mathbf{m}) = (m_1,\ m_2 - m_1,\ \dots,\ m_{r-1} - m_{r-2},\ -m_{r-1}),$$

and therefore

$$\mathcal{H}_{G^r,\alpha_{r,M}(\mathbf{m})} \simeq p_1^*\mathcal{H}_{G,m_1} \otimes \bigotimes_{i=2}^{r-1} p_i^*(\mathcal{H}_{G,m_i} \otimes \mathcal{H}_{G,m_{i-1}}^\vee) \otimes p_r^*\mathcal{H}_{G,m_{r-1}}^\vee.$$

Equivalently, this is the pullback by the abelian difference morphism $\alpha_{r,A} : A_G^r \to A_G^{r-1}$ of $\mathcal{H}_{G^{r-1},\mathbf{m}}$.

All these isomorphisms are compatible with the rigidifications, with the canonical Picard-zero metrics, and with the Raynaud/Kühne theta representatives after replacing sections by their tensor products or duals.

Proof. Pushing out a semiabelian extension by a character produces a $\mathbb{G}_m$-torsor over the abelian quotient. Equivalently, it produces a rigidified translation-invariant line bundle, hence an element of Pic^0. This is the construction recalled by the Raynaud–Bosch–Lütkebohmert pushout formalism, in the form used by FRSS, and in the line-bundle triple correspondence [15, Theorem 3.10].

The contracted product of $\mathbb{G}_m$-torsors corresponds to tensor product of line bundles. Applying this to the characters $m_1 + m_2$, $-m$, and 0 proves (a). If $\beta : G_1 \to G_2$ is a semiabelian homomorphism, the pushout of G_1 by $\beta_M(m)$ is the pullback, along β_A, of the pushout of G_2 by m; this is the universal property of pushouts of extensions, and gives (b). The product formula (c) is the same statement applied to the two projections.

For (d), one only computes characters:

$$\prod_{i=1}^{r-1} \chi^{m_i}(x_i x_{i+1}^{-1}) = \chi^{m_1}(x_1) \prod_{i=2}^{r-1} \chi^{m_i - m_{i-1}}(x_i) \chi^{-m_{r-1}}(x_r).$$

Combining this character identity with (a) and (c) gives the displayed tensor product. The final description as a pullback by $\alpha_{r,A}$ follows from (b).

The compatibility with theta representatives is the tensor-product compatibility in the Raynaud triple formalism and in the descent criterion for theta functions [15, Proposition 3.19 and Definition 3.20]. For the metrics, tensor products, duals, and pullbacks of canonical Picard-zero metrics again satisfy the defining rigidification and multiplication-isometry properties of the canonical Picard-zero metric. This functoriality is recorded in Lemma 94. □

Lemma 94 (Flat functoriality of canonical Picard-zero factors)**.** *Let $\phi : A_1 \to A_2$ be a homomorphism of abelian varieties, and let $\overline{H}$ be a rigidified Picard-zero line bundle on A_2 with its canonical adelic metric. Then $\phi^*\overline{H}$ is the canonical metrization of the rigidified Picard-zero line bundle ϕ^*H on A_1. At an archimedean place its Chern form is zero. Consequently any finite tensor product of such pulled-back canonical Picard-zero factors and their duals is flat at archimedean places and contributes no term to the Bedford–Taylor Chern-current products.*

Proof. The pullback ϕ^*H is again rigidified and algebraically trivial. The canonical Picard-zero metric is characterized by the rigidification at the origin and by the fact that the multiplication isomorphisms $[n]^*H \simeq H^{\otimes n}$ are isometries. Pulling these isometries back by ϕ gives the same defining property for $\phi^*\overline{H}$, so by uniqueness it is the canonical metric on ϕ^*H. At archimedean places, Proposition 4.24 identifies this metric with the flat hermitian metric attached to the rigidified Picard-zero bundle. Hence its Chern form is zero. Tensor products and duals preserve flatness, proving the final assertion. □

Theta-factor bookkeeping for operations. For products, torus inversion, quotient maps by connected stabilizers, and Kühne difference morphisms, the toric part of the quasi-canonical metric is normalized as in Lemma 92. The only role of the semiabelian theta factors here is to fix the sign and dual convention in the local coordinates. A character $m \in M_G$ gives the character-pushout Picard-zero bundle $\mathcal{H}_{G,m}$; replacing m by $-m$ replaces this bundle by its dual. Lemma 93 records the behavior of these Picard-zero factors under products, quotients, and Kühne difference morphisms, and Lemma 94 records their flatness at archimedean places. This bookkeeping is useful for comparing local formulas with Kühne's notation, but the push-forward identity used in equation (6.2) below is not derived from these theta factors; it comes directly from equidistribution of the relevant source and target Galois orbit measures.

Source of the pushforward identity. The pushforward identity used in Kühne's equation (6.2) is not an additional local Chern-measure projection formula. Let $\beta : Y \to X'$ be one of the product, quotient, or difference morphisms in the operation argument, and let (y_i) be a Y-generic small sequence such that $(\beta(y_i))$ is X'-generic and small. For every i, the finite Galois orbit measures satisfy the tautological identity

$$\beta_* \delta_{y_i,v} = \delta_{\beta(y_i),v}.$$

Applying the equidistribution theorem on Y and on X', and then passing to the weak limit, gives

$$\beta_* \mu_{Y,v} = \mu_{X',v}.$$

This argument is independent of the choice of toric compactification. The general-fan work above is needed to identify the limiting measures locally as finite Haar-slice measures and to run the small-box comparison, not to prove the pushforward identity itself.

Lemma 95 (Riemannian domination for finite Haar slices)**.** *Assume that X at the archimedean place v has the finite Haar-slice description supplied by Proposition 9.5. Then there is a smooth Riemannian metric $g_{X,v}$ on the real manifold $H(\mathbb{C}_v)$ such that, for every $\mathbf{m} \in \mathcal{A}_X$, every embedded real-analytic component*

$$M \subset X_{\mathbf{m},u_v} \setminus E_{\mathbf{m}},$$

and every open subset $U \subset M$, one has

$$a_{\mathbf{m}} \int_U \omega_{\mathbf{m},u_v} \wedge c_1(\rho^* \overline{N}_{H,v})^{d_0} \le \int_U \mathrm{vol}(g_{X,v}|_M).$$

The metric can be chosen simultaneously for the finite family $\mathcal{A}_X$.

Proof. All manifolds M under consideration lie in the open semiabelian analytic manifold $H(\mathbb{C}_v)$, which is smooth. Choose any smooth Riemannian metric g_0 on $H(\mathbb{C}_v)$. On each real-analytic component M, the form

$$a_{\mathbf{m}}\, \omega_{\mathbf{m},u_v} \wedge c_1(\rho^* \overline{N}_{H,v})^{d_0}$$

is a smooth positive density of top degree. Hence it can be written as

$$h_{\mathbf{m},M}\ \mathrm{vol}(g_0|_M)$$

with $h_{\mathbf{m},M}$ a positive smooth function on M. Since the family of frames and local real-analytic components is finite after the finite trivializing cover is fixed, we may choose a smooth positive function λ on $H(\mathbb{C}_v)$ whose restriction to every such M satisfies

$$\lambda^{\dim M} \ge h_{\mathbf{m},M}.$$

This is obtained by the usual extension and partition-of-unity argument on the paracompact real manifold $H(\mathbb{C}_v)$. Then

$$g_{X,v} := \lambda^2 g_0$$

has induced volume form $\lambda^{\dim M} \mathrm{vol}(g_0|_M)$ on M, and the displayed inequality follows. This is the part of Kühne's Lemma 5.3 which only uses the finiteness of the Haar-slice description, not the special $(\mathbb{P}^1)^t$ compactification. □

Lemma 96 (Kühne small-box contradiction for finite slice measures)**.** *Let $X \subseteq H_{\overline{K}}$ be irreducible of positive dimension and assume*

$$\mathrm{Stab}_{H_{\overline{K}}}(X) = \{e\}.$$

Let m be chosen so that the Kühne difference morphism

$$\alpha_m : X^m \longrightarrow H^{m-1}$$

is generically finite of degree 1 onto its image. Put

$$Y = \alpha_m(X^m).$$

Assume that, at some archimedean place v, the Kühne limiting measures on X, X^m, and Y have the finite Haar-slice descriptions supplied by Proposition 9.5, that the measure on X^m is the m-fold product of the measure on X, and that the pushforward identity

$$(\alpha_m)_*\mu_{X^m,v} = \mu_{Y,v}$$

holds on the dense open locus where α_m is an isomorphism. Then these assumptions are incompatible.

Proof. Let $U \subseteq X^m$ be a dense open subset on which α_m is an isomorphism onto its image. By the finite Haar-slice formula for Y, the measure $\mu_{Y,v}$ is a finite sum of positive smooth densities on embedded real-analytic manifolds. By Lemma 95, after pulling these manifolds back through $\alpha_m|_U^{-1}$, their contributions are dominated by the volume induced from a smooth Riemannian metric on $H(\mathbb{C}_v)^{m-1}$.

The product description of $\mu_{X^m,v}$ gives a finite sum of product density forms on products

$$M_1 \times \cdots \times M_m,$$

where each M_i is one of the real-analytic slice components for X. Lower-dimensional real-analytic subsets have zero mass for these smooth density forms. Hence one may choose a slice component M, a point $x \in M$ outside the other components and outside the exceptional locus, and a real-analytic chart

$$\varphi : (-1,1)^q \longrightarrow M, \qquad q = d + d_0,$$

with $\varphi(0) = x$, on which the corresponding density is strictly positive. The m-fold product chart

$$\varphi^m : ((-1,1)^q)^m \longrightarrow M^m \subset X^m$$

then has product density bounded below by a positive multiple of Euclidean volume on small boxes

$$B_\epsilon = \big((-\epsilon,\epsilon)^q\big)^m.$$

On the other hand, the differential of

$$\alpha_m \circ \varphi^m$$

annihilates the diagonal tangent subspace

$$\{(w,\dots,w) : w \in \mathbb{R}^q\} \subset (\mathbb{R}^q)^m,$$

because the difference map is constant along simultaneous translation of all coordinates. Therefore the Riemannian volume of $\alpha_m \circ \varphi^m(B_\epsilon)$ on each target slice is bounded by

$$O(\epsilon^q \ \mathrm{vol}(B_\epsilon)).$$

This is exactly Kühne's small-box estimate in the proof of [1, Proposition 6.1]; it uses only the existence of the diagonal kernel and the Riemannian domination of the target slice measures.

Choose a non-negative test function supported in the product chart and positive at $(x,\dots,x)$. The product source measure of B_ϵ is bounded below by

$$c \ \mathrm{vol}(B_\epsilon)$$

for some $c > 0$, while the pushforward side is bounded above by

$$C\,\epsilon^q \ \mathrm{vol}(B_\epsilon).$$

For sufficiently small ϵ these two estimates contradict the pushforward identity. Thus the finite slice descriptions and the pushforward identity cannot coexist when the stabilizer is trivial. □

9.6. **Kühne Transport Assembly.**

Lemma 97 (Smallness under semiabelian operations)**.** *Let $H_1, \dots, H_r, H'$ be semiabelian varieties equipped with quasi-canonical toric metrics on ample T-effective divisors and canonical metrics on their abelian quotients. Let*

$$\beta : H_1 \times \cdots \times H_r \longrightarrow H'$$

be a semiabelian homomorphism. Suppose that the critical vectors of the source metrics are $\mathbf{w}_1, \dots, \mathbf{w}_r$, and equip H' with a quasi-canonical metric whose critical vector is the image $\beta_{\mathrm{trop}}(\mathbf{w}_1, \dots, \mathbf{w}_r)$. If, for every k, $(x_{k,i})_i$ is small for the corresponding metric on H_k, then

$$\big(\beta(x_{1,i}, \dots, x_{r,i})\big)_i$$

is small for the chosen metric on H'.

In particular, this applies to quotients by connected semiabelian subgroups, to products, and to the Kühne difference morphisms. For a difference morphism applied to m copies with the same critical vector $\mathbf{w}$, the target critical vector is 0.

Proof. Normalize all toric minima to 0. For each source sequence, Proposition 71, applied to its quasi-canonical metric, gives adelic KR convergence of the valuation measures to the Dirac family at its critical vector. The proof of Proposition 70 also gives convergence of the abelian Néron–Tate height to 0.

Fix i, choose one finite extension over which the whole tuple $(x_{1,i}, \dots, x_{r,i})$ is defined, and take the uniform local Galois-orbit measure $\nu_{i,v}$ of that tuple. Its k-th marginal is precisely $\nu_{x_{k,i},v}$: the restriction map from the Galois orbit of the tuple to the orbit of $x_{k,i}$ is surjective and all its fibers have the same cardinality. Thus passage to a common defining field changes neither the normalized marginal measure nor any normalized height.

The tropicalization maps are group homomorphisms by Proposition 19. Hence the valuation measure of the image point is the pushforward, under the integral linear map β_{trop}, of $\nu_{i,v}$. A joint probability measure whose marginals converge in KR topology to Dirac measures converges in KR topology to the product Dirac measure: for the product norm,

$$W_1\big(\nu_i, \delta_{(w_1,\dots,w_r)}\big) \le \sum_{k=1}^{r} \int \|u_k - w_k\| \, d\nu_i.$$

Consequently the image valuation measures converge in KR topology to the Dirac family at $\beta_{\mathrm{trop}}(\mathbf{w}_1, \dots, \mathbf{w}_r)$; indeed, pushforward by the fixed integral linear map is Lipschitz for W_1, with constant $\|\beta_{\mathrm{trop}}\|$, uniformly at every place.

The local support functions for an ample quasi-canonical toric metric are continuous, piecewise linear, and of at most linear growth. Hence KR convergence implies convergence of their integrals. By the height identity in Lemma 67, the toric height of the image points tends to the toric minimum. On abelian quotients let $\beta_A : \prod_k A_{H_k} \to A_{H'}$ be the induced homomorphism. If N' is the target symmetric ample bundle and N_k are the source symmetric ample bundles, then $\beta_A^* N'$ is symmetric nef. In the finite-dimensional Néron–Severi space there is $C > 0$ such that $C \bigotimes_k \mathrm{pr}_k^* N_k - \beta_A^* N'$ is ample. The associated quadratic heights therefore satisfy

$$\widehat{h}_{N'}(\beta_A(a_1, \dots, a_r)) \le C \sum_{k=1}^{r} \widehat{h}_{N_k}(a_k).$$

The right-hand side tends to 0. The height decomposition now shows that the image sequence is small for the target quasi-canonical semiabelian metric.

For a quotient, take $r = 1$. For products use the exterior product metric. For the difference map, the induced tropical map sends $(\mathbf{w}, \dots, \mathbf{w})$ to 0, proving the final assertion. □

Lemma 98 (Subvariety-restricted variation package)**.** *Let $X \subseteq H_{\overline{K}}$ be irreducible, put $d = \dim X$ and $d_0 = \dim \rho(X)$, and let $\overline{X}$ be its closure in the chosen toric compactification. Form*

the auxiliary n-division tower $\varphi_n : H_n \to H$ with the same fan, and let Y be an irreducible component of $\varphi_n^{-1}(\overline{X}_{K_n})$. Write $\delta(Y) = \deg(\varphi_n|_Y : Y \to \overline{X}_{K_n})$, and restrict every auxiliary metrized line bundle to Y. Denote by $\overline{M}_n$ the corresponding auxiliary reference bundle on the compactification of H_n.

Then the degree asymptotic, measure compression, first variation, horizontal semipositivity correction, and arithmetic-volume estimate required in Kühne's Proposition 4.1 hold on Y, with constants depending on X and the fixed test metric but not on n, the component Y, or the field K_n. In particular, for a $\mathrm{Gal}(\mathbb{C}_v/K_v)$-invariant integrable continuous test function f on $\overline{X}^{an}_{\mathbb{C}_v}$, putting $f_{n,v'} = f \circ (\varphi_n|_Y)^{an}$ and

$$\mu_{Y,n,v'} = \frac{c_1(\overline{M}_{n,v'}|_Y)^d}{(M_n|_Y)^d},$$

one has

$$(\varphi_n|_Y)_*\mu_{Y,n,v'} \Longrightarrow \frac{c_1(\overline{H}(\overline{E})_v|_{\overline{X}})^{d-d_0} \wedge c_1(\rho^*\overline{N}_{H,v}|_{\overline{X}})^{d_0}}{\overline{H}(E)^{d-d_0} \cdot (\rho^* N_H)^{d_0} \cdot X}.$$

Moreover, for $|\lambda| \le n^{-1}$, the height first-variation remainder on Y is $O_{X,f}(|\lambda|^2 n)$; the horizontal correction restricted to Y is $O_X(n^{-2})$; and the quadratic arithmetic-volume defect for the restricted bundle is bounded with the same normalized field-extension convention. All heights, intersections, and volumes in these assertions are those of Y, not of the ambient compactification.

Remark 99. *The notation in Kühne's Lemmas 4.2–4.4 and the notation used here are not identical. Kühne's $\overline{Y}$ is our component Y, his d' is our d_0, his relative bundle $M|_{\overline{X}}$ is our $\overline{H}(E)|_{\overline{X}}$, and his $\overline{L}_n|_{\overline{Y}}$ is the restricted auxiliary bundle denoted here by $\overline{M}_n|_Y$. Thus his normalized measure*

$$\frac{c_1(\overline{L}_{n,v'}|_{\overline{Y}})^d}{(L_n|_{\overline{Y}})^d}$$

is exactly our $\mu_{Y,n,v'}$; this is a change of notation, not an additional normalization. Kühne's Lemma 4.3 requires continuity and $\mathrm{Gal}(\mathbb{C}_v/K_v)$-invariance, whereas Lemma 4.4 adds the condition that $\overline{O}_{\overline{X}}(f)$ be integrable. Our phrase "integrable continuous test function" combines these two requirements because both the measure limit and the first-variation estimate are used here. With this translation of notation, the invariance makes $f_{n,v'} = f \circ (\varphi_n|_Y)^{an}$ independent of the chosen identification $\mathbb{C}_{v'} \simeq \mathbb{C}_v$. It does not restrict the eventual weak-convergence statement: both the orbit measures and the limiting measure are Galois invariant, so an arbitrary continuous test function may be replaced by its Haar average under the compact local Galois group.

Proof. The argument is made on the cycle $[Y]$. The projection formula gives

$$(\varphi_n|_Y)_*[Y] = \delta(Y)[\overline{X}_{K_n}].$$

Consequently every degree and intersection in $(M_n|_Y)^d$ is an intersection on $\overline{X}$, multiplied by $\delta(Y)$ and the explicit power of n coming from $\varphi_n^*\overline{H}(E) = n\overline{H}_n(E)$. Terms containing more than d_0 factors from the abelian quotient vanish after restriction, and the exact restricted degree identity is

$$(M_n|_Y)^d = \delta(Y)\sum_{j=0}^{d_0}\binom{d}{j} n^{-(d-j)}\big(\overline{H}(E)|_{\overline{X}}\big)^{d-j} \cdot \big(\rho^* N_H|_{\overline{X}}\big)^j.$$

Its leading order is $\delta(Y)n^{-d+d_0}$, with positive coefficient

$$\binom{d}{d_0} c_1(\overline{H}(E)|_{\overline{X}})^{d-d_0} \cdot c_1(\rho^* N_H|_{\overline{X}})^{d_0} > 0.$$

The binomial factor occurs in both the local mixed-intersection numerator and the restricted top degree, so it cancels in the normalized measure ratio. Dividing the local intersection formula by the restricted top degree gives the displayed weak limit, exactly as in [1, Lemmas 4.2–4.3].

For first variation, expand the normalized height $h_{\overline{M}_n+\lambda\varphi_n^* O_{\overline{X}}(f)}(Y)$. The linear term is the integral against $\mu_{Y,n,v'}$. Every term of order at least two still contains $[Y]$; pushing it to $[\overline{X}]$ gives [1, equations (4.5)–(4.6)], hence $O_{X,f}(|\lambda|^2 n)$. Thus the projection-formula mechanism is applied to $[Y]$, rather than restricting an ambient numerical inequality. More explicitly, every numerator term carries the same factor $\delta(Y)$ as the restricted top degree; it cancels before the n-exponent is estimated.

The pointwise canonical lower bound holds on the auxiliary semiabelian variety and therefore on the closed points of Y. If X contains an X-generic small sequence, compatible lifts give a Y-generic small sequence; Zhang's inequalities then give the restricted height bound and the $O_X(n^{-2})$ correction, as in [1, Lemma 4.5]. Applying the quadratic arithmetic-volume package to the horizontally semipositive restricted bundle on Y, with the preceding restricted higher-order intersection estimates, gives the asserted defect by [1, Lemma 4.6]. The raw factor $[K_n : K]$ cancels under Lemma 40; this cancellation is separate from, and occurs after, cancellation of $\delta(Y)$. Every assertion is therefore an assertion on Y. □

The next proposition is the assembly point for the transport of Kühne's archimedean local-trivialization argument. It replaces the standard $(\mathbb{P}^1)^t$-compactification by the present toric compactification. It is useful to record the internal ingredients before giving the proof.

quasi-canonical shift and fan coordinates	$\rightsquigarrow$	local compact-torus Haar slices,
BPS quotient current and toric invariance	$\rightsquigarrow$	general-fan curvature–Haar formula,
products, quotients, and difference morphisms	$\rightsquigarrow$	orbit-measure functoriality,
character-pushout theta factors	$\rightsquigarrow$	Picard-zero sign/dual bookkeeping,
finite Haar slices plus equidistribution	$\rightsquigarrow$	Kühne small-box contradiction.

The proof below follows the published JEMS numbering. The local trivialization is [1, Section 3.5, Lemma 3.3]; subvariety equidistribution is [1, Proposition 4.1 and Lemmas 4.2–4.6]; the rank, Haar-current and domination inputs are [1, Lemmas 5.1–5.3]; difference geometry is [1, Lemma 6.2]; and the final contradiction is [1, Proposition 6.1, especially equation (6.2)].

Proposition 100 (Kühne local-trivialization transport package)**.** *Let H, $\rho : H \to A_H$, and*

$$\overline{M} = \overline{H}(\overline{E}) \otimes \rho^* \overline{N}_H$$

be as in Theorem 77, with $\overline{E}$ quasi-canonical. The Kühne local-trivialization and operation argument remains valid for this quasi-canonical toric compactification in the following precise sense.

(a) *For every irreducible subvariety $X \subseteq H_{\overline{K}}$ satisfying $\mu_{\overline{M}}^{\mathrm{ess}}(X) = \mu_{\overline{M}}^{\mathrm{ess}}(H)$, every X-generic sequence (x_i) satisfying*

$$h_{\overline{M}}(x_i) \longrightarrow \mu_{\overline{M}}^{\mathrm{ess}}(H),$$

and every archimedean place v, the Galois orbit measures of x_i on $X_{\mathbb{C}_v}^{an}$ converge to the analogue of Kühne's Proposition 4.1 measure

$$\mu_{X,v}^{\overline{M}} = \frac{c_1(\overline{H}(\overline{E})_v|_{\overline{X}})^{d-d_0} \wedge c_1(\rho^* \overline{N}_{H,v}|_{\overline{X}})^{d_0}}{\overline{H}(E)^{d-d_0} \cdot (\rho^* N_H)^{d_0} \cdot X},$$

where $d = \dim X$ and $d_0 = \dim \rho(X)$.

(b) *In the archimedean local trivializations, Kühne's Lemmas 5.1–5.3 admit the following general-fan replacement. The quasi-canonical shift is Lemma 79; the phase coordinates on arbitrary fan charts are supplied by Lemma 80; the rank and exceptional-set part is Lemma 82; the toric current computation is supplied by the BPS route through the corner-active polyhedral quotient factorization Proposition 9.1, Lemma 88, and Propositions 9.2*

and 9.3. The local Bedford–Taylor expansion is the corner-current route Proposition 90. Thus one obtains Proposition 9.5; the Riemannian volume domination is Lemma 95.

(c) *The preceding description is compatible with quotients by connected stabilizers, with products, and with the Kühne difference morphisms*

$$\alpha_m(x_1, \dots, x_m) = (x_1 x_2^{-1}, x_2 x_3^{-1}, \dots, x_{m-1} x_m^{-1}).$$

Smallness of quotient, product and difference-image sequences for the corresponding quasi-canonical target metrics is supplied by Lemma 97. The finite Haar-slice part of this compatibility is Proposition 9.6. The toric quasi-canonical normalization under these operations is Lemma 92; the Picard-zero theta factors only fix the local sign and dual convention, as recorded in Lemma 93 and Lemma 94. The pushforward identity used in [1, equation (6.2)] is obtained by pushing forward the finite Galois orbit measures and passing to the equidistribution limits on the source and on the target. Hence Lemma 96 applies in the trivial-stabilizer case.

Proof. We first apply the theta-factor convention stated at the beginning of the paper, since this is the only point at which a sign or dual could enter. In Kühne's Lemma 3.3 the local toric coordinate is written, after choosing a rational section on the abelian base, as a quotient of the semiabelian rational function by the corresponding theta function. In the notation of Theorem 4.1, the same coordinate is

$$\psi_{j,m}(y) = \frac{g_m(y)}{f_{m,v}(\overline{\pi}_j(y))}.$$

Here $m \in X^*(T_H)$, g_m is the rational semiabelian character section, and $f_{m,v}$ is the theta representative of the rational section of the character-pushout Picard-zero bundle $\mathcal{H}_{H,m}$. The opposite contravariant convention would write the same local expression with the dual bundle. In the present notation this is the harmless replacement $m \mapsto -m$, because $\mathcal{H}_{H,-m} \simeq \mathcal{H}_{H,m}^{\vee}$ by Lemma 93. Tensor products, duals, and pullbacks of these Picard-zero factors are flat by Lemma 94. Thus the convention chosen here fixes the signs in local formulas but changes neither the Chern currents nor the normalized measures used below.

We prove (a). The essential point is that every estimate is made after restriction to $\overline{X}$ and to an irreducible component of its inverse image in the auxiliary tower. Lemma 98 supplies this restricted chain; no estimate on the full ambient compactification is substituted for it. This is not a word-for-word invocation of [1, Proposition 4.1]. We check its published dependency chain [1, Lemmas 4.2–4.6]: geometric degree growth (Lemma 4.2), identification of the limiting mixed measure (Lemma 4.3), height first variation (Lemma 4.4), the horizontal semipositivity correction and height lower bound (Lemma 4.5), and the arithmetic-volume estimate (Lemma 4.6). The normalized height equality in (a) is exactly the equality needed when this chain is applied to X. The local coordinates $\phi_i^{(j)}$ are replaced by the fan-chart characters $\psi_{j,m}$ of Lemma 80; the unitary transition condition is exactly the valuation-zero overlap condition in Theorem 4.1; and the quasi-canonical metric

$$\Psi_{\overline{E},v}(u) = \Psi_E(u - u_v) - \gamma_v$$

is obtained from the canonical one by translating the tropical variable by u_v and adding a constant. The constant has no curvature contribution, and the adelic normalization of the constants is already included in the quasi-canonical normal form. The restricted geometric degree identities, mixed-measure limit, first variation, semipositivity correction, lower bound, and arithmetic-volume estimate on the inverse-image component are precisely the assertions of Lemma 98. The field-extension convention in every step is Lemma 40. Consequently every input of [1, Lemmas 4.2–4.6] has an explicit replacement, and their assembly gives the displayed measure. It is the mixed Chern measure of $\overline{H}(\overline{E})|_{\overline{X}}$ and $\rho^* \overline{N}_H|_{\overline{X}}$, with the same normalization by the algebraic intersection degree as in Kühne's formula.

We next prove (b). Kühne's Lemma 5.1 is the rank and exceptional-set statement for the maps defined by active toric coordinates. Its general-fan replacement is Lemma 82, with Lemma 80 providing the phase maps to compact tori and Lemma 79 translating the canonical compact torus to $\mathrm{val}_v^{-1}(u_v)$. Kühne's Lemma 5.2 is the calculation of the Chern current as Haar measure on the compact phase fibers. Here this is Proposition 9.5: the BPS quotient route gives the atomic tropical pushforward, and Lemma 86 lifts the atomic tropical measure to Haar measure on compact phase fibers. The optional Bedford–Taylor calculation through corner currents gives the same local formula. Finally, Kühne's Lemma 5.3 uses only finiteness of the slice description; its replacement is Lemma 95. Thus all three local lemmas survive unchanged after replacing the coordinate axes of $(\mathbb{P}^1)^t$ by the chosen fan-chart monomial coordinates.

It remains to prove (c). Products and inverses are monomial maps in the local toric coordinates, and the Kühne difference morphism is given on toric coordinates by

$$(z^{(1)},\dots,z^{(m)}) \longmapsto \big(z^{(1)}(z^{(2)})^{-1},\dots,z^{(m-1)}(z^{(m)})^{-1}\big).$$

Hence the Haar-slice part is functorial by Proposition 9.6. The toric quasi-canonical centers are carried by the same integral linear maps by Lemma 92; in particular the common center $(u_v,\dots,u_v)$ is sent to 0 by the difference map. The residual semiabelian theta factors are exactly the tensor products, duals, and pullbacks of the character-pushout Picard-zero factors described in Lemma 93. By the first paragraph of this proof, every dual which appears under products, inverses or difference morphisms is exactly the factor attached to the pulled-back character $-m$, hence to $\mathcal{H}_{H,-m} \simeq \mathcal{H}_{H,m}^{\vee}$. These factors have the same flatness property, so the local formulas have the same sign and dual conventions as Kühne's.

It remains only to identify the source of the pushforward identity. This identity is automatic from equidistribution. For a generic small product sequence $y_i \in X^m(\overline{K})$, the Galois orbit measures satisfy

$$(\alpha_m)_*\delta_{y_i,v} = \delta_{\alpha_m(y_i),v}.$$

Applying part (a) to X^m and to $\alpha_m(X^m)$, and passing to the weak limit, gives the normalized measure identity used in Kühne's equation (6.2). Thus no separate Chern-measure projection formula is required; the finite-slice contradiction is Lemma 96. □

Lemma 101 (Kühne–Zhang difference geometry)**.** *Let $X \subseteq H_{\overline{K}}$ be an irreducible subvariety with trivial stabilizer*

$$\mathrm{Stab}_{H_{\overline{K}}}(X) = \{e\}.$$

For all sufficiently large m, the difference morphism

$$\alpha_m : X^m \longrightarrow H^{m-1}, \qquad (x_1,\dots,x_m) \longmapsto (x_1x_2^{-1}, x_2x_3^{-1},\dots,x_{m-1}x_m^{-1})$$

is generically finite of degree 1 onto its image.

Proof. This is the semiabelian form of the difference-map lemma used by Zhang and Kühne. Kühne states it as [1, Lemma 6.2], referring to Zhang's proof in the abelian case. The proof is purely algebraic and uses only the group law and the triviality of the stabilizer: if two general points of X^m have the same successive differences, then their coordinates differ by a common translation; for m large enough this common translation must lie in $\mathrm{Stab}_{H_{\overline{K}}}(X)$, hence is the identity. □

Proof of Proposition 78. Let $Z \subsetneq H_{\overline{K}}$ and (z_i) satisfy the hypotheses of Proposition 78. We prove that Z is contained in a proper translate of a connected algebraic subgroup.

First perform the standard stabilizer reduction from Kühne's proof of [1, Proposition 6.1]. Let B be the identity component of $\mathrm{Stab}_{H_{\overline{K}}}(Z)$, and let $q : H \to H/B$ be the quotient. If $q(Z)$ is a point, then Z is contained in a translate of B, and we are done. Otherwise, replacing H by H/B, Z by $q(Z)$, and the sequence by its image, it is enough to treat the case

$$\mathrm{Stab}_{H_{\overline{K}}}(Z) = \{e\}.$$

Lemma 97 shows that the projected sequence is small for the quotient quasi-canonical metric whose critical vector is the image of the source critical vector. The projected sequence remains generic in the projected subvariety.

Choose m so that Lemma 101 applies to Z. As in Kühne's proof, choose a sequence of m-tuples

$$y_i = (z_{\phi_1(i)}, \dots, z_{\phi_m(i)}) \in Z^m$$

which is Z^m-generic after passing to a subsequence. Then $\alpha_m(y_i)$ is generic in $\alpha_m(Z^m)$. The product sequence is small for the exterior-product quasi-canonical metric, and Lemma 97 shows that the difference sequence is small for the quasi-canonical target metric on H^{m-1} with critical vector (0).

Apply Proposition 100 to Z^m and to $\alpha_m(Z^m)$. The pushforward identity used by Kühne in [1, equation (6.2)] follows directly from these two equidistribution statements. Indeed, for every i, the finite Galois orbit measures satisfy

$$(\alpha_m)_* \delta_{y_i,v} = \delta_{\alpha_m(y_i),v}.$$

Since y_i is generic in Z^m and $\alpha_m(y_i)$ is generic in $\alpha_m(Z^m)$, passing to the weak limits gives, for every compactly supported continuous test function f on $\alpha_m(Z^m)^{an}_{\mathbb{C}_v}$,

$$\int_{(Z^m)^{an}_{\mathbb{C}_v}} (f \circ \alpha_m)\, d\mu^{\overline{M}}_{Z^m,v} = \int_{\alpha_m(Z^m)^{an}_{\mathbb{C}_v}} f\, d\mu^{\overline{M}}_{\alpha_m(Z^m),v}.$$

By the real-analytic Haar description in Proposition 100, the measure on Z^m is the product of the translated compact-torus Haar measures attached to Z, while the measure on the image is described by the corresponding Haar volume on the image. The common quasi-canonical shift u_v only translates the compact tori and does not affect the tangent-space computation, the positivity of the volume forms, or the Haar invariance.

The hypotheses of Lemma 96 are therefore satisfied. That lemma is precisely Kühne's support and dimension comparison from [1, pp. 2119–2121, Section 5], rewritten in terms of the finite Haar-slice package above. It gives a contradiction in the trivial-stabilizer case.

Thus the trivial-stabilizer quotient case cannot occur unless the image is contained in a proper translate of a connected algebraic subgroup. Lifting back through the quotient $H \to H/B$ gives the same conclusion for the original Z. □

Lemma 102 (Quasi-canonical arithmetic T-effectivity)**.** *Let X_Σ be a proper toric variety with principal torus T_H, and let $\overline{E}$ be the quasi-canonical toric metric attached to an ample T_H-effective divisor E. Write the BGPS quasi-canonical normal form as*

$$\psi_{\overline{E},v}(z) = \Psi_E(z - w_v) - \gamma_v, \qquad \sum_v n_v w_v = 0.$$

Then the T_H-invariant section corresponding to $0 \in \Delta_E$ is a $\mu^{\mathrm{abs}}_{\overline{E}}(T_H)$-small invariant section in the sense of Definition 4.20; in particular $\overline{E}$ is arithmetically T_H-effective. More precisely,

$$\mu^{\mathrm{abs}}_{\overline{E}}(T_H) = \mu^{\mathrm{ess}}_{\overline{E}}(X_\Sigma) = \sum_v n_v \gamma_v.$$

Proof. By the BGPS quasi-canonical normal form, recorded in Proposition 121, the metric functions have the displayed shape. Let $s_0 = \chi^0 s_E$ be the invariant section of $O(E)$ corresponding to $0 \in \Delta_E$. The convention for toric metric functions is

$$\psi_{\overline{E},v}(\mathrm{val}_v(p)) = \log \|s_E(p)\|_v,$$

which is the same convention as in Section 3, where $\Psi_{\overline{E},v} = -g_{\overline{E},v}$. Since $s_0 = s_E$ on the principal torus, we have, by density of the principal torus and continuity of the toric metric,

$$\log \|s_0\|_{v,\sup} = \sup_{z \in N_{H,\mathbb{R}}} \psi_{\overline{E},v}(z) = \sup_{z \in N_{H,\mathbb{R}}} \Psi_E(z - w_v) - \gamma_v.$$

Because E is T_H-effective, $0 \in \Delta_E$, equivalently $\Psi_E \le 0$ on $N_{H,\mathbb{R}}$, and $\Psi_E(0) = 0$. Hence

$$\sup_z \Psi_E(z - w_v) = 0, \qquad -\log \|s_0\|_{v,\sup} = \gamma_v.$$

Thus s_0 is a small invariant section in the sense of Definition 4.20, with $c_v = \gamma_v$. The family $(\gamma_v)_v$ is finitely supported in the BGPS normal form.

It remains to identify the resulting sum with the toric minimum. For $x \in \Delta_E$, the local roof function is

$$\vartheta_{\overline{E},v}(x) = \inf_z \big(\langle z, x\rangle - \Psi_E(z - w_v) + \gamma_v\big).$$

Putting $y = z - w_v$ and using the canonical identity $\Psi_E^\vee(x) = 0$ on Δ_E, this becomes

$$\vartheta_{\overline{E},v}(x) = \langle w_v, x\rangle + \gamma_v.$$

Therefore

$$\vartheta_{\overline{E}}(x) = \left\langle \sum_v n_v w_v, x \right\rangle + \sum_v n_v \gamma_v = \sum_v n_v \gamma_v.$$

The toric successive-minima formula [18, Theorem 3.9 and Corollary 3.10] gives

$$\mu^{\mathrm{abs}}_{\overline{E}}(T_H) = \mu^{\mathrm{ess}}_{\overline{E}}(X_\Sigma) = \max_{x \in \Delta_E} \vartheta_{\overline{E}}(x) = \sum_v n_v \gamma_v.$$

This also shows that the constants $c_v = \gamma_v$ have the required global sum, so $\overline{E}$ is arithmetically T_H-effective. □

Lemma 103 (Height separation in the quasi-canonical case)**.** *Let*

$$\overline{M} = \overline{H}(\overline{E}) \otimes \rho^* \overline{N}_H$$

be as in Theorem 77. Assume that the toric part $\overline{E}$ is in the situation of Lemma 102. Then

$$\mu^{\mathrm{ess}}_{\overline{M}}(H) = \mu^{\mathrm{abs}}_{\overline{E}}(T_H), \qquad h_{\overline{H}(\overline{E})}(x) \ge \mu^{\mathrm{abs}}_{\overline{E}}(T_H), \qquad \hat{h}_{\overline{N}_H}(\rho(x)) \ge 0$$

for all $x \in H(\overline{K})$.

Proof. By Lemma 102, $\overline{E}$ is arithmetically T_H-effective. Repeating the invariant-section argument in Lemma 23, but without adding the abelian factor, gives

$$h_{\overline{H}(\overline{E})}(x) \ge \mu^{\mathrm{abs}}_{\overline{E}}(T_H)$$

for every $x \in H(\overline{K})$. The canonical height attached to the symmetric ample line bundle $\overline{N}_H$ is nonnegative, so

$$\hat{h}_{\overline{N}_H}(\rho(x)) \ge 0.$$

Finally, Proposition 4.28, applied to H and $\overline{E}$, gives

$$\mu^{\mathrm{ess}}_{\overline{M}}(H) = \mu^{\mathrm{abs}}_{\overline{E}}(T_H).$$

□

Proposition 104 (Base reduction for quasi-canonical Bogomolov)**.** *Assume that the toric part $\overline{E}$ is in the situation of Lemma 102. Assume also Proposition 78 for the semiabelian subvarieties obtained by pulling back abelian subvarieties of A_H. Then Theorem 77 holds for $(H, \overline{M})$.*

Proof. Let $Z \subsetneq H_{\overline{K}}$ be irreducible and let (z_i) be a Z-generic sequence with

$$h_{\overline{M}}(z_i) \to \mu^{\mathrm{ess}}_{\overline{M}}(H).$$

By Lemma 103,

$$\mu^{\mathrm{ess}}_{\overline{M}}(H) = \mu^{\mathrm{abs}}_{\overline{E}}(T_H), \qquad h_{\overline{H}(\overline{E})}(x) \ge \mu^{\mathrm{abs}}_{\overline{E}}(T_H), \qquad \hat{h}_{\overline{N}_H}(\rho(x)) \ge 0.$$

Since

$$h_{\overline{M}}(z_i) = h_{\overline{H}(\overline{E})}(z_i) + \hat{h}_{\overline{N}_H}(\rho(z_i))$$

and the first summand is bounded below by $\mu_{\overline{E}}^{\mathrm{abs}}(T_H)$, while the second is nonnegative, we have

$$\hat{h}_{\overline{N}_H}(\rho(z_i)) \to 0.$$

Let $C = \rho(Z)$. The projected sequence is C-generic: if $W \subsetneq C$ is closed, then $Z \cap \rho^{-1}(W)$ is a proper closed subset of Z, so z_i is eventually outside it.

By the abelian Bogomolov theorem, Theorem 113, C is a torsion translate $\tau + A_0$ of an abelian subvariety $A_0 \subseteq A_H$. If $A_0 \neq A_H$, then

$$\rho^{-1}(\tau + A_0)$$

is a proper translate of the connected semiabelian subgroup $\rho^{-1}(A_0)$, and it contains Z. This proves the desired conclusion in the proper-image case.

It remains to treat the case $A_0 = A_H$. In the general reduction, after translating by a torsion lift of τ and replacing H by $\rho^{-1}(A_0)$, this is exactly the dominant case covered by Proposition 78. Therefore Z is contained in a proper translate of a connected algebraic subgroup of $H_{\overline{K}}$. □

Proof of Theorem 77. After adding an M_K-constant, the quasi-canonical toric metric is in the normal form of Lemma 102. The dominant case is Proposition 78, proved above from the Kühne transport package. Proposition 104 then proves the asserted Bogomolov property for every irreducible proper subvariety $Z \subsetneq H_{\overline{K}}$. □

Proposition 105 (Bogomolov property implies the strict upgrade)**.** *Assume Theorem 77 for $(H, \overline{M})$. Then every strict $\overline{M}$-small sequence in $H(\overline{K})$ is H-generic. Consequently, if the generic equidistribution statement of Theorem 61 applies to $\overline{M}$, every strict $\overline{M}$-small sequence equidistributes toward the canonical probability measure.*

Proof. Let (x_i) be strict and $\overline{M}$-small. If it is not H-generic, then after passing to a subsequence and replacing its Zariski closure by an irreducible component, there is an irreducible proper subvariety $Z \subsetneq H_{\overline{K}}$ such that the subsequence is Z-generic. The subsequence is still $\overline{M}$-small in the ambient sense, so

$$h_{\overline{M}}(x_i) \longrightarrow \mu_{\overline{M}}^{\mathrm{ess}}(H).$$

Theorem 77 implies that Z is contained in a proper translate of a connected algebraic subgroup of $H_{\overline{K}}$, contradicting strictness. Hence every strict $\overline{M}$-small sequence is H-generic, and the generic equidistribution theorem applies. □

Theorem 106 (Quasi-canonical strong equidistribution)**.** *Let $\overline{M}$ be an adelic line bundle on a semiabelian variety H of the form*

$$\overline{M} = \overline{H}(\overline{E}) \otimes \rho^* \overline{N}_H,$$

where $\overline{E}$ is the quasi-canonical toric metric attached to an ample T_H-effective divisor and $\overline{N}_H$ is canonical on the abelian quotient. Then every strict $\overline{M}$-small sequence equidistributes at every archimedean place toward the canonical probability measure.

Proof. By Theorem 77 and Proposition 105, every strict $\overline{M}$-small sequence is H-generic. The quasi-canonical generic equidistribution theorem Theorem 61 then applies and gives convergence to the canonical probability measure at every archimedean place. □

Corollary 107 (Quasi-canonical Bogomolov and strictness)**.** *For the quasi-canonical semiabelian metrics under consideration and for the semiabelian subvarieties obtained after the base reduction in Proposition 104, Theorem 77 holds. Consequently the strict quasi-canonical equidistribution theorem Theorem 106 holds whenever the generic equidistribution theorem Theorem 61 applies.*

Proof. Proposition 100 is the source of the proof of the dominant bridge, Proposition 78. After the height-separation normalization in Lemma 102, Proposition 104 proves Theorem 77. Finally Proposition 105 upgrades ambient small strict sequences to generic sequences, and Theorem 61 gives their equidistribution. □

Corollary 108 (Restricted strict equidistribution for the paper's metric class)**.** *Let*

$$\overline{L} = \overline{G}(\overline{D}) \otimes \overline{\pi}^* \overline{N}$$

be an ambient adelic line bundle whose toric metric $\overline{D}$ satisfies the standing hypotheses of the Bogomolov part of the paper: D is ample and T-effective, and $\overline{D}$ is semipositive, monocritical, and arithmetically T-effective. Then for every minimal translate $Y = a + B$ arising in the restriction package, the restricted quasi-canonical line bundle $\overline{L}'_{a,B}$ satisfies the strict quasi-canonical equidistribution theorem, Theorem 106.

Proof. Let $X \subseteq G_{\overline{K}}$ be irreducible and let $Y = a + B$ be the minimal translate used in the restriction argument. The monocritical-to-quasi-canonical replacement Proposition 72, together with the restriction construction in Proposition 118, produces the restricted quasi-canonical metric $\overline{L}'_{a,B}$ on B. Smallness for the original metric is transferred to this restricted quasi-canonical metric by Corollary 119. Applying Proposition 100 to $\overline{L}'_{a,B}$ gives Theorem 106 by Corollary 107. This is exactly the strict equidistribution statement used in Theorem 128. □

Remark 109 (Scope of the phrase "general metric")**.** *The preceding corollary is the precise sense in which the Kühne-operation route proves the Bogomolov theorem for a general metric in this paper. The word "general" refers to the paper's controlled class of toric metrics: semipositive, monocritical, and arithmetically T-effective metrics on an ample T-effective toric divisor, together with their restricted quasi-canonical replacements. It does not mean an arbitrary semipositive toric metric. Extending the argument to such a larger class would require a new replacement or deformation theorem beyond Proposition 72.*

Remark 110. *Theorem 61 supplies the corresponding generic equidistribution statement, but not this strict statement. This distinction is essential in the restriction argument below: an X-generic sequence in a proper subvariety $X \subsetneq Y$ is not Y-generic, although it is strict in the minimal translate Y. The strict upgrade is supplied here by the Bogomolov theorem in the quasi-canonical metric, namely Theorem 77, through Proposition 105. The dominant case of that Bogomolov theorem is Proposition 78, whose proof is the direct Kühne local-trivialization transport package, Proposition 100. The relative toric fiber-specialness route in the appendix is therefore not part of the main proof. The reduction from the original monocritical metric to the quasi-canonical restricted metric is supplied by Proposition 72 and by the restriction package below.*

Proposition 111 (Strict equidistribution for the ambient metric)**.** *Let (x_i) be a strict $\overline{L}$-small sequence in $G(\overline{K})$. Then, for every archimedean place v, the Galois orbit measures $\delta_{x_i,v}$ converge to the canonical probability measure $\mu_{\overline{L},v}$. If $\mathbf{u} = (u_v)_v$ is the critical point of $\overline{D}$, then $\mu_{\overline{L},v}$ is supported on*

$$S_{u_v} := S_v \cdot e^{-u_v},$$

where S_v is the maximal compact subgroup of $G(\mathbb{C}_v)$.

Proof. Normalize the metric by an M_K-constant so that $\mu_G = 0$. Proposition 72 constructs the quasi-canonical toric metric

$$\Psi_{\overline{D}',v}(z) = \Psi_D(z - u_v)$$

and proves that every $\overline{L}$-small sequence is also small for

$$\overline{L}' = \overline{G}(\overline{D}') \otimes \pi^* \overline{N}.$$

Strictness is purely algebraic, so (x_i) is strict as a $\overline{L}'$-small sequence. Theorem 106 applies to $\overline{L}'$. The limiting measure is the Haar measure on S_{u_v}, which is the canonical measure associated with the original monocritical metric because Proposition 71 proves convergence of the valuation measures to δ_{u_v}. □

9.7. **Minimal Special Envelopes.**

Lemma 112 (Minimal envelope)**.** *Let $X \subseteq G_{\overline{K}}$ be irreducible. There is a unique minimal translate*

$$Y = a + B$$

of a connected algebraic subgroup of $G_{\overline{K}}$ containing X. If $X \neq Y$, then every X-generic sequence is strict as a sequence in Y.

Proof. Among all translates of connected algebraic subgroups containing X, choose one, say $Y = a + B$, of minimal dimension. Such a translate exists because G itself is one candidate. If $Y' = a' + B'$ is another minimal candidate, then $Y \cap Y'$ is nonempty and is a translate of the algebraic subgroup $B \cap B'$, possibly with finitely many connected components. Since X is irreducible and contained in this intersection, it is contained in one connected component, which is a translate of $(B \cap B')^0$. Minimality forces this component to have the same dimension as both Y and Y', and hence $Y = Y'$. This proves existence and uniqueness.

Assume $X \neq Y$. If $U \subsetneq Y$ is a translate of a connected subgroup, then $X \cap U$ is a proper closed subset of X; otherwise U would be a smaller translate containing X. An X-generic sequence therefore has only finitely many terms in U, which is exactly strictness in Y. □

Let $X \subseteq G_{\overline{K}}$ be irreducible and let

$$Y = a + B$$

be the minimal translate of a connected algebraic subgroup containing X. Put

$$T_B = B \cap T, \qquad A_B = \pi(B).$$

Let

$$\iota_N : N_{B,\mathbb{R}} \longrightarrow N_{\mathbb{R}}$$

be the injection of cocharacter spaces induced by $T_B \hookrightarrow T$.

9.8. **The Abelian Part Becomes Torsion.**

Theorem 113 (Bogomolov theorem for abelian varieties)**.** *Let A be an abelian variety over $\overline{K}$, equipped with a Neron–Tate height attached to an ample symmetric line bundle. If an irreducible subvariety $Z \subset A$ carries a Z-generic sequence whose Neron–Tate heights tend to 0, then Z is a torsion subvariety. In particular, if a translate $a + A_0$ of an abelian subvariety contains such a generic small sequence, then $a + A_0$ is a torsion translate of A_0. This is Zhang's Bogomolov theorem for abelian varieties [29, Theorem 2], building on Ullmo's curve case [28].*

Lemma 114 (Torsion base representative)**.** *Assume*

$$\mu_{\overline{L}}^{\mathrm{ess}}(X) = \mu_G.$$

Then, after replacing a by $a + b$ for a suitable $b \in B(\overline{K})$, one may assume that $\pi(a)$ is torsion in $A(\overline{K})$.

Proof. Choose an X-generic sequence (x_i) with

$$h_{\overline{L}}(x_i) \to \mu_G.$$

By Proposition 4.28 and the height decomposition

$$h_{\overline{L}}(x_i) = h_{\overline{G}(\overline{D})}(x_i) + \hat{h}_{\overline{N}}(\pi(x_i)),$$

with

$$h_{\overline{G(D)}}(x_i) \geq \mu_G, \qquad \hat{h}_{\overline{N}}(\pi(x_i)) \geq 0,$$

we obtain

$$\hat{h}_{\overline{N}}(\pi(x_i)) \to 0.$$

Let $C = \overline{\pi(X)} \subseteq \pi(a) + A_B$. The projected sequence is C-generic: the inverse image in X of every proper closed subset of C is a proper closed subset of X. Theorem 113 therefore shows that

$$C = \tau + A_0$$

for a torsion point τ and an abelian subvariety $A_0 \subseteq A_B$. If $A_0 \subsetneq A_B$, the identity component of $B \cap \pi^{-1}(A_0)$ is a proper connected algebraic subgroup of B, and a translate of it contains X. This contradicts the minimality of $Y = a + B$. Hence $A_0 = A_B$, so

$$\pi(a) + A_B = C = \tau + A_B.$$

Thus there is a torsion point $\tau \in A(\overline{K})$ such that

$$\pi(a) + A_B = \tau + A_B.$$

Since $\pi(B) = A_B$, we can choose $b \in B(\overline{K})$ with $\pi(b) = \tau - \pi(a)$. Then $\pi(a+b) = \tau$ is torsion, and replacing a by $a + b$ does not change $Y = a + B$. □

9.9. **Tropical Position of the Envelope.** From now on we choose a as in Lemma 114. Choose a torsion point $q \in G(\overline{K})$ above $\pi(a)$. Then there is a unique $t_a \in T(\overline{K})$ such that

$$a = q\, t_a.$$

For every place v, put

$$\alpha_v := \mathrm{val}_v(t_a) \in N_{\mathbb{R}}.$$

Lemma 115 (Valuations of translated subtori)**.** *Let $Y = q\, t_a B$ with q torsion over the abelian quotient and $t_a \in T(\overline{K})$. For the valuation map on the toric part used in Proposition 4.27, the valuation of every algebraic point of Y at a place v belongs to*

$$\alpha_v + \iota_N(N_{B,\mathbb{R}}) \subseteq N_{\mathbb{R}}, \qquad \alpha_v = \mathrm{val}_v(t_a).$$

Equivalently, after translating Y by $q\, t_a$ back to B, the ambient valuation differs from the intrinsic T_B-valuation by the fixed affine shift α_v.

Proof. It is enough to prove the assertion on the dense torus, since the valuation map used here is defined by local toric coordinates and is compatible with restriction to the toric part. Write a point of Y as

$$q\, t_a\, \iota(s), \qquad s \in T_B(\overline{K}).$$

By Lemma 21, the torsion lift q has zero toric valuation. The valuation on a split torus is additive under multiplication and functorial for the homomorphism $\iota : T_B \hookrightarrow T$. Hence

$$\mathrm{val}_v(q\, t_a\, \iota(s)) = \mathrm{val}_v(t_a) + \iota_N(\mathrm{val}_{B,v}(s)) = \alpha_v + \iota_N(\mathrm{val}_{B,v}(s)).$$

Since $\mathrm{val}_{B,v}(s) \in N_{B,\mathbb{R}}$, the result follows. The same computation in local trivialization charts shows that the statement is independent of the chosen chart. □

Lemma 116 (Critical vector lies in the tropical envelope)**.** *Assume $\mu_{\overline{L}}^{\mathrm{ess}}(X) = \mu_G$. Then for every place v,*

$$u_v - \alpha_v \in \iota_N(N_{B,\mathbb{R}}).$$

Equivalently, there is a unique $w_v \in N_{B,\mathbb{R}}$ such that

$$\iota_N(w_v) + \alpha_v = u_v.$$

Proof. Let (x_i) be an X-generic $\overline{L}$-small sequence. Proposition 71 gives adelic KR convergence of the ambient valuation measures:

$$(\mathrm{val}_v)_*\delta_{x_i,v} \longrightarrow \delta_{u_v}$$

for every v.

On the other hand, because $x_i \in Y = a + B = q\,t_a B$, the toric valuation of every point in the Galois orbit of x_i lies in the closed affine subspace

$$\alpha_v + \iota_N(N_{B,\mathbb{R}}) \subseteq N_{\mathbb{R}}.$$

This is Lemma 115; its independence of the local trivialization chart is part of Proposition 4.27. Since a weak limit of probability measures supported on a closed set is again supported on that closed set, the limit point u_v belongs to

$$\alpha_v + \iota_N(N_{B,\mathbb{R}}).$$

This proves the lemma. □

9.10. **The Restricted Quasi-canonical Model.** Let Σ_B be the fan on $N_{B,\mathbb{R}}$ obtained from the inverse images

$$\iota_N^{-1}(\sigma), \qquad \sigma \in \Sigma,$$

after removing redundancies and passing to the associated normal toric variety. Denote this toric variety by X_{Σ_B}.

Lemma 117 (Toric normalization of a subtorus closure)**.** *The toric variety X_{Σ_B} is the normalization of the closure of T_B in X_Σ. The inclusion $T_B \hookrightarrow T$ extends to a toric morphism*

$$\varphi_B : X_{\Sigma_B} \longrightarrow X_\Sigma$$

whose image is this closure and whose induced morphism to the image is finite. In particular, if D is an ample toric Cartier divisor on X_Σ, then $\varphi_B^* D$ is ample on X_{Σ_B}. If D is T-effective, then $\varphi_B^* D$ is T_B-effective.

Proof. The inverse images of the cones of Σ form a fan because inverse image commutes with taking faces and intersections. On an affine toric chart $U_\sigma = \operatorname{Spec} K[\sigma^\vee \cap M]$, the closure of $T_B \cap U_\sigma$ has coordinate semigroup given by the image of $\sigma^\vee \cap M$ in the character lattice M_B. Since $T_B \subseteq T$ is a saturated subtorus, the saturation of this image is

$$\left(\iota_N^{-1}\sigma\right)^\vee \cap M_B.$$

Thus the normalization of the affine closure is $\operatorname{Spec} K[(\iota_N^{-1}\sigma)^\vee \cap M_B]$. These affine normalizations glue over the fan Σ_B, giving X_{Σ_B}. This is the standard subtorus-closure construction in toric geometry; see [20, Section 1.4]. The induced morphism from the normal variety X_{Σ_B} to its image is the normalization map, hence finite.

The restriction of an ample line bundle to a closed subvariety is ample, and the finite pullback of an ample line bundle is ample. Therefore $\varphi_B^* D$ is ample. Finally, a non-zero T-eigensection of D restricts on the dense torus and pulls back to a non-zero T_B-eigensection, so T-effectivity descends to T_B-effectivity. □

The map

$$T_B \longrightarrow T, \qquad t \longmapsto t_a\,\iota(t),$$

extends, by composing φ_B with the action of t_a on X_Σ, to a morphism

$$\varphi_{a,B} : X_{\Sigma_B} \longrightarrow X_\Sigma.$$

Let

$$\overline{B} := B \times^{T_B} X_{\Sigma_B}.$$

Translation by $q\,t_a$ identifies $\overline{B}$ with the normalization of the closure of Y in $\overline{G}$.

Proposition 118 (Restriction of the metric)**.** *With the above notation,*

$$\tau_a^*\overline{L} = \overline{B}(\overline{D}_{a,B}) \otimes \pi_B^*\overline{N}_B,$$

where $\overline{N}_B$ is the canonical metrized line bundle obtained by restricting $\overline{N}$ to A_B, and $\overline{D}_{a,B}$ is the toric metrized divisor on X_{Σ_B} whose local metric functions are

$$\Psi_{\overline{D}_{a,B},v}(z) = \Psi_{\overline{D},v}\big(\iota_N(z) + \alpha_v\big), \qquad z \in N_{B,\mathbb{R}}.$$

Moreover, the quasi-canonical metric obtained in Proposition 72 restricts to the quasi-canonical toric metric

$$\Psi_{\overline{D}'_{a,B},v}(z) = \Psi_{D_B}(z - w_v),$$

where w_v is defined by Lemma 116. The divisor D_B is ample and T_B-effective, and

$$\sum_v n_v w_v = 0.$$

Hence $\overline{D}'_{a,B}$ is an admissible quasi-canonical toric metric for the restricted semiabelian variety B.

Proof. The algebraic statement follows from the construction of the normalized closure. On the toric fiber, the map is $t \mapsto t_a\iota(t)$, so the underlying toric divisor is the pullback of D to X_{Σ_B}. Its support function is

$$\Psi_{D_B}(z) = \Psi_D(\iota_N(z)).$$

By Lemma 117, the induced map from X_{Σ_B} to the closure of T_B is finite. Therefore the pullback of the ample toric divisor D is ample on X_{Σ_B}. The same lemma shows that a T-effective section of D pulls back to a non-zero T_B-eigensection, so D_B is T_B-effective.

The translation by the torsion lift q has zero toric valuation at every place by Lemma 21, and hence does not change the metric. The translation by t_a shifts the valuation by α_v. This gives

$$\Psi_{\overline{D}_{a,B},v}(z) = \Psi_{\overline{D},v}\big(\iota_N(z) + \alpha_v\big).$$

Since $\pi(a) = \pi(q)$ is torsion, translation by $\pi(a)$ preserves the canonical metric on A. Hence the abelian factor is simply

$$\pi_B^*\overline{N}_B.$$

For the quasi-canonical metric, Proposition 72 defines

$$\Psi_{\overline{D}',v}(y) = \Psi_D(y - u_v).$$

Pulling this back gives

$$\Psi_{\overline{D}'_{a,B},v}(z) = \Psi_D(\iota_N(z) + \alpha_v - u_v).$$

By Lemma 116,

$$\alpha_v - u_v = -\iota_N(w_v),$$

and therefore

$$\Psi_{\overline{D}'_{a,B},v}(z) = \Psi_D(\iota_N(z - w_v)) = \Psi_{D_B}(z - w_v).$$

This is precisely the quasi-canonical metric on X_{Σ_B} with critical vector $(w_v)_v$. Finally,

$$\sum_v n_v \iota_N(w_v) = \sum_v n_v(u_v - \alpha_v).$$

The critical vector $(u_v)_v$ satisfies the product formula, and $(\alpha_v)_v$ is the valuation vector of the algebraic point $t_a \in T(\overline{K})$, so both sums on the right vanish. Since ι_N is injective, $\sum_v n_v w_v = 0$. □

Corollary 119 (Smallness after restriction)**.** *Let (x_i) be an X-generic $\overline{L}$-small sequence and assume $\mu_{\overline{L}}^{\mathrm{ess}}(X) = \mu_G$. Then the corresponding sequence in $B(\overline{K})$ is small for*

$$\overline{L}'_{a,B} := \overline{B}(\overline{D}'_{a,B}) \otimes \pi_B^*\overline{N}_B.$$

Proof. Write $x_i = ab_i$ with $b_i \in B(\overline{K})$. The base part satisfies

$$\hat{h}_{\overline{N}_B}(\pi_B(b_i)) = \hat{h}_{\overline{N}}(\pi(x_i)) \to 0,$$

because $\pi(a)$ is torsion. By Proposition 71,

$$(\mathrm{val}_v)_* \delta_{x_i,v} \to \delta_{u_v}.$$

Equivalently, after subtracting the fixed translation α_v, the valuation measures of b_i in B converge to δ_{w_v}. The quasi-canonical metric $\overline{D}'_{a,B}$ has critical vector w_v, so the same Lipschitz/KR argument as in Proposition 72 gives

$$h_{\overline{B}(\overline{D}'_{a,B})}(b_i) \to 0.$$

Thus

$$h_{\overline{L}'_{a,B}}(b_i) \to 0 = \mu^{\mathrm{ess}}_{\overline{L}'_{a,B}}(\overline{B}),$$

which proves smallness. □

9.11. **Strict Equidistribution on the Envelope.**

Proposition 120 (Restricted strict equidistribution)**.** *Assume $\mu^{\mathrm{ess}}_{\overline{L}}(X) = \mu_G$. If $X \neq Y$, then an X-generic $\overline{L}$-small sequence in $X(\overline{K})$ is strict in Y, and its archimedean Galois orbit measures converge to a probability measure whose support is Zariski dense in Y.*

Proof. Strictness in Y is Lemma 112.

By Corollary 119, after translating the sequence to B, it is small for the quasi-canonical line bundle $\overline{L}'_{a,B}$. Hence Theorem 106 gives equidistribution on B; the hypotheses of that theorem are verified for $\overline{L}'_{a,B}$ by Proposition 118. Translating back by a gives the required convergence on Y.

At an archimedean place v, the limiting support is a translate of the maximal compact subgroup of $B(\mathbb{C}_v)$, with toric valuation w_v. The maximal compact subgroup of a complex semiabelian variety is Zariski dense: its image in the abelian quotient is the whole complex torus $A_B(\mathbb{C}_v)$, and its intersection with the torus part is $(S^1)^{\dim T_B}$, which is Zariski dense in T_B. Therefore its translate is Zariski dense in Y. □

9.12. **Special Points on the Envelope.**

Proposition 121 (BGPS bridge)**.** *Let $\overline{E}$ be a quasi-canonical semipositive toric metrized divisor on a proper toric variety with principal torus T_H. Then $\overline{E}$ is monocritical. More precisely, by the BGPS characterization of quasi-canonical toric metrics, the metric functions may be written in the form*

$$\psi_{\overline{E},v}(z) = \Psi_E(z - w_v) - \gamma_v$$

for a centered adelic vector $(w_v)_v$ and constants $(\gamma_v)_v$; this vector is the critical vector of $\overline{E}$. The implication from this normal form to monocriticality is the BGPS monocritical criterion. We use [16, Propositions 4.16 and 5.3] for these two facts.

If F/K is a finite extension, the scalar extension $\overline{E}_F$ is again monocritical, and the critical vector at a place w of F over v is the same vector w_v. This is the scalar-extension compatibility in [16, Proposition 4.17]. Consequently, a statement written below in terms of $T_H(\overline{K})$ and $\mathrm{val}_v(T_H(\overline{K}))$ is interpreted by applying the cited BGPS result over a sufficiently large finite field of definition and then viewing the resulting point over $\overline{K}$.

Theorem 122 (BGPS toric Bogomolov property)**.** *Let X_Σ be a proper toric variety with principal torus T, and let $\overline{E}$ be a monocritical toric metrized $\mathbb{R}$-divisor with big underlying divisor. If an irreducible subvariety $V \subset T_{\overline{K}}$ satisfies*

$$\mu^{\mathrm{ess}}_{\overline{E}}(V) = \mu^{\mathrm{ess}}_{\overline{E}}(X_\Sigma),$$

then V is a translate of a subtorus of $T_{\overline{K}}$. Moreover, for a translate $U = T_0 \cdot p$, with associated affine valuation spaces

$$A_{U,w} = \mathrm{val}_w(p) + N_{T_0,\mathbb{R}},$$

the translate U is $\overline{E}$-special if and only if the critical vector satisfies $u_{v(w)} \in A_{U,w}$ for every place w after a finite field of definition. This is the Bogomolov property and special-translate criterion of [16, Theorem 5.12 and Proposition 5.14].

Remark 123 (Why the toric theorem is not yet the semiabelian theorem)**.** *Theorem 122 is the correct toric source for a possible proof of Theorem 77. It does not by itself prove Theorem 77, because a subvariety of a semiabelian variety need not be contained in one torus fiber, and the quotient by the abelian part has to be controlled at the same time. A non-circular proof of Theorem 77 has to combine the abelian Bogomolov theorem for $\rho(Z)$, the toric BGPS theorem on the fiber directions, and a gluing statement saying that the resulting local translated-subtorus conditions come from a global translate of a connected semiabelian subgroup.*

Proposition 124 (Toric special-point criterion)**.** *Let $\overline{E}$ be a quasi-canonical toric metrized divisor on a torus T_H, with centered critical vector $(w_v)_v$. Then $T_H(\overline{K})$ contains a point whose height equals the toric absolute minimum if and only if*

$$w_v \in \mathrm{val}_v(T_H(\overline{K})) \otimes_{\mathbb{Z}} \mathbb{Q}$$

for every place v. When this condition holds, such a point may be chosen inside the torus and with valuation vector w_v at every v, after clearing denominators and, if necessary, replacing the ground field by a finite extension. This is the special-point criterion of [16, Proposition 5.13]; the construction in its proof produces $q^{1/\ell}$ from an S-unit q with $\mathrm{val}_v(q) = \ell w_v$. The passage from quasi-canonical metrics to the monocritical hypothesis of that proposition, and the harmless finite-extension interpretation, are recorded in Proposition 121.

Lemma 125 (Rational valuation exactness for subtori)**.** *Let $T_B \subseteq T$ be a saturated subtorus. For every place v, the valuation groups of algebraic points are exact after tensoring with $\mathbb{Q}$. In particular, if*

$$\eta \in \mathrm{val}_v(T(\overline{K})) \otimes_{\mathbb{Z}} \mathbb{Q}$$

and $\eta \in \iota_N(N_{B,\mathbb{R}})$, then the unique $\xi \in N_{B,\mathbb{R}}$ with $\iota_N(\xi) = \eta$ satisfies

$$\xi \in \mathrm{val}_v(T_B(\overline{K})) \otimes_{\mathbb{Z}} \mathbb{Q}.$$

This is the valuation form of exactness for the short exact sequence of algebraic tori

$$1 \to T_B \to T \to T/T_B \to 1,$$

using that the character lattice quotient is torsion-free.

Proof. Let

$$\Gamma_v = \mathrm{val}_v(\overline{K}^{\times}) \subseteq \mathbb{R}.$$

For a split torus with cocharacter lattice N, the valuation group of its algebraic points is

$$\mathrm{val}_v(T(\overline{K})) = N \otimes_{\mathbb{Z}} \Gamma_v \subseteq N_{\mathbb{R}},$$

and similarly

$$\mathrm{val}_v(T_B(\overline{K})) = N_B \otimes_{\mathbb{Z}} \Gamma_v \subseteq N_{B,\mathbb{R}}.$$

Since $T_B \subseteq T$ is saturated, the quotient N/N_B is torsion-free. Thus

$$0 \longrightarrow N_B \longrightarrow N \longrightarrow N/N_B \longrightarrow 0$$

remains exact after tensoring with the $\mathbb{Q}$-vector space $\Gamma_v \otimes_{\mathbb{Z}} \mathbb{Q}$. Hence

$$\big(N_B \otimes_{\mathbb{Z}} \mathbb{R}\big) \cap \big(N \otimes_{\mathbb{Z}} \Gamma_v \otimes_{\mathbb{Z}} \mathbb{Q}\big) = N_B \otimes_{\mathbb{Z}} \Gamma_v \otimes_{\mathbb{Z}} \mathbb{Q}$$

inside $N_{\mathbb{R}}$. This is exactly the asserted descent of rational valuation vectors from T to T_B. □

Proposition 126 (Special points restrict)**.** *Assume that $G(\overline{K})$ contains $\overline{L}$-special points and that $\mu_{\overline{L}}^{\mathrm{ess}}(X) = \mu_G$. Then the minimal envelope $Y = a + B$ contains an $\overline{L}$-special point.*

Proof. By Proposition 124, applied to the quasi-canonical replacement from Proposition 72, the existence of $\overline{L}$-special points on G is equivalent to

$$u_v \in \mathrm{val}_v(T(\overline{K})) \otimes_{\mathbb{Z}} \mathbb{Q}$$

for every v. Since $t_a \in T(\overline{K})$, also

$$\alpha_v = \mathrm{val}_v(t_a) \in \mathrm{val}_v(T(\overline{K})).$$

By Lemma 116,

$$u_v - \alpha_v = \iota_N(w_v).$$

Since $\alpha_v = \mathrm{val}_v(t_a)$, the vector $u_v - \alpha_v$ also lies in $\mathrm{val}_v(T(\overline{K})) \otimes_{\mathbb{Z}} \mathbb{Q}$. Applying Lemma 125 to the saturated subtorus $T_B \subseteq T$, the rational valuation condition descends to

$$w_v \in \mathrm{val}_v(T_B(\overline{K})) \otimes_{\mathbb{Z}} \mathbb{Q}.$$

Proposition 124 applied to $\overline{D}'_{a,B}$ gives a point $t_B \in T_B(\overline{K})$ whose valuations are $\mathrm{val}_v(t_B) = w_v$ and whose height realizes the toric minimum for the quasi-canonical restricted metric. For the original restricted metric $\overline{D}_{a,B}$, the local valuation argument is

$$\iota_N(w_v) + \alpha_v = u_v.$$

Hence the toric height of t_B for the restricted original metric equals the ambient toric absolute minimum:

$$h_{\overline{B}(\overline{D}_{a,B})}(t_B) = \mu_G.$$

Here the abelian canonical part contributes 0 at the minimum, so the toric absolute minimum is the same number as the ambient minimum μ_G. The base point at_B has torsion image in A, because $\pi(a)$ is torsion and t_B lies in the torus. Hence

$$h_{\overline{L}}(at_B) = \mu_G.$$

Thus $at_B \in Y(\overline{K})$ is $\overline{L}$-special. □

Lemma 127 (Special translates have minimal essential minimum)**.** *Let $p \in G(\overline{K})$ be an $\overline{L}$-special point and let $B \subseteq G_{\overline{K}}$ be a connected algebraic subgroup. Then*

$$\mu_{\overline{L}}^{\mathrm{ess}}(p + B) = \mu_G.$$

Proof. The global lower bound gives

$$\mu_{\overline{L}}^{\mathrm{ess}}(p + B) \geq \mu_G.$$

It remains to produce a Zariski-dense set of points of height μ_G in $p + B$.

By the height decomposition

$$h_{\overline{L}}(p) = h_{\overline{G}(\overline{D})}(p) + \hat{h}_{\overline{N}}(\pi(p)),$$

together with $h_{\overline{G}(\overline{D})}(p) \geq \mu_G$ and $\hat{h}_{\overline{N}} \geq 0$, the equality $h_{\overline{L}}(p) = \mu_G$ forces $\hat{h}_{\overline{N}}(\pi(p)) = 0$. Hence $\pi(p)$ is torsion. Choose a torsion lift $q \in G(\overline{K})$ of $\pi(p)$ and write

$$p = q\,t$$

with $t \in T(\overline{K})$. Corollary 22 gives

$$h_{\overline{D}}(t) = h_{\overline{L}}(p) = \mu_G.$$

Torsion points are Zariski dense in the connected semiabelian variety B. For every torsion point $\xi \in B(\overline{K})$, the point $q\xi$ is torsion in G, and

$$p\xi = (q\xi)t.$$

Applying Corollary 22 again gives

$$h_{\overline{L}}(p\xi) = h_{\overline{D}}(t) = \mu_G.$$

Thus $p+B$ contains a Zariski-dense set of points of height μ_G, proving the reverse inequality. □

9.13. **Restriction Package and Bogomolov.**

Theorem 128 (Restriction package sufficient for Bogomolov)**.** *Let $Y = a + B$ be the minimal translate of a connected subgroup containing X, and assume*

$$\mu_{\overline{L}}^{\mathrm{ess}}(X) = \mu_G.$$

Then:

(1) a may be chosen with $\pi(a)$ torsion.
(2) The ambient critical vector satisfies

$$u_v - \mathrm{val}_v(t_a) \in \iota_N(N_{B,\mathbb{R}})$$

for every v.
(3) The quasi-canonical metric from Proposition 72 restricts to a quasi-canonical metric on B, with critical vector w_v defined by

$$\iota_N(w_v) + \mathrm{val}_v(t_a) = u_v.$$

(4) If $X \neq Y$, any X-generic small sequence is strict in Y and equidistributes toward a measure with Zariski-dense support in Y.
(5) If G has $\overline{L}$-special points, then Y has $\overline{L}$-special points.

Proof. The assertions are Lemma 114, Lemma 116, Proposition 118, Proposition 120, and Proposition 126. □

Theorem 129 (Bogomolov for the paper's metric class)**.** *Let $X \subseteq G_{\overline{K}}$ be irreducible, and assume that the ambient toric metric $\overline{D}$ satisfies the standing hypotheses of the Bogomolov part: D is ample and T-effective, and $\overline{D}$ is semipositive, monocritical, and arithmetically T-effective. At every archimedean place, assume in addition either the standard $(\mathbb{P}^1)^t$-compactification used by Kühne, or, for every subvariety occurring in the difference-morphism argument, the three general-fan inputs of Subsection 9.5: the product-corner normal form, a primitive frame in the saturated active lattice, and absence of additional boundary mass. Equivalently, the conditional curvature–Haar slice formula 9.5, with its normalization and boundary assertions, is required for all those subvarieties.*

(1) If

$$\mu_{\overline{L}}^{\mathrm{ess}}(X) = \mu_G,$$

then X is special.
(2) If $G(\overline{K})$ contains $\overline{L}$-special points, then

$$\mu_{\overline{L}}^{\mathrm{ess}}(X) = \mu_G$$

if and only if X is $\overline{L}$-special.

Proof. By Proposition 100 and Corollaries 107 and 108, the restricted quasi-canonical strong equidistribution theorem applies to every restricted quasi-canonical semiabelian metric $\overline{L}'_{a,B}$ attached to a minimal translate occurring below.

Let $Y = a + B$ be the minimal special envelope of X.

Assume first that $\mu_{\overline{L}}^{\mathrm{ess}}(X) = \mu_G$. If $X \neq Y$, choose an X-generic $\overline{L}$-small sequence. By Theorem 106, Theorem 128 applies, and this sequence is strict in Y and equidistributes toward a measure with Zariski-dense support in Y. Since all terms of the sequence lie in X, every weak limit is supported on X_v^{an}. Hence X must be Zariski dense in Y, contradicting $X \neq Y$. Therefore $X = Y$, so X is special. This proves (1).

For (2), first suppose that $X = p + B$ with p $\overline{L}$-special. Lemma 127 gives

$$\mu_{\overline{L}}^{\mathrm{ess}}(X) = \mu_G.$$

Conversely assume $\mu_{\overline{L}}^{\mathrm{ess}}(X) = \mu_G$ and that $G(\overline{K})$ contains $\overline{L}$-special points. By (1), X equals its minimal special envelope $Y = a + B$. The same restricted strong equidistribution theorem and Theorem 128 then say that Y contains an $\overline{L}$-special point. Hence X is $\overline{L}$-special. □

Appendix A. Further Directions: A Relative Toric Route

The proof of Theorem 129 proceeds through the direct Kühne-operation transport package, Proposition 100, together with the quasi-canonical Bogomolov and strong equidistribution results proved in the main text. A natural continuation is to develop a relative version over the abelian quotient. We record the main ingredients of such a route.

(i) A relative toric fiber-specialness theorem. Smallness on the semiabelian variety should force the generic toric fiber, after quotienting by its maximal toric stabilizer, to be special in the corresponding function-field torus.

(ii) A global-to-generic-fiber height comparison. The global height gap on the semiabelian variety should control a positive generic-fiber height gap after restricting to a suitable open subset of the abelian base.

(iii) A function-field toric Bogomolov theorem with descent. The generic-fiber height gap should descend through the relevant torus torsor and identify the quotient sections which are forced by small points.

The first ingredient belongs to a relative equidistribution/Bogomolov problem over the abelian base; the second is a height-specialization problem; the third is a function-field toric Bogomolov/descent problem. Together they would give a fiberwise approach to specialness in semiabelian varieties, in which the generic toric fiber detects the toric component of a small sequence.

Such a project could produce a relative version of the present theorem, where small points are studied fiberwise over the abelian quotient and specialness is detected on the generic toric fiber.

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